

# Babet Manual

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## Babet — User manual

*Babet*, n.m. — a regional word from south-eastern France (Lyonnais, Forez, Dauphiné, Savoie, and neighbouring Swiss Romandy) meaning a pine cone. Small, light, full of seeds, and able to start a fire—like this binary.

Babet is a standalone Lua binary for Linux scripting and automation. This manual is organised by need and by module: the tables below describe both the technical chapter name and the features it contains.

Documentation for the stable and audited **Babet 2.9.2** release.

### Getting started

- [Getting started](#) — installation, first script, running a file or directory, creating an embedded executable, and using Babet through PATH.
- [Security](#) — threat model, actual protections, limitations, and least-privilege rules.
- [Cookbook](#) — complete recipes combining several modules.
- [Changelog](#) — complete 2.9.2 release notes and the history of earlier releases, migration, validation, and known limitations.

### Find a feature

| I need to...                                      | Chapter                                                | What it contains                                                                                                                                                         |
|---------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| create, remove, list, search, copy, or move files | <a href="#">FS — filesystem</a>                        | files, directories, paths, symlinks, listFiles, find, copyTree, permissions, and checksums                                                                               |
| atomically publish configuration or binary data   | <a href="#">writeFileAtomic — atomic writing</a>       | default no-overwrite, permissions, durability, path confinement, concurrency, and workers                                                                                |
| parse script arguments                            | <a href="#">Argparse — command line</a>                | flags, options, positional arguments, defaults, choices, and conversion                                                                                                  |
| run an external program                           | <a href="#">Exec — processes</a>                       | shell-free arguments, complete capture with exec, streaming with spawn, environment, cwd, and process control                                                            |
| chain several commands                            | <a href="#">Process pipelines</a>                      | complete capture or streaming, separate stderr streams, per-stage statuses, and process-group cleanup                                                                    |
| create or inspect/extract ZIP or TAR archives     | <a href="#">Archive — secure multi-format archives</a> | deterministic ZIP and plain/gzip/xz/bzip2/zstd TAR creation, listing, full testing, selective extraction, dry_run preview, anti-bomb limits, and atomic file publication |

| I need to...                                                | Chapter                                                      | What it contains                                                                                                                    |
|-------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| encode or decode binary data as Base64                      | <a href="#">Base64 – binary data</a>                         | standard or URL-safe alphabets, optional padding, strict canonical decoding, controlled whitespace, and an output limit             |
| compress or decompress one file                             | <a href="#">Compression – standalone streams</a>             | gzip, xz, bzip2, and zstd streams, content detection, output limits, concatenated members, integrity checks, and atomic publication |
| inspect the process, environment, machine, or Lua VM memory | <a href="#">SYS – process and machine</a>                    | runtime version, PID, hostname, uname, PATH, env, setenv, and Lua-state memory                                                      |
| send an HTTP request                                        | <a href="#">HTTP – web client</a>                            | GET/POST and other methods, query, validated headers, binary bodies, atomic downloads, redirects, TLS, timeout, and size limits     |
| open a TCP connection                                       | <a href="#">Socket – TCP</a>                                 | client, server, accept, binary streams, lines, EOF reads, timeouts, buffers, and limits                                             |
| encrypt TCP or perform STARTTLS                             | <a href="#">TLS – secure sockets</a>                         | direct TLS, STARTTLS, verification, CA, hostname, SNI, versions, timeout, and fail-closed state                                     |
| store SQL data locally                                      | <a href="#">SQLite – embedded database</a>                   | opening, execution, prepared statements, explicit BLOBs, and assisted transactions                                                  |
| encode or decode JSON                                       | <a href="#">JSON – structured data</a>                       | scalars, arrays, objects, null, empty arrays, pretty printing, and errors                                                           |
| read TOML configuration                                     | <a href="#">TOML – configuration</a>                         | decoding, TOML types, arrays, sections, dates, and parse errors                                                                     |
| watch a directory                                           | <a href="#">Inotify – file events</a>                        | watches, events, timeout reads, moves, cookies, and closing                                                                         |
| handle Unix signals                                         | <a href="#">SIGNAL – clean shutdown and reload</a>           | TERM/INT/HUP/USR1/USR2/PIPE, deferred callbacks, fixed order, coalescing, interruptible calls, and workers                          |
| parallelise Lua work                                        | <a href="#">WORKERS – OS threads, messages, and channels</a> | isolated Lua states, spawn, bounded lifecycle, inbox/outbox, and direct MPMC channels                                               |
| obtain or format time                                       | <a href="#">Time – clocks and durations</a>                  | realtime and monotonic clocks, sleep, ISO-8601, parsing, and duration formatting                                                    |
| look up a system account by name or UID                     | <a href="#">USER – system accounts</a>                       | NSS, get, exists, UID, primary GID, GECOS, home, shell, and resolution errors                                                       |
| split or transform strings                                  | <a href="#">Strings</a>                                      | split, separators, limits, character mode, and binary strings                                                                       |
| copy, merge, or mark tables                                 | <a href="#">Tables</a>                                       | mergeTables, deepCopyTable, cycles, and shared structures                                                                           |

| I need to... | Chapter                 | What it contains                                                  |
|--------------|-------------------------|-------------------------------------------------------------------|
| produce logs | <a href="#">Logging</a> | levels, threshold, destination, colours, and sink-error behaviour |

## Module reference

Each module has a standalone page under `modules/`. The expanded names below define each chapter's scope before you open it.

| Module                                                                          | Detailed scope                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Archive – secure ZIP, TAR, gzip, xz, bzip2, and zstd operations</a> | Create deterministic ZIP, TAR, gzip TAR, xz TAR, bzip2 TAR, or zstd TAR archives, then inspect, test, and selectively extract them with <code>dry_run</code> previews, bounded resources, confined paths, and atomic per-file publication. |
| <a href="#">Base64 – binary encoding and decoding</a>                           | Encode or decode binary Lua strings with standard or URL-safe alphabets, controlled padding, canonical validation, optional whitespace, and a <code>max_output</code> ceiling.                                                             |
| <a href="#">Compression – standalone gzip, xz, bzip2, and zstd streams</a>      | Compress or decompress one regular file with automatic content detection, bounded expansion, integrity verification, symlink refusal, and atomic output publication.                                                                       |
| <a href="#">Argparse – command-line arguments</a>                               | Declare flags, options, and positional values; generate help; validate choices; convert values.                                                                                                                                            |
| <a href="#">Exec – external programs and processes</a>                          | Run without a shell using <code>exec</code> , or progressively drive <code>stdin/stdout/stderr</code> using <code>spawn</code> .                                                                                                           |
| <a href="#">Process pipelines</a>                                               | Connect commands without a shell, using complete capture or streaming and per-stage statuses.                                                                                                                                              |
| <a href="#">FS – files, directories, paths, and attributes</a>                  | Existence and types, creation/removal, paths, listings, search, iterators, tree copy/move, symlinks, Unix modes, and checksums.                                                                                                            |
| <a href="#">writeFileAtomic – atomic and durable file writing</a>               | Publish a binary Lua string through a private temporary file and atomic rename, with explicit overwrite, exact permissions, <code>fsync</code> , path confinement, and worker support.                                                     |
| <a href="#">HTTP – web requests</a>                                             | URLs, methods, query, validated headers, binary bodies, responses, atomic file downloads, redirects, TLS verification, timeout, and limits.                                                                                                |
| <a href="#">Inotify – filesystem monitoring</a>                                 | Add/remove watches, read events, handle timeouts, moves, cookies, and closing.                                                                                                                                                             |
| <a href="#">JSON – encoding and decoding</a>                                    | Lua/JSON types, null, empty arrays, array marking, indentation, UTF-8, cycles, and limits.                                                                                                                                                 |
| <a href="#">Logging</a>                                                         | Levels, filtering, destination, colours, and variadic calls.                                                                                                                                                                               |
| <a href="#">SIGNAL – POSIX signals and clean shutdown</a>                       | Install, replace, or remove callbacks; ignore/restore; dispatch order, coalescing, interruptions, and multithread restrictions.                                                                                                            |
| <a href="#">Socket – TCP client and server</a>                                  | Connect, listen, accept, binary streams, block/line/EOF reads, shared buffers, addresses, closing, and timeouts.                                                                                                                           |

| Module                                                              | Detailed scope                                                                                                                                             |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">SQLite – embedded database</a>                          | Connections, WAL/timeout, direct SQL, reusable prepared statements, explicit BLOBs, iteration, and assisted transactions.                                  |
| <a href="#">Strings</a>                                             | Splitting, character mode, separators, split limits, and binary content.                                                                                   |
| <a href="#">SYS – process, machine, environment, and Lua memory</a> | Version constants, PID, hostname, uname, executable lookup, environment reads/writes, worker interaction, and Lua-state memory.                            |
| <a href="#">Tables</a>                                              | Deterministic merging, list/map keys, deep copy, cycles, and shared subtables.                                                                             |
| <a href="#">Time – clocks, ISO, and durations</a>                   | Realtime and monotonic clocks, sleep, ISO-8601 formatting/parsing, and duration parsing/rendering.                                                         |
| <a href="#">TLS – encrypted connections</a>                         | Direct TLS, STARTTLS, verification, CA, hostname, SNI, versions, deadlines, and fail-closed behaviour.                                                     |
| <a href="#">TOML – configuration files</a>                          | TOML decoding, scalars, arrays, tables, arrays of tables, dates/times, and diagnostics.                                                                    |
| <a href="#">USER – NSS system users</a>                             | Lookup by name or UID, existence, absence vs NSS errors, passwd fields, workers, and security limits.                                                      |
| <a href="#">WORKERS – OS threads, queues, and channels</a>          | Isolated Lua states, JSON transport, status, join(timeout), cooperative cancellation, inbox/outbox, direct MPMC channels, closing, GC, and deadlock traps. |

## Structure of a module page

Detailed pages are progressively aligned on this structure:

1. **Scope** — what the module covers and deliberately excludes;
2. **Internal contents** — direct access to each function group;
3. **API overview** — signatures and results;
4. **Detailed behaviour** — defaults, symlinks, recursion, limits, and side effects;
5. **Examples** — one example per important mode or option;
6. **Error contract** — raised errors and returned errors;
7. **Design decisions and limitations** — intentional choices and omissions.

The goal is to let a function be used without reading its source code or having to guess default values.

## PDF generation

The whole manual can be exported as one PDF:

```
cd docs
./build_doc.sh
```

English chapter order is defined in `manual_order_en.txt`.

## Getting started

### Download

Prebuilt binaries for x86\_64, aarch64 (RPi4), and armv6l (RPi0) are available on the [releases page](#).

## Build from source

Babet vendors all its dependencies (Lua, OpenSSL, SQLite, miniz, libarchive, zlib, XZ Utils/liblzma, bzip2/libbz2, Zstandard/libzstd, RE2, Abseil, nlohmann/json, cpp-httplib, tomlplusplus), so the only prerequisites on your system are :

- a C++23 compiler (recent GCC or Clang)
- CMake ( $\geq 3.22$ )
- wget, unzip, and xz

```
git clone https://github.com/Chipsterjulien/babet.git
cd babet
./build_local.sh           # downloads deps and compiles
                           # (~5 min on first run, faster afterwards)
./run_tests.sh            # offline harness – should finish with 0 FAIL
```

The build script downloads each dependency from a pinned upstream source, verifies its SHA256, then compiles it statically. Several dependencies have an explicit mirror or fallback URL. When a source has no mirror, the script prints the exact archive name to place manually in downloads/; the same SHA256 check still applies before extraction.

The resulting binary is at test/babet.

## Three ways to run a Lua script

Babet supports three launch modes :

### 1. Folder mode

Point the binary at a directory containing main.lua :

```
echo 'print("hello from Babet")' > main.lua
./test/babet .
```

require("X") in main.lua looks for X.lua in the same directory.

### 2. Embedded mode (single-file executable)

Package a script and its modules into a self-contained binary :

```
./test/babet --create-exe . myapp
./myapp           # runs main.lua from the embedded ZIP
```

Version-control metadata directories .git, .svn, and .hg are automatically excluded at any depth.

Internally, Babet appends a ZIP to its own binary, and reads main.lua (plus any required module) from that ZIP at runtime. The resulting myapp is fully self-contained — no Babet install needed on the target machine, just glibc.

### 3. Embedded via PATH (folder + auto-detect)

If Babet is on \$PATH and you chmod +x main.lua after adding a shebang line :

```
#!/usr/bin/env babet
print("hello")

chmod +x main.lua
./main.lua
```

Babet uses the script's own directory as the folder, so require("helpers") finds helpers.lua next to main.lua.

## Command-line invocation

In addition to the three launch modes above, Babet accepts a few standard flags :

| Flag                                     | Effect                                                                                           |
|------------------------------------------|--------------------------------------------------------------------------------------------------|
| -h, --help                               | Print the help text and exit 0.                                                                  |
| -V, --version                            | Print babet <version> and exit 0.                                                                |
| -c <dir> <out>, --create-exe <dir> <out> | Create a self-contained executable named <out> by embedding <dir> (which must contain main.lua). |

Any other argument starting with - is treated as an **unknown option** : Babet prints Unknown option: ... + a hint to use --help, and exits 1 instead of trying to interpret it as a directory name. Folders whose name legitimately starts with - can still be passed via ./-dirname (POSIX convention).

```
babet --version    # babet 2.9.2
babet --help      # full usage
babet --bogus     # Unknown option: --bogus
                  # Try 'babet --help' for more information.
```

The same version is also exposed to scripts as babet.VERSION (plus babet.VERSION\_MAJOR / VERSION\_MINOR / VERSION\_PATCH as integers) — see [sys](#).

## First script

Once the binary is built, try this :

```
-- main.lua
print("Babet says hi")
print("PID:", babet.pid())
print("Host:", babet.hostname())

local r, err = babet.http.get("https://example.com/")
if r then
    print("Status:", r.status)
else
    print("HTTP failed:", err)
end
```

Run it with ./test/babet . (or whichever mode you prefer).

## Where to look next

- Each module has its own page under modules/.
- The [user](#) module is a small, complete example of the documentation pattern used throughout — read it as the canonical reference.
- For date and duration utilities (ISO 8601 timestamps, human durations like "5m" or "2h30m"), see [time](#) — the babet.time.\* sub-table covers parsing and formatting.
- See [security.md](#) before exposing Babet scripts to anything that might receive untrusted input.

## Security model

**Babet runs trusted Lua scripts. It is not a sandbox.**

Scripts execute with the full Lua standard library (luaL\_openlibs), and Babet additionally exposes filesystem and process primitives such as exec, remove, rmdirAll and chdir. A script therefore has

the same privileges as the process running it : it can read, modify or delete files and run arbitrary commands. This is by design — like make, a shell script, or any build tool, Babet is meant to run code you control.

Only run `main.lua` and required modules that you trust. Do **not** use Babet to execute Lua from untrusted sources ; it provides no isolation against hostile code, and none is intended. If you need to run untrusted scripts, use a purpose-built Lua sandbox instead.

## What the hardening *does* protect against

The hardening work in Babet protects *legitimate* use against accidents and supply-chain tampering :

- **copyTree / moveTree confinement** : the destination root is opened once, then every create, copy and move is performed relative to that descriptor with `openat/mkdirat/renameat` and `O_NOFOLLOW/AT_SYMLINK_NOFOLLOW`. A pre-existing symlink, or one swapped during the operation, cannot redirect a write to an outside target. The `is_within()` guards additionally reject a destination resolving inside the source.
- **Confined archive creation and extraction**: `babet.archive` refuses symlinked creation-source and output/extraction-destination paths, selected unsupported source objects, non-UTF-8 selected source names, outputs inside the source tree, absolute archive-entry paths, `./..` components, backslashes, duplicates, ZIP symlinks, and special filesystem types. Input archives are opened once and must resolve to a regular file, excluding blocking FIFO sources. It walks the destination through descriptors with `O_NOFOLLOW`; each file is decompressed into a same-directory temporary, checked against anti-bomb limits, then published atomically. Creation-side include/exclude rules reuse the non-recursive safe-glob engine, prune excluded directories before opening, and enforce per-pattern, cumulative pattern-text, pattern-evaluation, matching-work, and pattern-count limits.
- **Standalone compression confinement**: `babet.compression` accepts only real regular source files, rejects symlink path components and same-inode destinations, pins and revalidates the source descriptor, bounds decompressed output, verifies codec integrity, rejects trailing junk, stages output beside the destination, and publishes atomically.
- **General atomic file writing**: `writeFileAtomic()` opens every parent without following symlinks, rejects `..`, a final symlink or non-regular destination, and implicit overwrite. Data is written to a private same-directory temporary file, permissions are applied, then the file and directory are synchronized around atomic publication.
- **Bounded process-group cleanup** in `babet.exec` also covers the `chdir/exec` launch phase, internal polling failures, and children that close all pipes while continuing to run. On timeout, `TERM` then `KILL` target the whole group; these error paths never end in an unbounded final `waitpid`.
- **Output limits** on `babet.exec` bound the amount of `stdout` and `stderr` captured into memory (default 10 MiB each, configurable), so a runaway subprocess can't OOM the Babet process.
- **Pinned local metadata updates**: `setAttributes` resolves its target once and keeps `chown`, `chmod`, and `rollback` attached to the same inode. `touch` creates with `O_EXCL`, never truncates an existing file, and pins an existing target before changing its timestamp.
- **Anonymous and pinned executable staging**: `--create-exe` unlinks its temporary ZIP immediately after `mkstemp` and lets `miniz` and the merge step reopen the still-live inode through `/proc/self/fd`, not through a replaceable temp path. The final merge pins the output directory and publishes with `renameat()`, so changing a parent symlink cannot redirect the executable.
- **Dependency checksums** : every vendored dependency is SHA256-pinned in `build_local.sh`. An empty hash refuses to build. A mismatch deletes the downloaded file and exits with an error. This protects against a compromised upstream or Wayback Machine archive (which is the fallback source).
- **TLS verification** is on by default (`verify=true`). Disabling it requires an explicit per-call `verify=false` — there is no silent fallback. TLS sockets accept extra trust through `ca_cert` or `ca_path`; HTTP exposes `ca_cert`. OpenSSL's `SSL_CERT_FILE` / `SSL_CERT_DIR` remain available. SNI is sent independently from verification when a DNS hostname is known.

- **HTTP line validation** rejects CR/LF in URLs and header values, and header names must follow the HTTP token grammar. User-controlled values cannot inject an extra header or request line through those fields.
- **Network memory limits** bound `socket:recv_line`, `socket:recv_all`, and `http.max_body_size`. After a socket-read timeout, already consumed bytes stay in one shared pending buffer so switching receive methods cannot lose or reorder the stream.
- **Bounded filename matching** provides a non-recursive safe glob engine for `babot.find` and archive creation, plus RE2-backed regular expressions for `babot.find`. Regex patterns are capped at 4096 bytes, compiled with a 1 MiB memory budget, and cannot trigger catastrophic backtracking. Archive filter lists add cumulative count, byte, and matching-work ceilings. RE2-unsupported constructs fail before traversal.

## What it does *not* protect against

- A malicious script. Babet has no sandbox. `babot.exec` does not invoke a shell, but allowing untrusted input to choose the executable or its arguments can still permit arbitrary program execution or option injection. Passing untrusted text to `load` gives arbitrary Lua code execution.
- A determined attacker who has obtained code execution on the machine running Babet. The hardening makes accidental mistakes loud, not adversarial attacks impossible.
- Long-tail OS-level issues (kernel exploits, container escapes, privilege escalation). Babet is an ordinary userland binary.

## Recommended pattern : least privilege

When running Babet as a service, treat it like any other unprivileged process :

*# As root, create a dedicated service user with no shell.*

```
useradd --system \
        --home-dir /var/lib/myapp \
        --create-home \
        --shell /usr/sbin/nologin \
        --comment "myapp Babet service" \
        myapp
```

*# Use `babot.user.exists("myapp")` in your install script to  
# verify the user is provisioned before launching the daemon.*

Combine with `systemd` unit file options like `User=myapp`, `PrivateTmp=yes`, `ProtectSystem=strict`, `NoNewPrivileges=yes`, and `CapabilityBoundingSet=` (empty unless you genuinely need a capability). The kernel does much more than Babet can to contain a misbehaving script ; let it.

## Archive — secure ZIP and TAR operations with `gzip`, `xz`, `bzip2`, or `zstd`

### Scope

The `babot.archive` submodule creates, inspects, and extracts **ZIP**, **TAR**, and **gzip-**, **xz-**, **bzip2-**, or **zstd-compressed TAR** archives without calling an external command:

```
babot.archive.create(source_or_sources, archive [, opts])
babot.archive.list(archive [, opts])
babot.archive.test(archive [, opts])
babot.archive.extract(archive, destination [, opts])
babot.archive.extractFile(archive, entry, destination [, opts])
```

ZIP creation, inspection, and extraction continue to use miniz. TAR creation, inspection, and extraction use the statically linked libarchive backend; gzip filtering is provided by the statically linked zlib backend, xz filtering uses the statically linked XZ Utils/liblzma backend, bzip2 filtering uses the statically linked bzip2/libbz2 backend, and zstd filtering uses the statically linked Zstandard/libzstd backend. Readers detect both the container and compression from the contents. For `create()`, `.tar` selects plain TAR, `.tar.gz` and `.tgz` select gzip-compressed TAR, `.tar.xz` or `.txz` select xz-compressed TAR, `.tar.bz2`, `.tbz2`, or `.tbz` select bzip2-compressed TAR, and `.tar.zst`, `.tar.zstd`, or `.tzst` select zstd-compressed TAR. `opts.format` can override the extension. Standalone compressed streams are handled separately by [babet.compression](#).

All five functions use the usual result contract:

```
local result, err = babet.archive.list("backup.zip")
if not result then
    io.stderr:write(err, "\n")
end
```

They return `(result, nil)` on success and `(nil, message)` on failure. Wrong arity, a destination that is not a string, or a source that is neither a string nor a table raises a Lua error. Invalid source lists and option tables return `(nil, message)`.

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### **babet.archive.create**

```
local result, err = babet.archive.create(
    source_or_sources,
    archive
    [, opts]
)
```

The first argument supports two distinct contracts:

- a **string** preserves the historical behavior: it must name an existing directory, and Babet archives its contents without including the source directory name itself;
- a **non-empty dense array** (1..n, no holes or extra keys) explicitly selects regular files and/or directories, possibly from unrelated locations. Each source is rooted at its final component: `/tmp/report.txt` becomes `report.txt`, while `/opt/project/docs` becomes `docs/` with all descendants. Absolute sources are accepted, but host path prefixes are never exposed in the archive.

In list mode, the final component must be stable: `.` and the filesystem root are rejected, as is every `..` component. Two sources with the same final component collide and fail, even when their descendants would differ. Repeating the same source therefore fails as well. Input list order does not affect the result: entries are always sorted by archive name.

In both modes traversal is descriptor-based and follows no symlink. A symlink in a top-level source path always fails. Within a traversed tree, a selected symlink, FIFO, socket, device, or other unsupported object fails the operation before the archive is published; an object removed by the filters is not opened. Entry names produced by `create()` must be valid UTF-8. Linux permits arbitrary bytes in filenames, but Babet refuses names that cannot be represented safely and consistently in both output formats.

## Output format

Without `opts.format`, `create()` uses these rules:

- a destination ending in `.tar` (case-insensitive) creates an uncompressed POSIX pax TAR archive;
- `.tar.gz` and `.tgz` (case-insensitive) create a gzip-compressed POSIX pax TAR archive;
- `.tar.xz` and `.txz` (case-insensitive) create an xz-compressed POSIX pax TAR archive;
- `.tar.bz2`, `.tbz2`, and `.tbz` (case-insensitive) create a bzip2-compressed POSIX pax TAR archive;
- `.tar.zst`, `.tar.zstd`, and `.tzst` (case-insensitive) create a zstd-compressed POSIX pax TAR archive;
- every other destination keeps the pre-2.6 historical behaviour and creates a ZIP archive, regardless of its extension.

`opts.format = "zip", "tar", "tar.gz", "tar.xz", "tar.bz2", or "tar.zst"` explicitly selects a backend. An explicit format always overrides the destination suffix. This means, for example, that `format = "zip"` may create a ZIP named `backup.tar.gz`, while `format = "tar"` may create an uncompressed TAR named `backup.tgz` or `backup.txz`, `backup.tbz2`, or `backup.tzst`. Format selection is based on the explicit option or content, not trust in the suffix.

Creation options:

| Option                           | Default           | Behaviour                                                                                       |
|----------------------------------|-------------------|-------------------------------------------------------------------------------------------------|
| <code>format</code>              | inferred          | strict string: "zip", "tar", "tar.gz", "tar.xz", "tar.bz2", or "tar.zst"                        |
| <code>compression_level</code>   | 6                 | ZIP/gzip/xz: 0 through 9; bzip2: 1 through 9; zstd: 0 through 19                                |
| <code>overwrite</code>           | false             | atomically replaces an existing regular archive                                                 |
| <code>deterministic</code>       | true              | stable entry order and fixed format-specific timestamps                                         |
| <code>include_directories</code> | true              | emits explicit directory entries, including empty directories                                   |
| <code>include</code>             | none              | dense array of case-sensitive safe-glob patterns selecting archive entry paths                  |
| <code>exclude</code>             | none              | dense array of case-sensitive safe-glob patterns removed after inclusion; exclusion always wins |
| <code>max_entries</code>         | 10000             | maximum output entries; hard ceiling 100000                                                     |
| <code>max_file_size</code>       | 256 * 1024 * 1024 | maximum source-file size; hard ceiling 8 GiB                                                    |

| Option                      | Default            | Behaviour                                    |
|-----------------------------|--------------------|----------------------------------------------|
| <code>max_total_size</code> | 1024 * 1024 * 1024 | maximum source bytes;<br>hard ceiling 64 GiB |

For plain TAR, an explicitly supplied `compression_level` is rejected. For ZIP, gzip-compressed TAR, and xz-compressed TAR, level 0 is valid and level 9 is the maximum. bzip2 accepts levels 1 through 9; level 0 is rejected because libbz2 block sizes start at 100 KiB. zstd accepts levels 0 through 19; Babet deliberately excludes the ultra levels 20 through 22.

All integer options require strict Lua integers. Boolean options require real Lua booleans. Unknown keys are rejected. `format` is case-sensitive and does not accept aliases.

## Include and exclude filters

`include` and `exclude` are optional dense arrays of non-empty Lua strings. An absent or empty `include` list keeps the historical behaviour and initially selects every supported entry. A non-empty `include` list selects an entry when at least one pattern matches. `exclude` is then applied and always wins, even when the same path also matches `include`.

Patterns are matched against the complete path that will be stored in the archive, using `/` as the separator. The historical string form therefore matches paths relative to the source directory, while `explicit-list` sources include their final basename (`docs/readme.md`, `report.txt`, and so on). Matching is anchored, byte-oriented, and case-sensitive:

- `*` matches zero or more bytes except `/`;
- `**` matches zero or more bytes including `/`;
- `?` matches exactly one byte except `/`;
- `\x` quotes the following byte `x`.

Directories are tested both as path and `path/`. Consequently `build/**` selects or excludes the `build/` directory itself as well as its descendants. A directly excluded directory is pruned before it is opened, so nothing below it is inspected or archived. When a deep file is included, `include_directories = true` still emits every required parent directory even if those parents do not directly match `include`. A matched empty directory is preserved only when directory entries are enabled. With `include_directories = false`, only selected regular files are emitted and parents remain implicit.

Filtering defines the selected archive plan. A symlink, FIFO, socket, device, or invalid UTF-8 filename that is excluded or does not match a non-empty `include` list is ignored; the same object still fails creation when selected. The explicit paths supplied as top-level sources are always validated first and must themselves remain regular files or real directories reached without symlinks. It is valid for all filters to select nothing: Babet then creates a valid empty ZIP or TAR.

The safe-glob engine has no recursive backtracking. Each pattern is limited to 4096 bytes. `include` and `exclude` together are limited to 256 patterns and 256 KiB of pattern text, one million pattern evaluations, and a fixed 100,000,000-cell matching-work budget per creation call. Exceeding any limit fails before publication. Pattern order does not affect selection or deterministic archive bytes.

All source paths are validated and each non-pruned directory tree is scanned before the temporary output file is created. `max_entries`, `max_file_size`, and `max_total_size` apply only to entries selected by the filters. Each selected archive name is limited to 4096 bytes and their sum to 64 MiB. Traversal still has fixed ceilings of 100000 sources/filesystem objects and 256 directory levels, independent of `include_directories`; descendants of a pruned directory are not visited. Files are then reopened from their pinned source descriptor with `O_NOFOLLOW`. Device, inode, size, modification time, and change time are checked before and after streaming; a file modified during creation fails the operation and removes the temporary output.

The destination parent must already exist and may not contain symlink or `..` components. The destination itself must be absent unless `overwrite = true`, and it may never be a symlink or direc-

tory. The archive cannot be created inside the historical source directory or any explicitly selected directory, and cannot directly replace a selected regular file.

The result includes the selected format:

```
{
  files = 12,
  directories = 3,
  bytes = 987654,
  sources = 1,
  path = "backup.tar",
  format = "tar",
  compression = "none",
  compression_level = nil,
  deterministic = true,
  include_patterns = 0,
  exclude_patterns = 0,
}
```

`sources` is the number of supplied paths: 1 for the historical single source-directory contract, or the explicit list length. `include_patterns` and `exclude_patterns` report the number of compiled patterns supplied for that call. For ZIP and gzip-, xz-, bzip2-, or zstd-compressed TAR, `compression_level` is the effective integer level. For plain TAR it is `nil`. Archive-level compression is "gzip", "xz", "bzip2", or "zstd" for compressed TAR and "none" otherwise. ZIP entry compression remains reported through the ZIP-specific per-entry metadata returned by `list()`.

With the default deterministic mode, identical directory contents, format, pinned dependency versions, and options produce byte-for-byte identical archives:

- ZIP entries use the fixed DOS timestamp 1980-01-01 00:00:00;
- TAR entries use Unix epoch zero, UID/GID 0, empty owner/group names, mode 0644 for files, and mode 0755 for directories;
- gzip-compressed TAR additionally writes a gzip header timestamp of zero, so the outer stream does not inject wall-clock time;
- xz streams contain no wall-clock timestamp, and the selected LZMA2 preset is fixed by `compression_level`;
- bzip2 streams contain no wall-clock timestamp, and their block size is fixed by `compression_level`;
- zstd frames contain no wall-clock timestamp, enable their frame checksum, and use the compression level fixed by `compression_level`.

With `deterministic = false`, sorted entry order remains stable. ZIP uses each source modification time, which must fit the DOS range 1980 through 2107 in the local timezone. TAR preserves the source modification time with nanosecond precision when representable by the platform and is not limited to the ZIP DOS range.

Ownership, ACLs, extended attributes, and source permission bits are not copied. TAR uses libarchive's restricted POSIX pax writer: ordinary names use portable ustar headers where possible, while long paths or metadata that need extension records are represented with pax headers. ZIP64 output is enabled automatically when the preflight indicates that classic ZIP fields may be exceeded. Small ZIP archives remain classic ZIP files.

The preflight and file checks prevent symlink substitution and detect changes to every file that is actually archived. Creation is not a filesystem snapshot: a file added after its directory was scanned may be absent from the result; a previously enumerated file that is then removed or renamed normally makes its secure reopen fail. Concurrent directory renames are not reported as a transactional conflict.

When `include_directories = false`, empty directories cannot be represented and are therefore omitted. In an explicit list, regular files still retain the selected directory basename as their prefix. Required parents are recreated implicitly during extraction.

Examples:

```
local zip, err = babet.archive.create("project", "project.zip", {
    compression_level = 9,
    overwrite = true,
})
assert(zip, err)

local selected, list_err = babet.archive.create({
    "bin/babet",
    "README.md",
    "docs",
}, "release.tar.zst", {
    compression_level = 19,
})
assert(selected, list_err)
-- Entries: babet, README.md, docs/, docs/...

local filtered, filter_err = babet.archive.create("project", "source-only.tar.zst", {
    include = { "src/**", "README.md" },
    exclude = { "src/generated/**", "**/*.tmp" },
})
assert(filtered, filter_err)

local tar
tar, err = babet.archive.create("project", "project.tar", {
    deterministic = true,
})
assert(tar, err)

local compressed
compressed, err = babet.archive.create("project", "project.tar.gz", {
    compression_level = 9,
})
assert(compressed, err)

local xz
xz, err = babet.archive.create("project", "project.tar.xz", {
    compression_level = 9,
})
assert(xz, err)

local bzip2
bzip2, err = babet.archive.create("project", "project.tar.bz2", {
    compression_level = 9,
})
assert(bzip2, err)

local zstd
zstd, err = babet.archive.create("project", "project.tar.zst", {
    compression_level = 19,
})
assert(zstd, err)

local named
named, err = babet.archive.create("project", "snapshot.data", {
    format = "tar.zst",
})
```

```
})
assert(named, err)
```

## Inspection/extraction options and anti-bomb limits

The following options are shared by `list()`, `test()`, `extract()`, and `extractFile()`. They apply to the **whole archive**, even when `extractFile()` requests only one entry: selecting a small file can never bypass global limits.

| Option                             | Default            | Accepted maximum       | Effect                                                                                                                                           |
|------------------------------------|--------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>max_entries</code>           | 10000              | 100000                 | maximum number of entries exposed by the archive                                                                                                 |
| <code>max_entry_size</code>        | 256 * 1024 * 1024  | 8 * 1024 <sup>3</sup>  | maximum expanded size of one non-directory entry                                                                                                 |
| <code>max_total_size</code>        | 1024 * 1024 * 1024 | 64 * 1024 <sup>3</sup> | maximum sum of expanded non-directory entry sizes                                                                                                |
| <code>max_path_length</code>       | 64 * 1024          | 1024 * 1024            | maximum raw entry-name length, in bytes, accepted for inspection                                                                                 |
| <code>max_total_name_bytes</code>  | 64 * 1024 * 1024   | 64 * 1024 * 1024       | maximum byte sum of all entry names; this option can tighten the fixed ceiling                                                                   |
| <code>max_compression_ratio</code> | 1000               | 1000000000             | per-entry ratio for ZIP; global expanded-file-bytes/archive-bytes ratio for gzip-, xz-, bzip2-, or zstd-compressed TAR; irrelevant for plain TAR |

The five integer limits must be **real, strictly positive Lua integers**. Floating-point values, even 10.0, booleans, numeric strings, zero, and values above the ceiling are rejected. `max_compression_ratio` must be finite and at least 1; NaN and infinity are rejected. An unknown key is always an error.

`max_path_length` bounds metadata accepted by Babet; it does not make a long path extractable. Extraction path policy remains stricter: a name longer than **4096 bytes** is reported by `list()` with `safe_path = false`, even when its length remains below `max_path_length`. With the default value, a 4097-byte name can therefore be inspected and diagnosed without ever being eligible for extraction. Callers may lower `max_path_length` to reject the archive before the Lua result table is built.

`max_total_name_bytes` already equals its fixed hard ceiling by default. It can be lowered for an untrusted archive, but not increased. `list()` reports actual usage as `total_name_bytes`. Allocations made by the miniz ZIP reader are also capped at 128 MiB per archive, so an excessive central directory or ZIP comment cannot trigger arbitrary allocation before per-entry checks.

TAR inspection is streaming: Babet reads every entry data stream without keeping it in memory, both to reach the following header and to detect truncation. For gzip TAR, bounded streaming validation with zlib also checks each member's CRC and ISIZE; standard zero padding is accepted, while corrupt trailers or foreign trailing bytes are rejected. For zstd TAR, an independent bounded libzstd pass requires complete frames, accepts valid concatenated frames, and rejects corruption, truncation, or foreign trailing bytes. xz and bzip2 filters remain fully consumed by libarchive.

For complete extraction, the output tree is additionally capped internally at 100000 unique directories, including implicit parents, and 64 MiB of cumulative normalised directory paths. These fixed ceilings are not configurable.

A non-empty entry announcing a zero compressed size is rejected. Directories do not count towards `max_entry_size`, `max_total_size`, or the compression ratio, but they do count towards `max_entries` and name budgets.

### One example per limit

Limit only the entry count:

```
local result, err = babet.archive.test("upload.zip", {
    max_entries = 2000,
})
assert(result, err)
```

Limit only one entry's expanded size:

```
local result, err = babet.archive.test("upload.zip", {
    max_entry_size = 64 * 1024 * 1024,
})
assert(result, err)
```

Limit only the announced total size:

```
local result, err = babet.archive.test("upload.zip", {
    max_total_size = 512 * 1024 * 1024,
})
assert(result, err)
```

Limit only raw name length:

```
local result, err = babet.archive.test("upload.zip", {
    max_path_length = 8192,
})
assert(result, err)
```

Limit only cumulative name bytes:

```
local result, err = babet.archive.test("upload.zip", {
    max_total_name_bytes = 2 * 1024 * 1024,
})
assert(result, err)
```

Limit only the compression ratio:

```
local result, err = babet.archive.test("upload.zip", {
    max_compression_ratio = 200,
})
assert(result, err)
```

Combine every protection for a third-party archive:

```
local result, err = babet.archive.test("upload.zip", {
    max_entries = 2000,
    max_entry_size = 64 * 1024 * 1024,
```

```

    max_total_size = 512 * 1024 * 1024,
    max_path_length = 8192,
    max_total_name_bytes = 2 * 1024 * 1024,
    max_compression_ratio = 200,
})
assert(result, err)

```

Creation or write-specific options such as `overwrite`, `preserve_permissions`, `format`, `dry_run`, `include`, or `exclude` are not accepted by `list()` or `test()`.

## babet.archive.list

```
local info, err = babet.archive.list(archive [, opts])
```

`archive` is the path to a ZIP, plain TAR, gzip-compressed TAR, xz-compressed TAR, bzip2-compressed TAR, or zstd-compressed TAR file. Format and compression are detected from **content**, never from the extension: a zstd TAR named `snapshot.bin` is recognised, while a file named `snapshot.tar.zst` containing something else is rejected. The usual `.tar`, `.tar.gz`, `.tgz`, `.tar.xz`, `.txz`, `.tar.bz2`, `.tbz`, `.tbz2`, `.tar.zst`, `.tar.zstd`, and `.tzst` suffixes are naming conventions only.

The source path may be a symlink to a regular file. Babet opens its target once and keeps that same descriptor for the whole scan. Directories, FIFOs, sockets, and devices are rejected before format parsing; non-blocking open notably prevents a FIFO with no writer from hanging the call.

Babet first tries the miniz ZIP reader. If the pinned file is not ZIP, the same descriptor is rewound and passed to libarchive's TAR reader, with only the built-in `none`, `gzip`, `xz`, `bzip2`, and `zstd` filters enabled. No external decompressor is executed. Split ZIP files and multi-volume archives are not supported. A standard empty archive is valid and returns `count = 0`.

`list()` creates **no file, directory, or temporary file**. It preserves the physical/logical entry order found in the archive: entries are not sorted, deduplicated, or rearranged. `entries[1]` is always the first encountered entry, `entries[2]` the second, and so on. This stable order lets `index`, `duplicate_of`, and `conflict_with` directly reference the returned array.

The function returns exactly two values: `(info, nil)` on success and `(nil, message)` on failure.

## Returned structure

```

{
  format = "zip",           -- "zip" or "tar"
  compression = "none",    -- "none", "gzip", "xz", "bzip2", or "zstd"
  entries = {
    {
      index = 1,
      name = "docs/readme.txt", -- raw name, binary Lua string
      path = "docs/readme.txt", -- normalised path when safe_path == true
      valid_utf8 = true,
      type = "file",
      size = 1234,

      compressed_size = 530,    -- ZIP only, otherwise nil
      crc32 = 305419896,       -- ZIP only, otherwise nil
      compression_method = 8,  -- ZIP only, otherwise nil
      encrypted = false,
      supported = true,

      safe_path = true,
      extractable = true,
      reason = nil,
    }
  }
}

```

```

        unix_mode = 420,                -- 0644, or nil
        mtime = 1750000000,            -- Unix seconds, or nil
        mtime_nsec = nil,              -- TAR only, or nil
        uid = nil,                     -- TAR only, or nil
        gid = nil,                     -- TAR only, or nil
        sparse = false,
        link_target = nil,

        duplicate = false,
        duplicate_of = nil,
        conflict = false,
        conflict_with = nil,
        conflict_reason = nil,
    },
},
count = 1,
total_size = 1234,
archive_size = 666,
total_name_bytes = 19,
duplicates = 0,
conflicts = 0,
zip64 = false,                        -- nil for TAR
}

```

format describes the container. For ZIP, compression is always "none" because compression is chosen per entry and reported by `compression_method`. For TAR, compression is "none", "gzip", "xz", "bzip2", or "zstd" according to the detected filter.

`count` is the number of elements in `entries`. `total_size` is the sum of announced non-directory entry sizes. `archive_size` is the size of the pinned archive file. `total_name_bytes` is the exact byte sum of all name fields. `zip64` is a boolean for ZIP and nil for TAR.

### Fields shared by every entry

- `index` is the 1-based Lua index in archive order;
- `name` preserves the raw name as a binary Lua string;
- `path` is the normalised extraction form. A directory's trailing `/` is removed. It is usable as a path only when `safe_path == true`, and may be empty for an unsafe entry;
- `valid_utf8` says whether name is strict valid UTF-8. Babet never converts, replaces, or normalises the bytes;
- `type` is file, directory, symlink, hardlink, fifo, character\_device, block\_device, socket, or unsupported;
- `size` is the announced expanded size. It is normally 0 for directories or special objects, according to the actual metadata read;
- `safe_path` judges the name only: relative path, `/` separator, non-empty components, no `.` or `..`, no Windows drive prefix, no NUL byte, a trailing `/` consistent with the type, and the fixed 4096-byte extractable-name limit;
- `extractable` judges the entry in isolation: safe path, allowed type, no encryption, and a supported method. It deliberately does not include a duplicate or collision with another entry; also inspect duplicate and conflict;
- `reason` is nil for an individually extractable entry, otherwise a stable reason such as symlink entries are refused or entry path contains `'.'` or `'..'`;
- `unix_mode` contains bits 0000 through 7777 when available, otherwise nil. Decimal 420 is octal 0644;
- `mtime` is modification time in seconds since the Unix epoch when available, otherwise nil;
- `mtime_nsec` is the nanosecond component of an available TAR timestamp. It is nil for ZIP;

- `uid` and `gid` are numeric TAR identifiers when explicitly present, otherwise `nil`. They are always `nil` for ZIP;
- `sparse` reports a sparse TAR file;
- `link_target` contains the raw target of a TAR symlink or hard link when exposed by `libarchive`, otherwise `nil`.

For ZIP, `mtime` comes from the entry's DOS timestamp converted by `miniz` to `time_t`. ZIP stores neither a time zone nor nanoseconds, so this is local calendar metadata and must be interpreted cautiously. For TAR, `mtime` may be negative or outside the DOS range, and `mtime_nsec` preserves available precision. Missing metadata is always represented by `nil`, never by an invented value.

## ZIP-specific fields

`compressed_size`, `crc32`, and `compression_method` mirror ZIP central directory metadata. `encrypted` reports the encryption flag and `supported` says whether `miniz` understands the compression method. These fields are useful for inspection, but `crc32` is not proof of integrity: `list()` does not inflate every ZIP payload. A readable central directory may therefore coexist with compressed data whose CRC failure is only found during reading or extraction. For complete validation of local headers, payloads, and CRCs, use [archive.test\(\)](#).

For TAR, `compressed_size`, `crc32`, and `compression_method` are `nil`, `encrypted` is `false`, and `supported` means the header was parsed. Any outer compression applies to the whole TAR stream and is reported by the global compression field.

## Binary names, unsafe paths, and normalisation

ZIP names are byte strings from the central directory. TAR names are native strings returned by `libarchive` after `ustar` prefix, GNU long-name, and `pax` path processing. Comparisons are exact, byte-oriented, and case-sensitive. No Unicode normalisation is applied: visually identical Unicode spellings remain distinct names. Invalid UTF-8 is preserved and reported by `valid_utf8 = false`; it is not automatically unsafe when all path rules pass.

Absolute paths, `.` or `..` components, doubled `/`, backslashes, `C:` drive prefixes, empty names, NUL bytes, and names longer than 4096 bytes are reported with `safe_path = false` and an explicit reason. `list()` continues with other entries while metadata limits remain satisfied. It never silently repairs an unsafe path.

## Duplicates and collisions

Diagnostics are attached to the **second and subsequent occurrences**:

- `duplicate = true` means an identical byte-for-byte name was already seen. `duplicate_of` is the first occurrence's index;
- `conflict = true` means this safe normalised output path cannot coexist with an earlier entry. `conflict_with` is the selected earlier index and `conflict_reason` is currently duplicate output path or file/directory path conflict;
- `duplicates` counts additional exact occurrences;
- `conflicts` counts additional occurrences that would make an extraction plan ambiguous or impossible.

An exact duplicate with a safe path is therefore both a duplicate and a conflict. Different raw names such as `docs/` and `docs` are not duplicates, but collide after normalisation and the latter receives `conflict = true`. An unsafe entry may be reported as a raw duplicate, but does not participate in output-path collision analysis because it has no safe extraction path.

`list()` does not turn these diagnostics into an error: inspecting a suspicious archive is its purpose. `extract()` still rejects duplicates and collisions before any write.

Diagnostic example:

```

local info, err = babet.archive.list("upload.zip")
assert(info, err)

for _, entry in ipairs(info.entries) do
    if not entry.safe_path then
        print(entry.index, entry.name, entry.reason)
    elseif entry.duplicate then
        print(entry.index, "duplicates entry", entry.duplicate_of)
    elseif entry.conflict then
        print(entry.index, entry.conflict_reason,
              "with entry", entry.conflict_with)
    end
end

```

## Usage examples

Print archive contents in stored order:

```

local info, err = babet.archive.list("backup.tar.zst")
assert(info, err)

print(info.format, info.compression, info.count)
for _, entry in ipairs(info.entries) do
    print(entry.index, entry.type, entry.path, entry.size)
end

```

Use optional metadata without assuming it exists:

```

local info, err = babet.archive.list("package.zip")
assert(info, err)

for _, entry in ipairs(info.entries) do
    if entry.mtime ~= nil then
        print(entry.name, os.date("%Y-%m-%d %H:%M:%S", entry.mtime))
    end
    if entry.unix_mode ~= nil then
        print(string.format("mode=%04o", entry.unix_mode))
    end
    if entry.uid ~= nil then
        print("uid/gid", entry.uid, entry.gid)
    end
end

```

Reject any ambiguous archive before further processing:

```

local info, err = babet.archive.list("upload.zip")
assert(info, err)

if info.duplicates ~= 0 or info.conflicts ~= 0 then
    error("ambiguous archive")
end

for _, entry in ipairs(info.entries) do
    if not entry.safe_path or not entry.extractable then
        error((entry.reason or "rejected entry") .. ": " .. entry.name)
    end
end

```

Handle an invalid UTF-8 name correctly:

```

local info, err = babet.archive.list("legacy.zip")
assert(info, err)

for _, entry in ipairs(info.entries) do
    if not entry.valid_utf8 then
        -- The string remains binary: display bytes instead of passing it to
        -- an interface that requires UTF-8 text.
        local bytes = {}
        for i = 1, #entry.name do
            bytes[#bytes + 1] = string.format("%02x", entry.name:byte(i))
        end
        print(entry.index, table.concat(bytes))
    end
end

```

Use `list()` in a worker:

```

local worker, err = babet.workers.spawn([
    local info, list_err = babet.archive.list(worker.args.archive, {
        max_entries = 5000,
    })
    if not info then error(list_err) end
    return {
        format = info.format,
        compression = info.compression,
        count = info.count,
        duplicates = info.duplicates,
        conflicts = info.conflicts,
    }
]), { archive = "backup.tar.zst" })
assert(worker, err)

local joined, result = worker:join()
assert(joined, result)
print(result.format, result.count)

```

The function and all fields behave identically in the main Lua state, folder mode, embedded mode, and workers.

## What `list()` validates — and what it does not guarantee

For ZIP, Babet validates the structure required to open and read the central directory, announced counts and sizes, paths, and limits. It does not read every compressed file payload, so a payload CRC failure may remain invisible here.

For TAR and compressed TAR, reaching the next header requires consuming each entry's data; Babet therefore detects many truncations, invalid headers, and outer-stream errors. This complete read does not change the API's role: `list()` returns an inventory, whereas `archive.test()` turns full consumption and aggregate safety rules into a strict verdict.

A technically readable archive containing unsafe paths, links, or special objects is deliberately returned with diagnostics. An unknown format, an unreadable structure, an exceeded limit, or a truncated TAR stream returns `(nil, message)`.

## `babet.archive.test`

```

local result, err = babet.archive.test(archive [, opts])

```

`test()` fully verifies an archive **without extracting, creating, or writing anything**. It accepts the same content-detected formats as `list()`: ZIP, plain TAR, gzip TAR, xz TAR, bzip2 TAR, and zstd TAR. It accepts exactly the same six anti-bomb limits documented above and rejects every other option.

The contract is deliberately strict: success means that the archive is both **technically valid** and **entirely safe under Babet's extraction rules**. A readable archive containing an unsafe path, duplicate, collision, symlink, hard link, sparse file, special object, encrypted entry, or unsupported ZIP method therefore returns `(nil, message)`. `test()` does not return a warnings table or `safe = false`; use `list()` when a suspicious archive must be inventoried and diagnosed without being accepted.

The source path follows the same contract as `list()`: a symlink to a regular file is accepted, while directories, FIFOs, sockets, and devices are rejected without blocking. The opened file is pinned by descriptor for the whole test. The filename extension never chooses the backend.

## Technical verification

For ZIP, Babet checks, among other things:

- EOCD and ZIP64 metadata, rejecting split and multi-volume archives;
- the central directory and declared limits;
- the local header of **every** entry, including empty files and directories;
- local/central filename identity, flags, method, sizes, CRC, local ZIP64 fields, and signed or unsigned data descriptors;
- every local range, overlap between local entries, and overlap with the central directory;
- complete decompression of every non-directory entry, the actual output size, and its CRC.

For TAR, the reader consumes all entry data and checks headers, available TAR checksums, sizes, truncation, padding, and trailing data. For gzip-, xz-, bzip2-, or zstd-compressed TAR, the compression stream is also consumed to its end; integrity errors, truncated members or frames, and foreign trailing bytes are rejected according to the codec and backend guarantees.

A correctly formed empty archive is valid. A read error, unknown format, ambiguous structure, exceeded limit, or disagreement between metadata and data returns `(nil, message)`.

## Safety verification

After technical validation, Babet applies the same rules as real extraction to entries in archive order:

- a non-empty relative path using only `/`, with no `.` or `..`, duplicate separators, backslash, drive prefix, or NUL byte;
- a maximum extractable path length of 4096 bytes;
- ordinary regular files and directories only;
- no symlink, hard link, FIFO, socket, device, unsupported object, or sparse file;
- no exact duplicate, output-path duplicate, or file/directory conflict;
- no encrypted ZIP entry or unsupported ZIP method.

The first diagnostic is deterministic: Babet keeps archive order and stops at the first failing check in the phases above. Success therefore guarantees that `files + directories == entries`.

## Returned structure

```
{
  format = "tar",
  compression = "zstd",
  entries = 42,
  files = 35,
  directories = 7,
  total_size = 12345678,
```

```

    archive_size = 3456789,
    total_name_bytes = 812,
    zip64 = nil,
}

```

- format is "zip" or "tar";
- compression is "none", "gzip", "xz", "bzip2", or "zstd";
- entries, files, and directories count verified entries;
- total\_size is the sum of expanded regular-file sizes;
- archive\_size is the byte size of the pinned archive file;
- total\_name\_bytes is the sum of raw entry-name lengths;
- zip64 is a boolean for ZIP and nil for TAR.

The function returns exactly (result, nil) on success and (nil, message) on failure.

## Usage examples

Simply verify an archive before keeping it:

```

local result, err = babet.archive.test("upload.zip")
if not result then
    io.stderr:write("archive rejected: ", err, "\n")
    return 1
end

```

```

print(result.format, result.entries, result.total_size)

```

Verify a compressed TAR even when its extension is misleading:

```

local result, err = babet.archive.test("snapshot.bin")
assert(result, err)
assert(result.format == "tar")
assert(result.compression == "zstd")

```

Tighten only the entry count:

```

local result, err = babet.archive.test("upload.zip", {
    max_entries = 500,
})
assert(result, err)

```

Tighten only the per-file size:

```

local result, err = babet.archive.test("upload.zip", {
    max_entry_size = 16 * 1024 * 1024,
})
assert(result, err)

```

Tighten only the total size:

```

local result, err = babet.archive.test("upload.zip", {
    max_total_size = 128 * 1024 * 1024,
})
assert(result, err)

```

Tighten only entry-name length:

```

local result, err = babet.archive.test("upload.zip", {
    max_path_length = 4096,
})
assert(result, err)

```

Tighten only the cumulative name budget:

```

local result, err = babet.archive.test("upload.zip", {
    max_total_name_bytes = 1024 * 1024,
})
assert(result, err)

```

Tighten only the compression ratio:

```

local result, err = babet.archive.test("upload.zip", {
    max_compression_ratio = 100,
})
assert(result, err)

```

Combine all limits for an untrusted archive:

```

local result, err = babet.archive.test("upload.zip", {
    max_entries = 500,
    max_entry_size = 16 * 1024 * 1024,
    max_total_size = 128 * 1024 * 1024,
    max_path_length = 4096,
    max_total_name_bytes = 1024 * 1024,
    max_compression_ratio = 100,
})
assert(result, err)

```

Distinguish diagnostic inspection from a strict verdict:

```

local info, list_err = babet.archive.list("upload.zip")
assert(info, list_err)

local verified, test_err = babet.archive.test("upload.zip")
if not verified then
    print("contents visible but archive rejected:", test_err)
end

```

Use test() in a worker:

```

local worker, err = babet.workers.spawn([[
    local result, test_err = babet.archive.test(worker.args.archive, {
        max_entries = 2000,
        max_total_size = 512 * 1024 * 1024,
    })
    if not result then error(test_err) end
    return result
]], { archive = "backup.tar.zst" })
assert(worker, err)

local joined, result = worker:join()
assert(joined, result)
print(result.format, result.compression, result.entries)

```

The return contract and behaviour are identical in folder mode, embedded mode, embedded-via-PATH mode, and workers.

## **babet.archive.extract**

```

local result, err = babet.archive.extract(
    archive,
    destination
    [, opts]
)

```

archive may be a ZIP, plain TAR, gzip-compressed TAR, xz-compressed TAR, bzip2-compressed TAR, or zstd-compressed TAR. Format and compression are detected from contents rather than the filename extension.

Accepted options:

| Option                             | Default            | Behaviour                                                                       |
|------------------------------------|--------------------|---------------------------------------------------------------------------------|
| <code>overwrite</code>             | <code>false</code> | allows atomic replacement of an existing regular file                           |
| <code>dry_run</code>               | <code>false</code> | previews extraction and verifies selected data without modifying the filesystem |
| <code>preserve_permissions</code>  | <code>false</code> | preserves ordinary Unix rwx bits when available                                 |
| <code>include</code>               | <code>none</code>  | dense array of safe globs selecting normalised internal paths                   |
| <code>exclude</code>               | <code>none</code>  | dense array of safe globs removed after inclusion; exclusion always wins        |
| <code>max_entries</code>           | 10000              | maximum entries scanned; hard ceiling 100000                                    |
| <code>max_entry_size</code>        | 256 * 1024 * 1024  | maximum expanded size of one entry; hard ceiling 8 GiB                          |
| <code>max_total_size</code>        | 1024 * 1024 * 1024 | maximum sum of announced sizes; hard ceiling 64 GiB                             |
| <code>max_path_length</code>       | 64 * 1024          | maximum raw-name length; hard ceiling 1 MiB                                     |
| <code>max_total_name_bytes</code>  | 64 * 1024 * 1024   | maximum sum of name lengths; hard ceiling 64 MiB                                |
| <code>max_compression_ratio</code> | 1000               | maximum expanded/compressed ratio, from 1 through 1000000000                    |

Booleans must be real Lua booleans, integer limits must be strict positive Lua integers, and `max_compression_ratio` must be finite. Unknown keys, excess arguments, and creation-only options are rejected. `dry_run`, `include`, and `exclude` are accepted by `extract()` only, not by `list()`, `test()`, or `extractFile()`.

### Safe-glob selection

`include` and `exclude` reuse exactly the same bounded engine as `archive.create()`. There is no second matcher and no extraction-specific glob dialect.

An absent or empty `include` list preserves historical behaviour and initially selects every entry. A non-empty `include` list retains an entry when at least one pattern matches. `exclude` is applied afterwards and always wins, even when the same path also matches `include`.

Patterns are anchored against the entry's complete **normalised internal path**, with `/` as the separator. Matching is byte-oriented and case-sensitive:

- `*` matches zero or more bytes except `/`;
- `**` matches zero or more bytes including `/`;

- ? matches exactly one byte except /;
- \x quotes the following byte x.

A directory is tested both as path and path/. Excluding src/generated or src/generated/\*\* therefore excludes that directory and prunes its complete subtree, even when the archive has no explicit directory entry. Only parents needed by retained files are created; an unselected explicit directory is not created. An empty directory is restored only when its explicit entry is selected.

The same bounds as creation apply: each pattern is limited to 4096 bytes; include and exclude together are capped at 256 patterns and 256 KiB of pattern text, one million pattern evaluations, and a fixed 100,000,000-cell matching-work budget per call. The normalised graph used to propagate directory exclusions is additionally capped at 100000 unique directory paths and 64 MiB of cumulative path text. Exceeding a limit fails before publication. Pattern order does not change selection.

### Read-only simulation with dry\_run

With `dry_run = true`, Babet builds the same plan as a real extraction but does not call any creation, write, `chmod`, `rename`, `unlink`, or publication operation. The archive is pinned and scanned, the same limits and filters are applied, selected entries are validated, and the destination is traversed read-only with `openat/fstatat` and `O_NOFOLLOW`. No destination root, implicit parent, temporary file, or final entry is created.

Destination checks retain the real extraction policy:

- a missing root is accepted and reported by `would_create_destination = true`;
- an existing root or parent that is a symlink or a non-directory object is rejected;
- a missing selected entry counts in `would_create`;
- an existing regular file counts in `would_overwrite` only with `overwrite = true`; with `overwrite = false`, the call fails exactly like real extraction;
- an explicitly selected directory that already exists counts in `would_skip`, because real extraction preserves it and does not `chmod` it;
- a file/directory conflict or a symlink final target remains an error.

`would_create`, `would_overwrite`, and `would_skip` count only **explicit selected archive entries**. Their sum therefore equals entries. The destination root and required implicit parents are not included; the root has its own boolean. When active filters select nothing, behaviour remains identical to real extraction: the destination is not inspected and `would_create_destination` is false. Conversely, an unfiltered empty archive previews creation of its root when that root is missing.

The simulation genuinely verifies the data that would be extracted. For ZIP, each selected regular file is inflated to a null consumer and its CRC is checked; unselected ZIP payloads remain untouched, as in real selective extraction. For TAR, the second pass consumes the complete plain or compressed stream and compares every header with the initial plan without forwarding bytes to the filesystem. `archive.test()` remains the API for checking every ZIP payload, including ones that extraction would skip.

The result is a **snapshot**. Another process may change the destination after `dry_run` returns, so the preview is never a guarantee that a later extraction will succeed or produce the same counters. Real extraction repeats all checks and pins the required descriptors before publication.

### Validation, skipped entries, and integrity

Babet first pins the archive file and scans **every entry and the metadata needed for the extraction plan**, applying global limits before opening the destination. Selection is then computed on normalised names. Only retained entries participate in the output plan:

- a duplicate, file/directory collision, link, or special object found only outside the selection does not block extraction;
- a selected unsafe entry is still rejected before any write;

- with a non-empty include list, an unsafe path has no normalised form and cannot match, so it remains unselected;
- with exclude only, historical “select everything” behaviour remains active: an unsafe path cannot be sanitised by an exclusion pattern and still fails the call;
- existing destination collisions outside the selection are neither replaced nor modified.

This policy permits extraction of a safe part of a mixed archive without presenting exclude as a path-repair mechanism. Entry, size, name, and compression-ratio limits still apply to the complete scanned archive, not just the retained files.

For ZIP, skipped payloads are not inflated, so their CRC is not checked by selective extraction. For TAR, the reader must advance through the stream to reach later headers, including through an outer compressed stream, but `extract()` is not the complete verification API. Call `archive.test()` first when integrity of **every** entry, including skipped data, must be guaranteed.

Before publication, Babet rejects retained non-extractable entries, duplicate retained outputs, retained file/directory conflicts, and incompatible existing destination objects. Selected files are staged in the secure destination and then published atomically. For TAR, the same pinned file is reread from the beginning and every header is compared with the first-pass plan.

When active filters select nothing, the call succeeds without creating the destination. Without filters, an empty archive keeps historical behaviour and creates the destination root directory.

## Return value

The function returns exactly `(result, nil)` or `(nil, message)`:

```
{
  entries = 28,      -- entries actually selected
  files = 24,        -- selected regular files
  directories = 4,  -- selected explicit directory entries
  skipped = 15,      -- archive entries not selected
  bytes = 987654,    -- expanded bytes of selected files
  path = "restore",
}
```

With `dry_run = true`, the same table additionally contains:

```
{
  dry_run = true,
  would_create = 20,
  would_overwrite = 3,
  would_skip = 5,
  would_create_destination = false,
}
```

These additional fields are absent from a real extraction so its historical contract remains unchanged. `directories` does not count implicit parents created to reach a file. `entries == files + directories` after a successful extraction because selected links and special objects are rejected. In a successful simulation, `would_create + would_overwrite + would_skip == entries`. `skipped` is 0 when no filter is active.

## Examples

Preview extraction to a missing destination:

```
local plan, err = babet.archive.extract("backup.zip", "restore", {
  dry_run = true,
})
assert(plan, err)
assert(plan.dry_run == true)
```

```
assert(plan.would_create_destination == true)
assert(not babet.fileExists("restore"))
```

Preview allowed replacements in an existing destination:

```
local plan, err = babet.archive.extract("backup.tar.zst", "restore", {
    dry_run = true,
    overwrite = true,
    preserve_permissions = true,
})
assert(plan, err)
print(plan.would_create, plan.would_overwrite, plan.would_skip)
```

Without overwrite = true, an existing final file remains an error:

```
local plan, err = babet.archive.extract("backup.zip", "restore", {
    dry_run = true,
})
assert(plan == nil and type(err) == "string")
```

Combine simulation, selection, and limits:

```
local plan, err = babet.archive.extract("project.tar.xz", "output", {
    dry_run = true,
    include = { "src/**", "docs/**", "README.md" },
    exclude = { "src/generated/**", "**/*.tmp" },
    overwrite = true,
    preserve_permissions = true,
    max_entries = 20000,
    max_total_size = 2 * 1024 * 1024 * 1024,
})
assert(plan, err)
assert(plan.would_create + plan.would_overwrite + plan.would_skip
    == plan.entries)
assert(not babet.fileExists("output"))
```

Extract only two precise areas:

```
local result, err = babet.archive.extract("project.tar.zst", "output", {
    include = {
        "src/**",
        "README.md",
    },
})
assert(result, err)
print(result.entries, result.skipped)
```

Exclude temporary files while retaining everything else:

```
local result, err = babet.archive.extract("backup.zip", "restore", {
    exclude = {
        "**/*.tmp",
        "cache/**",
    },
})
assert(result, err)
```

Combine inclusion and exclusion, with exclusion taking priority:

```
local result, err = babet.archive.extract("project.tar.xz", "output", {
    include = {
        "src/**",
    },
})
```

```

        "docs/**",
        "README.md",
    },
    exclude = {
        "src/generated/**",
        "**/*.tmp",
    },
    overwrite = false,
    preserve_permissions = true,
    max_total_size = 2 * 1024 * 1024 * 1024,
})
assert(result, err)

```

Use ? and quote a literal \* byte:

```

local result, err = babet.archive.extract("assets.zip", "assets", {
    include = {
        "icons/icon-?.png", -- icon-a.png, but not icon-large.png
        "docs/file\\*.txt", -- a name that literally contains an asterisk
    },
})
assert(result, err)

```

Handle an empty selection without creating a directory:

```

local result, err = babet.archive.extract("package.zip", "unused", {
    include = { "does-not-exist/**" },
})
assert(result, err)
assert(result.entries == 0 and result.skipped > 0)
assert(not babet.fileExists("unused"))

```

Use filters inside a worker:

```

local worker, err = babet.workers.spawn([[
local result, extract_err = babet.archive.extract(
    worker.args.archive,
    worker.args.destination,
    {
        include = { "docs/**", "README.md" },
        exclude = { "docs/drafts/**" },
    }
])
if not result then error(extract_err) end
return result
]], {
    archive = "package.tar.gz",
    destination = "documentation",
})
assert(worker, err)

```

```

local joined, result = worker:join()
assert(joined, result)
print(result.files, result.skipped)

```

Preview inside a worker without creating the destination:

```

local worker, err = babet.workers.spawn([[
local plan, extract_err = babet.archive.extract(
    worker.args.archive, worker.args.destination, {

```

```

        dry_run = true,
        include = { "manifest.json", "docs/**" },
    })
if not plan then error(extract_err) end
return plan
]], {
    archive = "package.tar.gz",
    destination = "preview-only",
})
assert(worker, err)

local joined, plan = worker:join()
assert(joined, plan)
assert(plan.dry_run == true)
assert(not babet.fileExists("preview-only"))

```

The contract, limits, and selection are identical in folder mode, embedded mode, embedded-via-PATH mode, and workers.

## **babet.archive.extractFile**

```

local result, err = babet.archive.extractFile(
    archive,
    entry,
    destination
    [, opts]
)

```

archive may be a ZIP, plain TAR, gzip-compressed TAR, xz-compressed TAR, bzip2-compressed TAR, or zstd-compressed TAR; detection uses the file contents, not the extension. ZIP stays on miniz. TAR uses the same pinned-source and two-pass libarchive reader as complete extraction.

entry is the exact **raw name** exposed by `list()`. Lookup is case-sensitive and fails when the name is absent or occurs more than once. For TAR, GNU long names and pax path records are matched after libarchive has decoded the TAR headers, exactly as they appear in `list().entries[i].name`.

Only a regular-file entry can be selected. Sparse TAR files, symlinks, hard links, directories, and special types are refused when selected. The entry path is not reproduced: the content is written exactly to the caller-supplied destination path.

```

local result, err = babet.archive.extractFile(
    "package.tar",
    "assets/logo.bin",
    "cache/current-logo.bin",
    { overwrite = true }
)

```

Result:

```

{
    bytes = 4096,
    path = "cache/current-logo.bin",
    entry = "assets/logo.bin",
}

```

Anti-bomb limits are still computed over the whole archive. For TAR, the first pass consumes and validates every data stream. The second pass compares every header with the inspected plan and consumes the archive to its end before the selected temporary file is published. Unselected regular-file data is discarded, and an unrelated unsafe path, special type, or sparse file does not prevent

selecting a safe regular file. Malformed or truncated data anywhere in the archive still makes the operation fail without publishing the destination.

## Path rules

For complete extraction, every entry name must satisfy all of these rules:

- between 1 and 4096 bytes;
- no NUL byte;
- no \;
- no leading /;
- no drive prefix such as C: in the first component;
- no empty component (a//b);
- no . or .. component;
- a trailing / only for an actual directory entry.

Paths remain relative to the destination. Babet never converts a Windows name into a Linux path and never silently “cleans” traversal: the archive is rejected. With `extractFile()`, the same rules apply to the selected entry name; unselected unsafe names do not become destination paths.

Duplicates are detected after removing the trailing / from directory names. An archive containing two identical outputs, or node as a file followed by node/child, is rejected before extraction.

## Destination protection

Security does not rely only on string comparison. Babet walks the destination component by component through directory descriptors and refuses to follow symlinks:

- a symlink in the destination root path is rejected;
- a symlinked parent below the destination is rejected;
- a symlink final target is rejected, even with `overwrite = true`;
- a file may only replace a regular file;
- a directory entry may only use an existing or newly created directory.

Existing directories are accepted when they are real directories and remain unchanged. Babet does not alter their permissions.

The destination path itself is trusted script input and may be absolute or relative. The .. rules above apply to untrusted names stored inside the archive.

## Atomic publication and cleanup

`create()` writes the complete ZIP, plain TAR, gzip-compressed TAR, or xz-compressed TAR to a unique mode-0600 temporary file in the destination directory. Its short internal name does not repeat the output basename, so a valid destination close to the NAME\_MAX limit remains creatable. After the miniz ZIP central directory or libarchive TAR stream is finalized, Babet flushes and `fsyncs` the file, changes it to 0644, publishes it atomically, and `fsyncs` the parent directory. Without `overwrite` it uses a no-clobber hard link publication; with `overwrite` it uses same-directory rename. Every error before publication removes the temporary and leaves the previous destination unchanged. If the final parent-directory `fsync` itself fails, the new archive has already been atomically published but the call returns an error because crash durability could not be confirmed.

Each extracted file is progressively written to a unique temporary file in its final directory with mode 0600. Temporary names are selected so they can never collide with an output declared by the archive. Writes are bounded by the announced size. For ZIP, miniz validates decompression and CRC. For TAR, libarchive provides streaming blocks and Babet validates their order, bounds, and final announced size. For gzip-wrapped TAR, Babet additionally validates every gzip member’s CRC and ISIZE with `zlib` before publication. The temporary file then receives its final permissions.

All files of a complete extraction are prepared before publication starts. A read, decompression, or CRC error therefore publishes no file. `extractFile()` likewise keeps its selected file temporary until the complete archive has passed the verified read. Temporary files are removed, as are newly created empty directories **below** the destination. Destination-root path components that had to be created may remain present.

Publication is atomic **per file**:

- without overwrite, Babet uses creation that fails if the target exists;
- with overwrite, the same-directory temporary atomically replaces the target regular file.

The complete operation is not a multi-file transaction. A very late system error during publication or final directory permission changes may happen after previous files have been published. Those files may remain, and an old file already replaced is not restored.

For extraction, atomic publication prevents a partially visible file but does not promise crash durability: extracted files and directories are not fully fsynced. `create()` follows the stronger file-and-parent synchronization sequence described above.

## Permissions

Defaults:

|                                        |                         |
|----------------------------------------|-------------------------|
| files                                  | 0644                    |
| final directories                      | 0755                    |
| temporary files                        | 0600                    |
| archive directories during preparation | 0700                    |
| created destination-path components    | 0755 (subject to umask) |

With `preserve_permissions = true`, Babet preserves only the nine ordinary owner/group/other bits (0777) from Unix modes announced by the ZIP or TAR. `setuid`, `setgid`, and sticky bits are always stripped.

Permissions of a newly created directory are applied only after its contents have been extracted, so a restrictive final mode such as 0000 cannot block the current operation. Existing directories are never `chmod`d.

UIDs, GIDs, ACLs, extended attributes, and timestamps are not restored.

## Supported entry types

| Type                               | <code>create()</code>  | <code>list()</code>     | <code>extract()</code>       | <code>extractFile()</code>   |
|------------------------------------|------------------------|-------------------------|------------------------------|------------------------------|
| regular file                       | supported              | inspected               | supported, except sparse TAR | supported, except sparse TAR |
| directory                          | supported when enabled | inspected               | supported                    | not selectable               |
| sparse TAR file                    | never created          | identified in TAR       | rejected                     | rejected if selected         |
| symlink                            | rejected if selected   | identified              | rejected                     | rejected                     |
| hard link                          | never created          | identified in TAR       | rejected                     | rejected if selected         |
| FIFO, socket, device, unknown type | rejected if selected   | identified when exposed | rejected                     | rejected                     |
| encrypted entry                    | never created          | identified in ZIP       | rejected                     | rejected                     |

| Type                           | create()      | list()     | extract() | extractFile() |
|--------------------------------|---------------|------------|-----------|---------------|
| unknown ZIP compression method | never created | identified | rejected  | rejected      |

During creation, a symlink or special filesystem object that is removed by the filters is ignored rather than opened. Directory entries are emitted only when `include_directories = true`.

For regular-file creation, ZIP, gzip, and xz accept compression levels 0 through 9; bzip2 accepts 1 through 9; zstd accepts 0 through 19; plain TAR stores the source bytes without compression.

A ZIP entry that resembles a hard link but is encoded as a regular file is treated as an independent regular file; no link is created.

## Errors and known limitations

Common failures include:

- missing or unsafe source directory, source symlink, non-UTF-8 source name, unsupported source object, invalid creation format, ZIP-only option used for TAR, or out-of-range ZIP timestamp;
- output archive inside the source, unsafe destination parent, or destination symlink;
- missing, unreadable, non-regular, unsupported, or malformed archive file;
- truncated TAR data, non-TAR trailing data, a non-regular TAR entry carrying a payload, or a change detected between inspection and extraction;
- inconsistent metadata or invalid CRC;
- exceeded anti-bomb limit;
- absolute path, traversal, duplicate, or path conflict;
- encrypted entry, symlink, or unsupported type;
- existing destination without `overwrite = true`;
- symlink or unsafe type in the destination;
- insufficient disk space, permissions, or another I/O error.

The module is synchronous: a call returns only after creation, inspection, or extraction has completed. It does not expose progress, cancellation, or a timeout yet.

The archive file must not be modified concurrently. ZIP truncation or modification will normally cause a read, decompression, or CRC error. TAR is streamed through libarchive; concatenated TAR streams are scanned to their end, and truncated or non-TAR trailing data is rejected. Neither backend provides a concurrent snapshot. The destination must not be actively reorganised by another process either. Traversal rejects symlinks and rechecks types before publication, but an actor that continuously renames directories can make the operation fail and turn temporary-file cleanup into a best-effort action.

Plain, gzip-, xz-, bzip2-, and zstd-compressed TAR creation is supported. Standalone compression streams remain intentionally outside this archive API and are handled by [babet.compression](#).

Multi-disk archives are not a supported target. ZIP64 is accepted when miniz can read it, while still being subject to the configured limits.

## babet.base64 — binary Base64 encoding and decoding

The `babet.base64` module implements the RFC 4648 Base64 alphabets without starting an external program. Inputs and outputs are **binary** Lua strings: NUL bytes, non-UTF-8 data, and every value from 0 through 255 are preserved exactly.

The decoder is deliberately strict. It validates the selected alphabet, padding placement, final-quantum length, and unused bits in the last symbol. A non-canonical representation is therefore not silently accepted.

## Module contents

- [API](#)
- [Standard encoding](#)
- [URL-safe alphabet](#)
- [Padding](#)
- [Strict decoding](#)
- [ASCII whitespace](#)
- [Output limit](#)
- [Binary data and files](#)
- [Using the module in a worker](#)
- [Error contract](#)
- [Limits and design choices](#)

## API

| Function                                        | Contract                      |
|-------------------------------------------------|-------------------------------|
| <code>babet.base64.encode(data [, opts])</code> | Returns text, nil or nil, err |
| <code>babet.base64.decode(text [, opts])</code> | Returns data, nil or nil, err |

data and text must be actual Lua strings. Numbers are never implicitly converted to text.

Encoding options:

```
{  
  url_safe = false,  
  padding = true,  
}
```

Decoding options:

```
{  
  url_safe = false,  
  allow_unpadded = false,  
  ignore_whitespace = false,  
  max_output = nil,  
}
```

Unknown options are rejected. Boolean options require actual Lua booleans; max\_output must be a non-negative integer.

## Standard encoding

Without options, encode() uses the standard Base64 alphabet: letters, digits, +, and /, with canonical = padding when the final group contains fewer than three bytes.

```
local encoded, err = babet.base64.encode("hello")  
assert(encoded, err)  
assert(encoded == "aGVsbG8=")
```

The classic RFC 4648 vectors are preserved:

```

assert(babet.base64.encode("") == "")
assert(babet.base64.encode("f") == "Zg==")
assert(babet.base64.encode("fo") == "Zm8=")
assert(babet.base64.encode("foo") == "Zm9v")
assert(babet.base64.encode("foob") == "Zm9vYg==")
assert(babet.base64.encode("fooba") == "Zm9vYmE=")
assert(babet.base64.encode("foobar") == "Zm9vYmFy")

```

encode() adds no line breaks and does not produce 76-column MIME Base64.

## URL-safe alphabet

url\_safe = true selects the RFC 4648 URL/file-name-safe alphabet: - replaces +, and \_ replaces /.

```

local binary = string.char(251, 255)
local encoded, err = babet.base64.encode(binary, {
    url_safe = true,
})
assert(encoded, err)
assert(encoded == "-_8=")

```

Decoding must select the same alphabet:

```

local decoded, err = babet.base64.decode("-_8=", {
    url_safe = true,
})
assert(decoded, err)
assert(decoded == string.char(251, 255))

```

The alphabets are not mixed automatically:

```

assert(babet.base64.decode("-_8=") == nil)
assert(babet.base64.decode("+/8=", { url_safe = true }) == nil)

```

This strict choice prevents the same input from being accepted in multiple forms when a protocol requires one precise alphabet.

## Padding

### Producing output without =

padding = false removes only trailing = characters. The useful content is unchanged.

```

local encoded, err = babet.base64.encode("hello", {
    padding = false,
})
assert(encoded, err)
assert(encoded == "aGVsbG8")

```

When the source length is a multiple of three, no padding is needed even with the default option:

```

assert(babet.base64.encode("foo") == "Zm9v")

```

### Accepting unpadded input

The decoder rejects an incomplete final group without = by default:

```

local decoded, err = babet.base64.decode("aGVsbG8")
assert(decoded == nil)
assert(type(err) == "string")

```

Enable allow\_unpadded explicitly when the protocol uses this form:

```

local decoded, err = babet.base64.decode("aGVsbG8", {
    allow_unpadded = true,
})
assert(decoded, err)
assert(decoded == "hello")

```

allow\_unpadded does not relax other rules: a one-symbol final group remains truncated, and non-zero unused bits remain invalid.

### Combined URL-safe, unpadded example

This form is common in protocol identifiers and some JSON fields:

```

local source = string.char(0, 1, 2, 251, 255)

local token, err = babet.base64.encode(source, {
    url_safe = true,
    padding = false,
})
assert(token, err)

local restored
restored, err = babet.base64.decode(token, {
    url_safe = true,
    allow_unpadded = true,
})
assert(restored, err)
assert(restored == source)

```

### Strict decoding

The decoder checks, among other rules:

- every character belongs exactly to the selected alphabet;
- = appears only at the end;
- no more than two padding characters are present;
- padded length is a multiple of four;
- an unpadded tail contains two or three symbols, never one;
- unused bits in the final symbol are zero.

These inputs are therefore rejected:

```

local invalid = {
    "%%%", -- characters outside the alphabet
    "AA=A", -- padding in the middle
    "A==", -- too much padding
    "Zg=", -- invalid padded length
    "Zh==", -- non-zero unused bits
    "Zm9=", -- non-zero unused bits
}

for _, text in ipairs(invalid) do
    local value, err = babet.base64.decode(text)
    assert(value == nil)
    assert(type(err) == "string")
end

```

Positions reported by invalid character, invalid padding, or non-zero trailing bits errors are one-based byte positions in the original input string.

## ASCII whitespace

Whitespace is an invalid character by default:

```
local value, err = babet.base64.decode("Zm9v\nYmFy")
assert(value == nil)
```

`ignore_whitespace = true` ignores the six classic ASCII whitespace bytes: space, tab, line feed, vertical tab, form feed, and carriage return.

```
local value, err = babet.base64.decode(" Zm9v\tYmFy\r\n", {
    ignore_whitespace = true,
})
assert(value, err)
assert(value == "foobar")
```

The option removes no other Unicode or control byte. It also does not permit bad padding or the wrong alphabet.

When whitespace is ignored, an error position still refers to the original string, including whitespace.

## Output limit

`max_output` bounds the exact decoded result size. The limit is checked after complete lexical and canonical Base64 validation, then before the decoded output string is allocated.

```
local value, err = babet.base64.decode("Zm9v", {
    max_output = 2,
})
assert(value == nil)
assert(err == "base64: decoded output exceeds max_output")
```

The limit is inclusive:

```
local value, err = babet.base64.decode("Zm9v", {
    max_output = 3,
})
assert(value, err)
assert(value == "foo")
```

`max_output = 0` accepts only empty output:

```
assert(babet.base64.decode("", { max_output = 0 }) == "")
assert(babet.base64.decode("Zg==", { max_output = 0 }) == nil)
```

Set a protocol-appropriate limit for untrusted Base64 values:

```
local png, err = babet.base64.decode(response.value, {
    max_output = 32 * 1024 * 1024,
})
assert(png, err)
```

The limit applies to decoded output, not to Base64 text length.

## Binary data and files

Base64 imposes no UTF-8 requirement. Lua strings containing arbitrary bytes are accepted:

```
local original = "\0\1\2abc\255"
local encoded = assert(babet.base64.encode(original))
local decoded = assert(babet.base64.decode(encoded))
assert(decoded == original)
```

To encode a file:

```
local file = assert(io.open("screenshot.png", "rb"))
local data = assert(file:read("*a"))
assert(file:close())
```

```
local text, err = babet.base64.encode(data)
assert(text, err)
```

To write a decoded value:

```
local data, err = babet.base64.decode(text, {
    max_output = 32 * 1024 * 1024,
})
assert(data, err)
```

```
local ok
ok, err = babet.writeFileAtomic("screenshot-restored.png", data, {
    overwrite = true,
    permissions = tonumber("600", 8),
})
assert(ok, err)
```

`babet.writeFileAtomic()` publishes the decoded bytes from a private same-directory temporary file and does not expose a partially written image.

## Using the module in a worker

The submodule is registered in every worker Lua state:

```
local job = assert(babet.workers.spawn([[
    local encoded = assert(babet.base64.encode("hello"))
    local decoded = assert(babet.base64.decode(encoded))
    return decoded == "hello"
]]))
```

```
local ok, result = job:join()
assert(ok and result == true)
```

The current `worker.args`, `result`, and `worker-queue` transport still uses JSON and rejects strings containing a NUL byte. Base64 can turn binary data into transportable text, at a size cost of roughly one third.

```
local encoded = assert(babet.base64.encode(binary_data))
local job = assert(babet.workers.spawn([[
    return assert(babet.base64.decode(worker.args.encoded))
]], { encoded = encoded })))
```

If the decoded result contains a NUL, it cannot be returned directly through the current JSON worker transport. The worker must process it, write it to a file, or re-encode it before returning it.

## Error contract

### Call errors: Lua exception

Programming errors raise a Lua error:

- wrong argument count;
- non-string data or text;
- non-table options;
- non-string option key;

- unknown option;
- wrongly typed boolean option;
- non-integer or negative max\_output.

```
local ok, err = pcall(function()
    babet.base64.decode("Zm9v", { max_output = -1 })
end)
assert(ok == false)
```

### Data errors: nil, err

Invalid Base64 or an exceeded limit returns nil, err:

```
local value, err = babet.base64.decode("%%%")
assert(value == nil)
assert(type(err) == "string")
assert(err:find("base64:", 1, true) == 1)
```

The main message forms are:

- base64: invalid character at byte N;
- base64: invalid padding at byte N;
- base64: truncated input;
- base64: non-zero trailing bits at byte N;
- base64: decoded output exceeds max\_output;
- base64: out of memory;
- base64: encoded output too large or decoded output too large.

The internal ... mismatch and unexpected internal error diagnostics are defensive guards. They should not occur for ordinary valid or invalid user input; report them as a Babet defect if they appear.

Only depend on documented reasons when code needs to distinguish a limit from invalid data. Allocation wording and positional detail may be enriched later.

## Limits and design choices

The module processes a complete string in memory. It does not currently provide:

- streaming encoders or decoders;
- direct file input/output;
- MIME line wrapping;
- automatic standard-versus-URL-safe alphabet detection;
- a permissive mode accepting non-canonical padding or trailing bits.

These exclusions are deliberate. The API stays small, deterministic, and suited to Selenium screenshots, tokens, JSON payloads, small binary resources, and exchanges with external tools.

## Compression - standalone gzip, xz, bzip2, and zstd streams

### Scope

babet.compression compresses or decompresses the bytes of one regular file. It handles **standalone streams**, not archives:

- gzip (.gz by convention);
- xz (.xz);
- bzip2 (.bz2);
- Zstandard (.zst).

A standalone stream has no portable entry name, directory tree, permissions, or archive meta-data. Use `babet.archive` for ZIP files or TAR archives, including `.tar.gz`, `.tar.xz`, `.tar.bz2`, and `.tar.zst`.

## API overview

```
local ok, err = babet.compression.compress(
    source, destination, format [, opts]
)

local ok, err = babet.compression.decompress(
    source, destination [, opts]
)
```

Both functions return:

- `true`, `nil` on success;
- `nil`, "message" on an operational or option error.

Wrong required argument types, embedded NUL bytes, and incorrect arity raise a Lua error. Unknown option names are rejected.

### `compress(source, destination, format [, opts])`

`source`, `destination`, and `format` are strict Lua strings. Paths must not be empty or contain an embedded NUL byte.

`format` must be exactly one of:

"gzip" "xz" "bzip2" "zstd"

The filename extension is not inspected or added automatically. This is valid:

```
local ok, err = babet.compression.compress(
    "payload.bin",
    "payload.data",
    "zstd"
)
```

The optional table accepts:

| Field                  | Type        | Default            | Meaning                                                  |
|------------------------|-------------|--------------------|----------------------------------------------------------|
| <code>overwrite</code> | boolean     | <code>false</code> | Atomically replace an existing regular destination file. |
| <code>level</code>     | Lua integer | format-specific    | Explicitly select the compression level.                 |

`level` is a strict Lua integer. Floats, numeric strings, and out-of-range values are rejected before the source file is opened. Accepted ranges and the values used when the field is absent are:

| Format | Minimum | Maximum | Default |
|--------|---------|---------|---------|
| gzip   | 0       | 9       | 6       |
| xz     | 0       | 9       | 6       |
| bzip2  | 1       | 9       | 9       |
| zstd   | 1       | 22      | 3       |

A higher level generally favours compression ratio at the cost of CPU time and sometimes additional memory. For gzip, level 0 creates a valid stream without DEFLATE compression. The highest xz presets can consume substantial memory and therefore must always be selected explicitly.

### Example

```
local ok, err = babet.compression.compress(
    "database.dump",
    "database.dump.zst",
    "zstd",
    { overwrite = true, level = 10 }
)

if not ok then
    io.stderr:write(err, "\n")
    return 1
end
```

### decompress(source, destination [, opts])

The compression format is detected from the input bytes, not from the extension. Renaming file.gz to file.bin therefore does not change how it is decoded. Uncompressed data and unsupported formats are rejected.

The optional table accepts:

| Field           | Type        | Default | Meaning                                                  |
|-----------------|-------------|---------|----------------------------------------------------------|
| overwrite       | boolean     | false   | Atomically replace an existing regular destination file. |
| max_output_size | Lua integer | 1 GiB   | Maximum number of decompressed bytes accepted.           |

max\_output\_size must be between 1 and 68719476736 bytes (64 GiB). The limit is checked before each write. If it is exceeded, the temporary file is removed and the destination is not published.

### Example

```
local ok, err = babet.compression.decompress(
    "database.dump.zst",
    "database.dump",
    {
        overwrite = true,
        max_output_size = 4 * 1024 * 1024 * 1024,
    }
)

if not ok then
    io.stderr:write(err, "\n")
    return 1
end
```

## Concatenated streams and integrity

Valid concatenated gzip members, xz streams, bzip2 members, and zstd frames are decoded in order into the same destination file.

The decoders verify the mandatory integrity fields and any optional checksum present in the stream. Babet-generated zstd frames enable the optional content checksum. An external zstd frame without that field remains readable, but the format itself then cannot detect every possible payload bit flip. Babet rejects:

- truncated streams;
- corrupted checksums or frames;
- arbitrary trailing bytes after the final valid member/frame;
- inputs whose magic bytes do not identify a supported format.

## Filesystem and publication guarantees

The module follows a fail-closed file policy:

- the source must be a real regular file;
- a symlink source is rejected;
- source and destination parent paths must already exist and are opened component by component with `O_NOFOLLOW`;
- ... components in those parent paths are rejected;
- a destination symlink or non-regular destination is rejected;
- source and destination must not refer to the same inode, including through a hard link;
- the source descriptor is pinned and its size and timestamps are checked again before publication;
- output is streamed into a mode-0600 temporary file in the destination directory;
- the completed file is synced, changed to mode 0644, and atomically published;
- with `overwrite = false`, publication cannot replace a destination that appeared concurrently;
- temporary files are removed on failures before publication.

The module never uses a shell and never derives the destination filename from compressed metadata.

## Binary data and memory use

Input and output are byte streams. NUL bytes and arbitrary binary content are preserved.

Compression and decompression use bounded 64 KiB working buffers. The full input or output is not loaded into the Lua state or into one C++ string. `max_output_size` additionally bounds decompression expansion. The xz decoder has a 256 MiB internal memory ceiling and the zstd decoder refuses windows above 128 MiB.

## Workers

`babet.compression` is registered in every worker Lua state. Different workers may process different files concurrently. Normal filesystem collision rules still apply if several workers target the same destination.

## Error examples

```
local ok, err = babet.compression.compress(
    "input.bin", "input.bin.gz", "zip"
)
-- nil, "compression: format must be 'gzip', 'xz', 'bzip2', or 'zstd'"

ok, err = babet.compression.decompress(
```

```

    "large.gz", "large.bin", { max_output_size = 1024 }
)
-- nil, "compression: decompressed output exceeds opts.max_output_size"

```

A failed operation should always be handled before using the destination.

## argparse — command-line arguments

argparse is a pure-Lua module bundled with Babet. It builds a declarative parser for flags, valued options, and positional arguments:

```

local argparse = require("argparse")
local parser = argparse("my-script", "Optional description")

```

The module never calls `print` or `os.exit`. Help and errors are returned to the script, which decides how to display them and which exit code to use.

### Module contents

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### Loading and metadata

```

local argparse = require("argparse")

print(argparse._VERSION)      -- "babet argparse 1.1.0"
print(argparse._DESCRIPTION)  -- English description

```

`require("argparse")` returns a callable table. The constructor accepts zero, one, or two arguments:

```

argparse()                -- displayed program: "prog"
argparse("tool")           -- program name
argparse("tool", "Description") -- name and description

```

`prog` and `description` must be strings when supplied. An extra argument or a wrong type raises a Lua error: this is a programming error, not a command-line error.

## API

| Method                                      | Result   | Purpose                                  |
|---------------------------------------------|----------|------------------------------------------|
| <code>parser:flag(spec [, opts])</code>     | parser   | Declare a flag with no value.            |
| <code>parser:option(spec [, opts])</code>   | parser   | Declare an option that consumes a value. |
| <code>parser:argument(name [, opts])</code> | parser   | Declare a positional argument.           |
| <code>parser:parse([src])</code>            | res, err | Parse src or the global arg table.       |
| <code>parser:get_usage()</code>             | string   | Generate the help text.                  |

Builder methods are chainable:

```
local parser = argparse("copy")
    :flag("-v --verbose")
    :option("-o --output")
    :argument("source")
```

Arities are strict. For example, `parser:option("-o", {}, "extra")` and `parser:get_usage(true)` raise a Lua error.

## Declaring flags and options

spec is one string containing one or more whitespace-separated names:

```
parser:flag("-v")
parser:flag("--verbose")
parser:flag("-v --verbose")
parser:option("-o --output")
```

Each name must:

- start with one or two dashes;
- contain at least one character after the dashes;
- contain no =;
- not be duplicated in the same spec or parser.

The following names are reserved and cannot be declared:

- -: treated as a literal positional;
- --: ends option parsing;
- -h and --help: built-in help.

A triple-dash name such as `---verbose` is rejected as well.

All validation happens before the parser is modified. If a builder error is caught with `pcall`, no partial option or alias remains registered.

## Declaring positionals

```
parser:argument("input")
parser:argument("output", { required = false })
```

The name must be a non-empty string with no whitespace and no leading dash. Positionals are assigned strictly from left to right.

A positional is required by default. It becomes optional when:

- `required = false` is specified; or

- a default is supplied without an explicit required field.

All required positionals must precede optional ones. The builder therefore rejects this ambiguous declaration:

```
parser
  :argument("optional", { required = false })
  :argument("required") -- Lua error
```

## The opts table

opts must be a table or nil. Unknown fields and fields with a wrong type raise a Lua error immediately. This catches a typo such as hlep instead of help.

### Common fields

| Field    | Type             | Effect                              |
|----------|------------------|-------------------------------------|
| help     | string           | Text appended by get_usage().       |
| default  | any Lua value    | Value used when the item is absent. |
| dest     | non-empty string | Key used in the result table.       |
| required | boolean          | Require the item to be present.     |

These four fields are accepted by flag, option, and argument.

### Value-only fields

| Field   | Type                   | Effect                                 |
|---------|------------------------|----------------------------------------|
| choices | dense array of strings | Allow only selected raw values.        |
| convert | function               | Convert the raw string to a Lua value. |

choices and convert are accepted by option and argument, but not by flag, since a flag consumes no value.

The builder copies the choices array. Mutating the original table later does not change the parser.

### Destination

Without dest, the destination is:

1. the first long name without --, when present;
2. otherwise the first short name without its first -.

```
parser:option("-o --output") -- destination: "output"
parser:flag("-v")           -- destination: "v"
```

A destination must be unique across the parser. help and usage are reserved for the built-in help result.

```
parser:flag("-q", { dest = "quiet" })
```

## Argument source

### Global arg table

With no argument, parse() reads the global arg table at indices 1, 2, and so on up to the first hole:

```
local result, err = parser:parse()
```

Index 0 and negative indices are ignored. `parse(nil)` is equivalent to `parse()`.

Every encountered value must be a string. If `arg` is not a table, the parser uses an empty argument list.

## Explicit array

For tests or programmatically built input:

```
local result, err = parser:parse({ "-v", "input.txt" })
```

The table must be a dense array indexed from 1 to `n`, with no holes, no extra keys, and string values only. An invalid source is a parsing error and returns `(nil, err)`; it does not raise.

## Recognized forms

A valued option accepts:

```
--long value
--long=value
-s value
-s=value
```

An empty inline value is valid:

```
parser:parse({ "--output=" }) -- output == ""
```

Options remain active after a positional until `--` is encountered:

```
input.txt --verbose
```

`--` ends option parsing. Every following token becomes positional, even if it starts with `-`:

```
parser:parse({ "--", "-5" })
```

Before `--`, a token such as `-5` is interpreted as an option name. A negative number used as an option value needs no delimiter:

```
parser:parse({ "--count", "-5", "input.txt" })
```

The token following a valued option is always consumed literally, even when it looks like an option or equals `--`:

```
parser:parse({ "--output", "--verbose" })
-- output == "--verbose"; verbose is not enabled
```

A lone `-` is positional. Repeated options are accepted and the last value wins. A repeated flag simply remains true.

The following forms are not supported:

- combined short options, such as `-abc` for three flags;
- attached short values, such as `-ofile.txt`;
- variadic positionals.

A literal declared name `-abc` is still a normal option when the spec contains that exact name; it is never split.

## Defaults, choices, and conversion

A user-supplied value is processed in this order:

1. check choices against the raw string;
2. call `convert` with that string.

```
parser:option("-n --count", {
    choices = { "1", "2", "3" },
    convert = tonumber,
})
```

Every choice is a string. If the raw value is absent from the array, `parse()` returns `(nil, err)` without calling the converter.

A converter may:

- return any non-nil value, including false, to succeed;
- return nil to fail with the generic invalid value message;
- return nil, "reason" to supply a reason;
- raise, which is caught and reported as conversion error.

default values are already prepared Lua values: they are neither checked by choices nor passed through convert. They keep their type and, for a table, their identity. An option with `required = true` remains required even when a default exists.

For a flag:

- when absent, it equals default if supplied, otherwise false;
- when present, it always equals true.

An absent option or positional with a nil default has no observable key in the result table, as expected for Lua tables.

## parse results

### Normal success

```
local result, err = parser:parse()
-- result: table
-- err:    nil
```

The table is indexed by declared destinations.

### Built-in help

Before `--`, `-h` or `--help` stops parsing immediately and returns:

```
{
    help = true,
    usage = "...",
}, nil
```

Required options, missing positionals, and later tokens are not validated. After `--`, `-h` and `--help` are ordinary positionals.

Help is a success and is not merged into a normal result table. Destinations `help` and `usage` are reserved to avoid ambiguity in calling code.

### User-input error

Command-line errors return `(nil, message)`:

```
unknown option '--bad'
option '--output' requires a value
flag '-v' does not take a value
missing required option '--output'
missing required argument 'input'
argument 'mode': invalid choice 'other'
```

```
option '--count': invalid value
unexpected argument 'extra'
```

parse() catches errors raised by convert, so a user value never propagates the converter's exception.

## Programming error

The constructor and builder raise a Lua error for:

- wrong arity;
- wrong type;
- an empty or unusable spec;
- a reserved name or destination;
- a duplicate name or destination;
- an unknown opts field;
- an invalid choices array;
- a required positional after an optional positional.

Wrong parse arity is also a programming error. In contrast, a malformed src table returns (nil, err).

## Complete example

```
local argparse = require("argparse")

local parser = argparse("convert", "Convert a file.")
:flag("-v --verbose", { help = "Show more detail" })
:option("-o --output", { default = "out.txt" })
:option("-n --count", {
    choices = { "1", "2", "3" },
    convert = tonumber,
    default = 1,
})
:argument("input")

local args, err = parser:parse()
if args and args.help then
    print(args.usage)
    return
end
if not args then
    io.stderr:write("error: ", err, "\n")
    io.stderr:write(parser:get_usage(), "\n")
    os.exit(2)
end

print(args.input, args.output, args.count, args.verbose)
```

## Text produced by get\_usage

get\_usage() returns a string and writes nothing. The section labels and the built-in help message are fixed in English: Usage, Arguments, Options, and show this help. There is no localization, column-width, or visual-group API yet.

The text lists positionals and declared help strings, but it does not automatically show defaults, choices, or the required status of options.

## Current limitations

The module does not expose:

- subcommands;
- variadic positionals (nargs);
- mutually exclusive groups;
- repeatable options accumulated into an array;
- combined short options;
- help localization or customization;
- a way to disable built-in help.

## EXEC - external programs, capture, and streaming

Babet provides two complementary APIs for starting an external program without an implicit shell:

- `babet.exec` optionally sends complete input, captures stdout and stderr in memory, then waits for completion;
- `babet.spawn` immediately returns a controllable process object so streams can be read and written progressively without automatic RAM accumulation.

EXEC covers:

- looking up a program through PATH;
- building argv safely, without shell parsing;
- selecting a working directory for the child process;
- merging a custom environment with Babet's environment;
- sending a binary string to standard input;
- capturing binary stdout and stderr;
- applying one timeout to launch, I/O, and final wait;
- enforcing a separate memory limit for each output stream;
- terminating the child's process group after a timeout.

`babet.exec` does not provide direct file redirection: it captures stdout and stderr in memory. For a controllable process, `babet.spawn` lets you configure stdin, stdout, and stderr independently as pipe, inherit, or null, and redirect stdout and stderr to a file.

`babet.exec` also does not provide an implicit command interpreter. To connect several programs without a shell, use `babet.pipeline()` or `babet.spawnPipeline()`.

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## Essential conventions

### No implicit shell

This call:

```
local result, err = babet.exec("printf", { "%s\n", "hello world" })
```

starts `printf` directly. Every `args` element becomes a separate `argv` slot. Spaces, `$`, `*`, `;`, `>`, quotes, and other shell characters are not interpreted.

This avoids most quoting mistakes and command-injection issues.

### A non-zero exit is not a launch error

A program that starts successfully and exits with code 1, 2, or 127 still produces a result table. The caller decides which codes are acceptable.

```
local result, err = babet.exec("grep", { "needle", "data.txt" })
assert(result, err)
```

```
if result.code == 0 then
    print("match found")
elseif result.code == 1 then
    print("no match")
else
    error(result.stderr)
end
```

`assert(babet.exec(...))` only checks that Babet could launch and monitor the program. It does not check `result.code`.

### Data strings are binary-safe

`opts.stdin`, `result.stdout`, and `result.stderr` are binary-safe Lua strings. They may contain NUL bytes and do not need to be UTF-8 text.

Strings passed to POSIX APIs that require C strings - `cmd`, `args` elements, `cwd`, and environment names and values - cannot contain a NUL byte.

## API overview

```
local result, err = babet.exec(cmd, args?, opts?)
```

| Element                      | Type                   | Default                     | Purpose                       |
|------------------------------|------------------------|-----------------------------|-------------------------------|
| <code>cmd</code>             | strict string          | required                    | program name or path          |
| <code>args</code>            | string sequence        | no arguments                | <code>argv[1]..argv[n]</code> |
| <code>opts.cwd</code>        | string                 | inherited current directory | child working directory       |
| <code>opts.env</code>        | string-to-string table | inherited environment       | variables to add or replace   |
| <code>opts.stdin</code>      | binary string          | stdin closed immediately    | data sent to the child        |
| <code>opts.timeout</code>    | finite number > 0      | no limit                    | global budget in seconds      |
| <code>opts.max_output</code> | integer 1..2 GiB       | 10 MiB                      | cap per captured stream       |

Main outcomes:

| Situation                                      | Return value                                                  |
|------------------------------------------------|---------------------------------------------------------------|
| program launched and completed                 | <code>(result, nil)</code>                                    |
| program launched then timed out                | <code>(result, nil)</code> with <code>timed_out = true</code> |
| non-zero exit status                           | <code>(result, nil)</code> with <code>code ~= 0</code>        |
| program not found or invalid <code>cwd</code>  | <code>(nil, err)</code>                                       |
| invalid <code>args</code> or <code>opts</code> | <code>(nil, err)</code>                                       |
| missing or non-string <code>cmd</code>         | raised Lua error                                              |

## Signature and arguments

### `cmd` and program lookup

`cmd` must be an actual Lua string.

```
local result, err = babet.exec("git", { "status", "--short" })
```

When `cmd` contains no `/`, Babet asks the C library to search for it through the Babet process's `PATH`.

```
local result = assert(babet.exec("python3", { "--version" }))
```

When `cmd` contains `/`, it is treated as a direct path.

```
local result = assert(babet.exec("/usr/bin/id", { "-u" }))
```

A relative path containing `/` is resolved in the child's directory after `opts.cwd` has been applied.

```
local result = assert(babet.exec("./tool", {}, {  
    cwd = "/opt/my-app/bin",  
}))
```

**PATH and `opts.env` nuance** The child may receive a replacement `PATH` through `opts.env`. However, the current Linux implementation uses `execvpe`, whose initial lookup of `cmd` uses the Babet process's `PATH`, not `opts.env.PATH`.

```
local result, err = babet.exec("my-private-tool", {}, {
    env = { PATH = "/opt/private/bin" },
})
```

This does not guarantee that `/opt/private/bin/my-private-tool` will be found. Use a direct path:

```
local result, err = babet.exec("/opt/private/bin/my-private-tool", {}, {
    env = { PATH = "/opt/private/bin" },
})
```

The replacement PATH is still visible **inside** the child and to commands it starts later.

### args and shell-free execution

args is optional. It must be a dense 1..n sequence of strings.

```
local result = assert(babet.exec("cp", {
    "--",
    "file with spaces.txt",
    "destination/",
}))
```

Babet reads only the Lua sequence part, from 1 through #args. For predictable behavior:

- use consecutive indexes starting at 1;
- do not leave holes;
- do not store metadata in the same table;
- use strings only.

*-- Correct*

```
local args = { "-c", "print('ok')" }
```

*-- Invalid: numeric element*

```
local result, err = babet.exec("lua", { "-e", 42 })
-- result == nil; err says all arguments must be strings
```

argv[0] is set to cmd automatically. The first args element becomes argv[1].

### Using a shell explicitly

Operators such as `|`, `>`, `&&`, `;`, variables such as `$NAME`, and globs are only interpreted when you explicitly start a shell.

```
local result = assert(babet.exec("sh", {
    "-c",
    "printf '%s\\n' \"$HOME\" | sed 's/^#home=#'",
}))
```

Values embedded in shell source must then follow shell quoting rules. For untrusted data, pass values through args, stdin, or env instead of concatenating them into the `-c` script.

## Options

The third argument is optional. When present, it must be a table. Unknown fields are currently ignored; a typo such as `timeot = 5` therefore does not raise an error. Use the exact names below.

### opts.cwd

Sets the child's current directory without changing the Babet process's current directory.

```

local result, err = babet.exec("pwd", {}, {
    cwd = "/var/tmp",
})
assert(result, err)
print(result.stdout) -- /var/tmp

```

The child performs `chdir` before launching the program. It affects:

- relative paths used by the program;
- a relative cmd containing `/`;
- relative entries in the `PATH` used for lookup.

If the directory does not exist or is inaccessible, Babet returns `(nil, err)`. The parent current directory is unchanged.

### **opts.env**

`env` is a table whose keys and values must be strings. It is **merged** with the Babet process's environment.

```

local result = assert(babet.exec("sh", {
    "-c", "printf '%s' \"$APP_MODE\"",
}, {
    env = { APP_MODE = "production" },
}))

```

```
assert(result.stdout == "production")
```

Rules:

- a supplied key replaces the inherited variable of the same name;
- an omitted key remains inherited;
- `""` creates or keeps a defined-but-empty variable;
- there is no option to remove a variable completely;
- a key must be non-empty and cannot contain `=`;
- keys and values reject NUL bytes;
- non-string keys and values are rejected.

Defined-but-empty is different from absent:

```

local result = assert(babet.exec("sh", { "-c", [[
    if [ "${TOKEN+x}" = x ] && [ -z "$TOKEN" ]; then
        printf 'defined-empty'
    fi
]] }, {
    env = { TOKEN = "" },
}))

```

```
assert(result.stdout == "defined-empty")
```

The environment is built in the parent before fork, avoiding `setenv` calls in a child of a multithreaded process.

### **opts.stdin**

`stdin` is one string sent completely to the program's standard input.

```

local result = assert(babet.exec("sha256sum", { "-" }, {
    stdin = "hello world\n",
}))
print(result.stdout)

```

It is binary-safe:

```
local payload = "A\0B\0C"
local result = assert(babet.exec("cat", {}, { stdin = payload }))
assert(result.stdout == payload)
```

When stdin is absent or nil, Babet closes the input pipe immediately, so the child sees EOF. An empty string also produces EOF after the normal pipe setup.

Babet writes stdin while reading both output pipes, preventing deadlocks with a program that produces substantial output before consuming all its input.

### **opts.timeout**

timeout is a finite, strictly positive number of seconds.

```
local result = assert(babet.exec("sleep", { "30" }, {
    timeout = 0.5,
}))
```

```
assert(result.timed_out)
```

The budget starts before launch preparation and covers:

1. environment preparation;
2. fork;
3. the child chdir;
4. the exec call;
5. stdin transfer;
6. stdout and stderr capture;
7. the final process wait.

Internal resolution is one millisecond. Values below 0.001 therefore act as an almost immediate deadline and should not be used for precise timing.

The value must fit within INT\_MAX milliseconds, about 24.8 days. Zero, negative, NaN, infinite, and larger values return (nil, err).

After expiry once the child has launched:

1. Babet sends SIGTERM to the process group;
2. it allows up to 2 seconds for a clean exit;
3. it sends SIGKILL if needed;
4. it continues draining output for a bounded window.

### **opts.max\_output**

max\_output limits retained bytes **for each stream**.

```
local result = assert(babet.exec("yes", {}, {
    timeout = 0.2,
    max_output = 4096,
}))
```

```
assert(#result.stdout == 4096)
assert(result.stdout_truncated)
```

Values:

- default: 10 \* 1024 \* 1024, or 10 MiB;
- minimum: 1 byte;
- maximum: 2 \* 1024 \* 1024 \* 1024, or 2 GiB;
- type: integral Lua number.

The cap is independent: up to 10 MiB of stdout **and** 10 MiB of stderr by default.

After reaching the cap, Babet keeps reading and discarding excess bytes. This is necessary to avoid filling the pipe and blocking the child. The first bytes are retained; the final bytes are lost.

## Result table

After a successful launch, the first return value is always a table with six fields:

```
{
  stdout = "...",
  stderr = "...",
  code = 0,
  timed_out = false,
  stdout_truncated = false,
  stderr_truncated = false,
}
```

| Field            | Type          | Meaning                                              |
|------------------|---------------|------------------------------------------------------|
| stdout           | binary string | first retained bytes from standard output            |
| stderr           | binary string | first retained bytes from standard error             |
| code             | integer       | normal code, signal convention, or -1 if unavailable |
| timed_out        | boolean       | Babet initiated termination after deadline expiry    |
| stdout_truncated | boolean       | stdout exceeded max_output                           |
| stderr_truncated | boolean       | stderr exceeded max_output                           |

## Normal exit code

When the program calls `exit(n)` or returns from `main`, `code` is the POSIX status `0..255`.

```
local result = assert(babet.exec("sh", { "-c", "exit 7" }))
assert(result.code == 7)
assert(not result.timed_out)
```

## Signal termination

When the process is terminated by a signal, Babet uses:

```
code = 128 + signal_number
```

For example, `SIGKILL` is normally 9 on Linux, giving `code == 137`. This is a useful convention but there is no separate result field containing the signal.

## Timeout

`timed_out == true` means Babet reached its deadline and started termination. `code` depends on how the child eventually stopped:

- an application-defined code if it handles `SIGTERM` and exits cleanly;
- often 143 for `SIGTERM`;
- often 137 after `SIGKILL`;
- exceptionally -1 if status could not be obtained within bounded cleanup.

Do not test only `code == 137` to detect a timeout; use `timed_out`.

Output produced before expiry is retained up to `max_output`.

## Truncation

`stdout_truncated` and `stderr_truncated` only mean that the corresponding stream exceeded `max_output`.

```
if result.stdout_truncated then
    print("stdout is incomplete: increase max_output or redirect to a file")
end
```

In an extremely pathological kernel/process-group case, a descendant may keep a pipe open after SIGKILL and force Babet to abandon draining rather than block forever. This rare cause is not represented by the truncation flags, which are reserved for the memory cap.

## Complete examples

### Basic capture

```
local result, err = babet.exec("git", { "rev-parse", "HEAD" })
assert(result, err)

if result.code ~= 0 then
    error("git failed: " .. result.stderr)
end

local commit = result.stdout:gsub("%s+$", "")
print(commit)
```

### Arguments containing spaces

```
local filename = "July 2026 report.txt"
local result = assert(babet.exec("printf", {
    "name=<%s>\n",
    filename,
}))

assert(result.stdout == "name=<July 2026 report.txt>\n")
```

No additional quoting is needed around `filename`: the Lua table already represents the argument boundary.

### Handling a non-zero code

```
local result, err = babet.exec("diff", {
    "--brief",
    "config.old",
    "config.new",
})
assert(result, err)

if result.code == 0 then
    print("files are identical")
elseif result.code == 1 then
    print("files differ")
else
    error("diff failed: " .. result.stderr)
end
```

### Capturing stderr

```
local result = assert(babet.exec("sh", {  
    "-c", "printf out; printf err >&2",  
}))
```

```
assert(result.stdout == "out")  
assert(result.stderr == "err")
```

### Sending text or binary stdin

```
local csv = "name,score\nAlice,18\nBob,15\n"  
local result = assert(babet.exec("sort", {}, { stdin = csv }))  
print(result.stdout)  
  
local bytes = string.char(0x00, 0x01, 0xFE, 0xFF)  
local result = assert(babet.exec("cat", {}, { stdin = bytes }))  
assert(result.stdout == bytes)
```

### Changing the environment

```
local result = assert(babet.exec("sh", { "-c", [[  
    printf 'mode=%s lang=%s' "$APP_MODE" "$LANG"  
]] }, {  
    env = {  
        APP_MODE = "test",  
        LANG = "C",  
    },  
}))
```

```
print(result.stdout)
```

Only supplied keys are replaced. Other variables, including HOME, remain inherited.

### Changing the working directory

```
local result = assert(babet.exec("find", {  
    ".", "-maxdepth", "1", "-type", "f",  
}, {  
    cwd = "/var/log",  
}))
```

```
print(result.stdout)
```

The main script's current directory is unchanged.

### Bounding execution time

```
local result, err = babet.exec("sh", {  
    "-c", "printf started; sleep 30",  
}, {  
    timeout = 1.0,  
})  
assert(result, err)  
  
if result.timed_out then  
    print("command exceeded its budget")
```

```

    print("available output:", result.stdout)
end

```

### Bounding captured output

```

local result = assert(babet.exec("sh", {
    "-c", "yes log-line | head -n 100000",
}, {
    max_output = 64 * 1024,
}))

if result.stdout_truncated then
    print("only the first 64 KiB were retained")
end

```

For gigabytes, do not capture into memory. Explicitly start a shell and redirect to a carefully selected file:

```

local result = assert(babet.exec("sh", {
    "-c", "my-command > /var/tmp/my-command.log 2>&1",
}, {
    timeout = 600,
}))

```

### Building an application helper

```

local function run_checked(cmd, args, opts)
    local result, err = babet.exec(cmd, args, opts)
    if not result then
        return nil, err
    end
    if result.timed_out then
        return nil, string.format("%s: timeout", cmd)
    end
    if result.code ~= 0 then
        local detail = result.stderr ~= "" and result.stderr or result.stdout
        return nil, string.format(
            "%s: exit code %d: %s",
            cmd,
            result.code,
            detail
        )
    end
    return result
end

local result, err = run_checked("git", { "status", "--short" }, {
    cwd = "/srv/project",
    timeout = 10,
})
assert(result, err)
print(result.stdout)

```

### Controllable processes with `babet.spawn`

`babet.spawn` uses the same shell-free launch engine as `babet.exec`, but does not wait for program completion. It returns a userdata representing the child:

```

local process, err = babet.spawn("yt-dlp", {
    "--newline",
    "https://example.invalid/video",
})
assert(process, err)

```

By default, stdin, stdout, and stderr use three non-blocking pipes, preserving the historical behaviour exactly. Each stream may instead be inherited or connected to /dev/null; output streams may be redirected directly to files, and stderr may be merged into stdout.

## Signature and options

```

local process, err = babet.spawn(command, args?, opts?)

```

| Element             | Type                          | Default               | Purpose                       |
|---------------------|-------------------------------|-----------------------|-------------------------------|
| command             | strict string                 | required              | program name or path          |
| args                | dense string array            | no arguments          | argv[1]..argv[n]              |
| opts.cwd            | string                        | inherited directory   | child working directory       |
| opts.env            | string-to-string table        | inherited environment | variables to add/replace      |
| opts.launch_timeout | finite number > 0             | unlimited wait        | bounds the chdir + exec phase |
| opts.stdin          | "pipe", "inherit", or "null"  | "pipe"                | standard-input source         |
| opts.stdout         | mode or file table            | "pipe"                | standard-output destination   |
| opts.stderr         | mode, "stdout", or file table | "pipe"                | standard-error destination    |

Unknown options are rejected. spawn does not accept the timeout and max\_output options of exec: lifetime is controlled with wait()/terminate(), and piped outputs are read progressively.

launch\_timeout covers preparation and launch only, until the new executable is established. It is not a total process-lifetime limit. A redirection-file open failure happens before fork(), so no child is created in that case.

PATH lookup, paths containing /, environment merging, and shell-free execution follow the same rules as exec.

## Redirection modes

**"pipe"** The default mode creates a non-blocking pipe between parent and child:

```

local process = assert(babet.spawn("cat", {}, {
    stdin = "pipe",
    stdout = "pipe",
    stderr = "pipe",
}))

```

```

assert(process:write("hello\n"))
assert(process:close_stdin())
print(assert(process:read_stdout(64 * 1024, 1)))

```

write, close\_stdin, read\_stdout, and read\_stderr are available only for streams configured as "pipe".

**"inherit"** The child directly reuses Babet's corresponding standard descriptor:

```
local process = assert(babet.spawn("make", { "-j4" }, {
    stdin = "inherit",
    stdout = "inherit",
    stderr = "inherit",
}))
assert(process:wait())
```

Use this when a program should interact with the terminal or respect an external redirection applied to Babet itself.

**"null"** The stream is connected to /dev/null:

```
local process = assert(babet.spawn("quiet-program", {}, {
    stdin = "null", -- immediate EOF in the child
    stdout = "null",
    stderr = "null",
}))
```

**File redirection** stdout and stderr accept a strict table:

```
{
    file = "/var/tmp/application.log", -- required, no NUL byte
    append = true, -- false by default: truncate
    permissions = tonumber("600", 8), -- 0600 by default, creation only
}
```

- append = false truncates an existing regular file;
- append = true writes with O\_APPEND;
- permissions must be an integer from 0000 to 0777 and applies only when Babet creates the file;
- the final destination must not be a symbolic link;
- after opening, Babet uses fstat() to require a regular file; directories, FIFOs, sockets, and devices are rejected;
- the descriptor is prepared before fork() and retained only by the child.

stdout-only example:

```
local process = assert(babet.spawn("generate-report", {}, {
    stdout = {
        file = "report.log",
        permissions = tonumber("640", 8),
    },
}))
```

stderr-only append example:

```
local process = assert(babet.spawn("worker", {}, {
    stderr = {
        file = "errors.log",
        append = true,
    },
}))
```

**Merging stderr into stdout** stderr = "stdout" duplicates descriptor 2 onto the already configured stdout. It works with a pipe, inheritance, /dev/null, or a file:

```
local process = assert(babet.spawn("driver", {}, {
    stdout = {
```

```

        file = "driver.log",
        append = true,
    },
    stderr = "stdout",
)))

```

For one combined log file, this form is preferable to two independent file tables because it uses one descriptor and one write offset.

## Process methods

| Method                                                                                | Return                                | Purpose                                |
|---------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------|
| <code>process:read_stdout([max_bytes(data, nil) or (nil, reason) [, timeout]])</code> |                                       | read stdout when it is a pipe          |
| <code>process:read_stderr([max_bytes(data, nil) or (nil, reason) [, timeout]])</code> |                                       | read stderr when it is a pipe          |
| <code>process:write(data [, timeout])</code>                                          | (bytes_written, nil) or (nil, reason) | write stdin when it is a pipe          |
| <code>process:close_stdin()</code>                                                    | (true, nil) or (nil, reason)          | idempotently close piped stdin         |
| <code>process:is_running()</code>                                                     | boolean or (nil, err)                 | check state without blocking           |
| <code>process:pid()</code>                                                            | integer                               | return the assigned PID                |
| <code>process:wait([timeout])</code>                                                  | (result, nil) or (nil, reason)        | wait without killing on timeout        |
| <code>process:terminate([grace_period])</code>                                        | (result, nil) or (nil, err)           | SIGTERM, then SIGKILL if needed        |
| <code>process:kill()</code>                                                           | (result, nil) or (nil, err)           | SIGKILL the process group              |
| <code>process:close()</code>                                                          | (true, nil)                           | clean descriptors and any active child |

Short reasons are:

- "timeout": no progress before the deadline;
- "closed": closed pipe or EOF;
- "interrupted": handled Babet interruption;
- "not\_piped": the method targets a stream not configured as "pipe".

Other diagnostics use a `process:` or `spawn:` prefix.

The table returned by `wait`, `terminate`, and `kill` contains:

```

{
  code = 0,           -- normal code, or 128 + signal
  exited = true,     -- normal exit/_exit termination
  signaled = false,   -- terminated by a signal
  signal = nil,       -- signal number when signaled == true
}

```

A second `wait()` after completion returns the same cached status.

## Reading stdout and stderr

Pipe reads are binary-safe. By default they read at most 64 KiB and are non-blocking (`timeout = 0`). The maximum size per call is 16 MiB.

```

local chunk, err = process:read_stdout(64 * 1024, 0.25)

if chunk then
    io.write(chunk)

```

```
elseif err == "timeout" then
    -- nothing became available for 250 ms
elseif err == "closed" then
    -- final EOF on stdout
elseif err == "not_piped" then
    -- stdout is inherited, null, or redirected to a file
else
    error(err)
end
```

"closed" is EOF for that stream, not necessarily process completion. When stdout and stderr are both pipes, drain both: reading only one while the other fills can block the child. File, inherited, and null outputs do not fill a parent-side pipe.

## Writing stdin

With `stdin = "pipe"`, write accepts a binary Lua string including NUL bytes. The non-blocking pipe may accept only part of the string, so resume from the returned byte count.

```
local function write_all(process, data)
    local offset = 1
    while offset <= #data do
        local written, err = process:write(data:sub(offset), 1)
        assert(written, err)
        offset = offset + written
    end
end
```

```
write_all(process, payload)
assert(process:close_stdin())
```

An empty string returns (0, nil). After `close_stdin`, write returns (nil, "closed"). With `stdin = "inherit"` or `stdin = "null"`, write and `close_stdin` return (nil, "not\_piped").

## Waiting, terminating, and closing

`wait()` without an argument waits for completion. `wait(timeout)` returns (nil, "timeout") without sending a signal or invalidating the object.

`terminate(grace_period)` sends SIGTERM to the whole group and then SIGKILL if needed. The default grace period is two seconds. `kill()` sends SIGKILL directly.

`close()` is idempotent. It terminates and reaps an active child with bounded waits, then closes every descriptor still owned by the parent. Lua `__close` and garbage collection perform the same cleanup.

## Combined example: launching a WebDriver or daemon

```
local process, err = babet.spawn("bin/geckodriver", {
    "--port=4444",
}, {
    stdin = "null",
    stdout = {
        file = "logs/geckodriver.log",
        append = true,
        permissions = tonumber("600", 8),
    },
    stderr = "stdout",
})
```

```

assert(process, err)

local result, wait_err = process:wait(0)
if not result and wait_err == "timeout" then
    print("geckodriver is running with PID", process:pid())
end

```

No shell is involved, arguments remain separate, stdin cannot wait for input, and logs cannot fill a parent-side pipe.

### Combined example: streaming loop

```

local process = assert(babet.spawn("yt-dlp", {
    "--newline",
    "-f", "bestvideo+bestaudio/best",
    url,
}))

local stdout_open, stderr_open = true, true
while stdout_open or stderr_open do
    if stdout_open then
        local data, read_err = process:read_stdout(64 * 1024, 0.1)
        if data then
            io.write(data)
            io.flush()
        elseif read_err == "closed" then
            stdout_open = false
        elseif read_err ~= "timeout" then
            error(read_err)
        end
    end

    if stderr_open then
        local data, read_err = process:read_stderr(64 * 1024, 0)
        if data then
            io.stderr:write(data)
            io.stderr:flush()
        elseif read_err == "closed" then
            stderr_open = false
        elseif read_err ~= "timeout" then
            error(read_err)
        end
    end
end

local result, wait_err = process:wait(5)
assert(result, wait_err)
process:close()

```

### Limits and deadlock prevention

- `wait()` does not drain outputs configured as "pipe". For verbose children, drain the pipes, redirect logs to a file, use "inherit"/"null", or use `babet.exec` for small bounded output.
- Large simultaneous input and output require alternating writes and reads.
- A `babet.spawn()` object cannot be connected after launch to another process; use `babet.spawnPipeline()` for a linear pipeline.
- File tables are not supported for stdin in this version.

- Babet does not create parent directories for redirection files.
- Redirection files are opened before `fork()`. If a later `fork()` or launch error occurs, a file may already have been created or truncated even though no usable process is returned.
- `close()` and `terminate()` target the launch-time process group. A descendant that deliberately changes group or session may escape that control.

## Error contract

### Raised Lua errors

Only the first argument's signature raises:

```
babet.exec()    -- raises: missing command
babet.exec(42) -- raises: cmd must be a string
```

Use `pcall` only when an invalid signature can come from uncontrolled data.

### Returned (nil, err) errors

The following validations return (nil, err) rather than raising:

- args is neither a table nor nil;
- an args element is not a string;
- opts is neither a table nor nil;
- cwd, stdin, timeout, max\_output, or env has the wrong type;
- an environment key is empty, contains =, or contains NUL;
- an environment value contains NUL;
- timeout is zero, negative, non-finite, or too large;
- max\_output is zero, negative, fractional, or above 2 GiB;
- program not found or not executable;
- invalid cwd;
- pipe, fork, poll, waitpid, or another internal system failure.

After a successful `fork()`, Babet retains ownership of descriptors and of the process group until ownership is transferred to a `spawn()` object or final reaping completes for `exec()`. If an internal C++ exception occurs in that window—for example while allocating an output buffer—the emergency path closes descriptors, immediately sends SIGKILL to the group, and attempts to reap the child within a short bounded window before returning nil, err. Exception cleanup does not use `terminate()`'s grace period, which is reserved for explicitly requested shutdown.

```
local result, err = babet.exec("echo", "hello")
assert(result == nil)
print(err) -- args must be a table
```

### Situations returning a table

These are not API errors:

- non-zero program exit;
- output written to stderr;
- timeout during or after launch;
- output exceeding max\_output.

## Processes, signals, and security

### Process group

The child creates its own process group. On timeout, Babet signals the group, not only the direct PID. This normally covers descendants started by a shell or by the program.

A descendant that deliberately calls `setuid` or changes process group can escape this strategy. `babet.exec` is not a sandbox and does not replace `systemd`, `cgroups`, `namespaces`, or resource limits.

### Interaction with `babet.signal`

`babet.exec` is not signal-aware. A handled signal received while the call is running does not return (`nil`, `"interrupted"`). The C handler records a pending flag; the Lua callback can only be dispatched after control returns to the Lua VM, normally through `SIGNAL`'s instruction hook.

Use `opts.timeout` to bound a long-running command. For externally requested cancellation, design another protocol or run the command under a supervisor.

### Memory

The script already holds `stdin` in memory, and `stdout/stderr` are captured in memory up to the selected limits. The defaults permit about 20 MiB of captured output in total, in addition to application and Lua-string allocations.

### Design and limitations

- Linux/POSIX only: the implementation uses `fork`, `execvpe`, `poll`, process groups, and signals.
- No implicit shell.
- One `exec` call still represents one command; native pipelines use `babet.pipeline()` or `babet.spawnPipeline()`.
- One complete `stdin` string, no streaming producer.
- Complete capture returned at the end, no line callback.
- No native `stdin/stdout/stderr` redirection to files or descriptors.
- No separate signal field in the result table.
- No full environment replacement or unset operation: `env` merges and an empty string remains a defined variable.
- Unknown `opts` fields are ignored.
- Initial `PATH` lookup currently follows the Babet process environment even when `opts.env.PATH` is replaced.
- `babet.signal` callbacks are not dispatched during the call.

See [Security](#) for the general security model.

### Process pipelines

Babet provides two complementary APIs for connecting programs through POSIX pipes without invoking a shell:

- `babet.pipeline()` runs the whole pipeline and captures its streams;
- `babet.spawnPipeline()` immediately returns a streaming process object.

In both cases, each stage's `stdout` is connected directly to the next stage's `stdin`. The parent only owns the first stage's `stdin`, the last stage's `stdout`, and one separate `stderr` stream per stage.

### Shared command syntax

```
local commands = {
  { "printf", { "orange\napple\norange\n" } },
  { "grep", { "orange" } },
  { "sort", { "-u" } },
}
```

`commands` must be a dense array containing 2 to 32 stages. Each stage has one of these forms:

```
{ command }
{ command, args }
{ command, args, opts }
```

To provide opts without arguments, explicitly use { command, {}, opts }; sparse arrays are rejected.

- command is a non-empty string without a NUL byte;
- args is a dense array of strings without NUL bytes;
- opts.cwd sets this stage's working directory;
- opts.env adds or replaces environment variables for this stage.

No string is parsed by a shell: globbing, redirection, variables, quoting and operators such as |, && or > have no special meaning.

Global cwd and env options provide defaults. A local cwd replaces the global one. Local environment entries are merged last and win on duplicate names. Every command is validated before the first fork().

As with exec() and spawn(), when a command contains no /, its initial lookup uses the Babet process's PATH, not a PATH replaced through opts.env or a stage-local environment. To launch a tool available only through that new PATH, use its explicit path or change Babet's PATH before launching.

## babet.pipeline() — complete execution

### Signature

```
result, err = babet.pipeline(commands [, opts])
```

Example:

```
local result, err = babet.pipeline({
  { "printf", { "orange\napple\norange\n" } },
  { "grep", { "orange" } },
  { "sort", { "-u" } },
}, {
  timeout = 10,
  max_output = 16 * 1024 * 1024,
})

assert(result, err)
print(result.stdout)
```

### Options

| Option     | Type                                             | Description                                                          |
|------------|--------------------------------------------------|----------------------------------------------------------------------|
| cwd        | string                                           | Default working directory for every stage.                           |
| env        | string-to-string table                           | Variables merged with Babet's environment.                           |
| stdin      | binary string                                    | Data progressively written to the first stage, then stdin is closed. |
| timeout    | finite positive number, at most 10 <sup>12</sup> | Global timeout in seconds, including launch, I/O and final wait.     |
| max_output | integer from 1 to 2 GiB                          | Separate limit for final stdout and each stderr; default: 10 MiB.    |

Unknown keys are rejected. stdin, stdout and stderr are binary-safe and may contain NUL bytes. Timeout resolution is one millisecond; values above 10<sup>12</sup> seconds are rejected.

## Result

Once every stage has launched, a non-zero exit or signal is a process result, not an API error. Babet returns:

```
{
  stdout = "...",
  stderr = { "stage 1 stderr", "stage 2 stderr", ... },
  stages = {
    {
      launched = true,
      code = 0,
      exited = true,
      signaled = false,
      signal = nil,
    },
    -- ...
  },
  code = 0,
  all_succeeded = true,
  failed_index = nil,
  timed_out = false,
  stdout_truncated = false,
  stderr_truncated = { false, false, ... },
}
```

code is the last stage's code, matching Unix pipeline convention. It is the normal exit status or 128 + signal\_number. In the pathological case where a status remains unavailable after bounded timeout cleanup, it is -1 for that stage. all\_succeeded is true only when every stage succeeded. failed\_index identifies the first non-zero stage even when the last one succeeds.

A non-zero intermediate stage does not artificially stop the other programs. Normal POSIX pipe behaviour applies; an upstream producer may receive SIGPIPE after a downstream consumer closes its input.

## Timeout, large streams and truncation

pipeline()'s timeout covers launch, stdin writes, stdout/stderr draining and the final wait. If it expires before every stage completes chdir and exec, the call returns nil, err with the message "pipeline: launch timed out" after rolling back stages already created. After a complete launch, expiry returns a result table with timed\_out == true: Babet closes stdin, sends SIGTERM to every stage process group, then SIGKILL to survivors, and reaps direct children.

Descriptors are non-blocking and driven with poll(). The first stdin, final stdout and all stderr streams progress concurrently, preventing the usual large-volume deadlocks. Once max\_output is reached, extra bytes are discarded but the stream continues to be drained, and the matching truncation flag is set.

## babet.spawnPipeline() — streaming

### Signature

```
pipeline, err = babet.spawnPipeline(commands [, opts])
```

Global options:

| Option         | Type                   | Description                                                                                                        |
|----------------|------------------------|--------------------------------------------------------------------------------------------------------------------|
| cwd            | string                 | Default working directory for every stage.                                                                         |
| env            | string-to-string table | Variables merged with Babet's environment.                                                                         |
| launch_timeout | finite positive number | Deadline covering only creation and complete launch of all stages, at most INT_MAX milliseconds (about 24.8 days). |

`spawnPipeline()` accepts no `stdin`, `timeout`, or `max_output` option: the script streams data and applies its own limits. Success returns a userdata and `nil`; validation or launch failure returns `nil`, `err` after cleaning up any stage that was already created.

```
local pipeline, err = babet.spawnPipeline({
    { "cat" },
    { "tr", { "a-z", "A-Z" } },
})
assert(pipeline, err)

local written, write_err = pipeline:write("hello\n")
assert(written, write_err)
assert(pipeline:close_stdin())

local chunks = {}
while true do
    local data, read_err = pipeline:read_stdout(64 * 1024, 1)
    if data then
        chunks[#chunks + 1] = data
    elseif read_err == "closed" then
        break
    elseif read_err ~= "timeout" then
        error(read_err)
    end
end

local status, wait_err = pipeline:wait(5)
assert(status, wait_err)
assert(status.code == 0)
print(table.concat(chunks))
```

## Read methods

```
data, err = pipeline:read_stdout([max_bytes [, timeout]])
data, err = pipeline:read_stderr(stage [, max_bytes [, timeout]])
```

- `max_bytes` defaults to 64 KiB and must be between 1 and 16 MiB;
- `timeout` is in seconds, defaults to zero, and must be finite and non-negative;
- `stage` ranges from 1 to the number of stages;
- invalid arity, stage index, or `max_bytes` raises a Lua error, matching `babet.spawn()` methods;
- an invalid timeout returns `nil`, `err`;
- success returns a non-empty binary string and `nil`;
- no immediately available data returns `nil`, "timeout";
- EOF, or reading an already closed stream, returns `nil`, "closed".

Each call reads at most one chunk. There is no hidden capture: the script must retain, limit, or discard received data itself. Stderr streams remain independent, so every stream capable of filling its pipe must be drained.

## Writing and closing stdin

```
count, err = pipeline:write(data [, timeout])
ok, err = pipeline:close_stdin()
```

`write()` accepts a binary string and may write only part of it; `count` is the number of bytes actually transmitted. An empty string returns `0`, `nil`. A timeout leaves the object reusable. Stdin closed by the script or downstream process returns `nil`, "closed" without delivering SIGPIPE to Babet itself.

`close_stdin()` closes the first stage's input and is idempotent. Call it after the last byte, otherwise a program waiting for EOF may never terminate.

Complete-write pattern:

```
local offset = 1
while offset <= #data do
    local count, err = pipeline:write(data:sub(offset), 1)
    if count then
        offset = offset + count
    elseif err ~= "timeout" then
        error(err)
    end
    -- Drain stdout and stderr during large transfers as well.
end
assert(pipeline:close_stdin())
```

## State and identifiers

```
running = pipeline:is_running([stage])
pids = pipeline:pids()
```

Without an index, `is_running()` is true while at least one direct stage has not been reaped. With an index, it describes only that stage. It refreshes statuses through `waitpid(..., WNOHANG)` without consuming the final result: `wait()` remains available afterwards.

`pids()` returns direct PIDs in stage order. They are diagnostic values; scripts must not assume that a PID remains valid after the pipeline exits.

## Waiting and final status

```
status, err = pipeline:wait([timeout])
```

Without a timeout, `wait()` waits for every direct child. With a finite non-negative timeout, `nil`, "timeout" leaves the pipeline intact and reusable. Once every status is known, subsequent calls are idempotent and return the same result:

```
{
    stages = {
        { launched = true, code = 0, exited = true,
          signaled = false, signal = nil },
        -- ...
    },
    code = 0,
    all_succeeded = true,
    failed_index = nil,
}
```

`wait()` **does not drain** stdout or stderr. A pipeline that fills a pipe may therefore block before exiting. Streams must be read while it runs, exactly as with `babet.spawn()`.

## Explicit termination

```
status, err = pipeline:terminate([grace_period])
status, err = pipeline:kill()
```

`terminate()` sends SIGTERM to every process group, gives processes `grace_period` seconds—2 seconds by default—to exit, then sends SIGKILL to survivors. The grace period must fit within INT\_MAX milliseconds, about 24.8 days. `kill()` sends SIGKILL immediately. Both close stdin and return the same status table as `wait()` after direct children are reaped.

Each stage owns its own process group. Before reaping the direct process, Babet removes descendants that still remain in that group, including normal completion where a command exits without waiting for a background child. The leader PID therefore remains reserved until signalling and cannot be recycled for an unrelated process.

For `terminate()`, every group member receives SIGTERM at the beginning of the grace period. Direct leaders that have already exited deliberately remain unreaped for the whole period: their PID, which is also used as the PGID, cannot be recycled before the optional final SIGKILL sent to the group. A stage that deliberately creates a new session or process group escapes this POSIX guarantee. A script that knows its processes are cooperative may select a shorter grace period, for example `pipeline:terminate(0.05)`.

## close(), GC and Lua <close>

```
ok, err = pipeline:close()
```

`close()` is idempotent. It closes descriptors, attempts SIGTERM, uses SIGKILL when necessary, and reaps direct children. After closure, reads and writes return `nil`, `"closed"`, and `is_running()` returns `false`. A fully collected status remains available from `wait()`.

The same cleanup is used by `__gc` and `__close`. Recommended use:

```
local pipeline <close> = assert(babet.spawnPipeline({
    { "producer" },
    { "consumer" },
}))
```

GC is a safety net, not a synchronization mechanism. Use `wait()`, `terminate()`, or `close()` explicitly when cleanup timing matters.

## Launch errors and reuse

Validation, pipe, fork, chdir, exec, and `launch_timeout` failures return `nil`, `err`. When a stage fails during launch, every already-created stage is terminated and reaped; Lua never receives a partially valid object. The launcher itself enforces the public 32-stage limit independently of the Lua binding's validation.

After the first `fork()`, and throughout synchronous `pipeline()` capture, Babet retains an emergency owner for descriptors and direct children. If an internal C++ exception occurs, emergency cleanup closes streams, immediately sends SIGKILL only to groups whose leader has not already been reaped, and then performs a bounded reap attempt. `terminate()`'s grace period is not used during stack unwinding. For `spawnPipeline()`, resources are transferred to the userdata only after launch has completed successfully.

Non-destructive `"timeout"` and `"interrupted"` errors allow another call. `"closed"` means the relevant stream or object can no longer perform that operation. Other system errors are returned as text and should be handled as such.

# FS — files, directories, paths, searching, copying, permissions, and checksums

The functions in this chapter are exposed directly on the global `babet` table. There is no `babet.fs` subtable: this flat namespace is historical and is now part of the stable API.

The FS module covers:

- checking whether a path exists and what type it has;
- creating, atomically writing, and removing files and directories;
- changing the current working directory;
- lexical path manipulation;
- listing, iterating, and recursively searching a tree;
- copying and moving directory trees;
- symbolic links;
- Unix permissions, owners, and groups;
- CRC-32, MD5, SHA, and BLAKE2 checksums.

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## General conventions

### Paths

Paths are Lua strings and may be relative or absolute.

Babet does not automatically perform:

- ~ expansion;
- variable expansion such as \$HOME;
- glob expansion such as \*.lua;
- general . and .. normalization in lexical path helpers.

For example:

```
local home = assert(babet.env("HOME"))
local config = babet.joinPath(home, ".config", "my-app")
```

A NUL byte in a path is always rejected. Depending on the function, a bad signature raises a Lua error, while a filesystem failure is generally returned as (nil, err).

### Symbolic links

There is no single symlink rule that applies to every function:

- isFile, isDir, fileSize, getMode, getAttributes, and checksum functions follow a valid link to its target;
- remove removes the link itself, including a dangling link;
- rmdir and rmdirAll reject a symlink root;
- symlinkAttr changes the owner and group of the link itself;
- listFiles, createFileIterator, and find do not recurse into a symlink to a directory;
- copyTree and moveTree reject a symlink source or destination root, but handle links found inside the tree.

Each function documents its exact behavior below.

### Result order

listFiles, createFileIterator, and find preserve the order supplied by the filesystem. That order is not sorted and must not be treated as stable.

Sort explicitly when order matters:

```
local files = assert(babet.listFiles("documents", true))
table.sort(files)
```

### Return values

The most common convention is:

```
local value, err = function_name(...)
if not value then
    print(err)
end
```

A successful action generally returns (true, nil). A few pure functions and iterator methods use a different shape; those exceptions are called out in their sections.

## API overview

### Check and inspect

| Function                               | Successful result                                                     |
|----------------------------------------|-----------------------------------------------------------------------|
| <code>babet.fileExists(path)</code>    | (boolean, nil) — true only for a regular file                         |
| <code>babet.isFile(path)</code>        | (boolean, nil) — same functional predicate as <code>fileExists</code> |
| <code>babet.isDir(path)</code>         | (boolean, nil) — true for a directory                                 |
| <code>babet.fileSize(path)</code>      | (integer, nil) — regular-file size in bytes                           |
| <code>babet.currentDir()</code>        | (string, nil) — current working directory                             |
| <code>babet.getMode(path)</code>       | (integer, nil) — Unix mode 0000..07777                                |
| <code>babet.getAttributes(path)</code> | (table, nil) — {mode, owner, group}                                   |

### Manipulate paths

| Function                                 | Successful result |
|------------------------------------------|-------------------|
| <code>babet.getBasename(path)</code>     | (string, nil)     |
| <code>babet.getFilename(path)</code>     | (string, nil)     |
| <code>babet.getExtension(path)</code>    | (string, nil)     |
| <code>babet.getPath(path)</code>         | (string, nil)     |
| <code>babet.joinPath(a, b, ...)</code>   | string            |
| <code>babet.joinPath({a, b, ...})</code> | string            |

### Act on the filesystem

| Function                                              | Successful result                                                                     |
|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <code>babet.touch(path)</code>                        | (true, nil)                                                                           |
| <code>babet.writeFileAtomic(path, data, opts?)</code> | (true, nil) — atomic and durable binary publication; <a href="#">detailed chapter</a> |
| <code>babet.mkdir(path)</code>                        | (true, nil) — recursive and idempotent                                                |
| <code>babet.remove(path)</code>                       | (true, nil) — regular file or symlink                                                 |
| <code>babet.rmdir(path)</code>                        | (true, nil) — real, empty directory                                                   |
| <code>babet.rmdirAll(path)</code>                     | (true, nil) — real directory, recursive removal                                       |
| <code>babet.rename(source, destination)</code>        | (true, nil)                                                                           |
| <code>babet.link(target, linkpath)</code>             | (true, nil) — creates a symlink                                                       |
| <code>babet.chdir(path)</code>                        | (true, nil)                                                                           |

### List and search

| Function                                                | Successful result |
|---------------------------------------------------------|-------------------|
| <code>babet.listFiles(path, recursive?)</code>          | (table, nil)      |
| <code>babet.createFileIterator(path, recursive?)</code> | (iterator, nil)   |
| <code>babet.find(path, opts?)</code>                    | (table, nil)      |

### Copy and move

| Function                                                             | Successful result                   |
|----------------------------------------------------------------------|-------------------------------------|
| <code>babet.copy(source, destination)</code>                         | <code>(true, nil)</code> — one file |
| <code>babet.copyTree(source, destination, continue_on_error?)</code> | <code>(true, nil)</code> — one tree |
| <code>babet.moveTree(source, destination)</code>                     | <code>(true, nil)</code> — one tree |

## Change attributes

| Function                                                | Successful result        |
|---------------------------------------------------------|--------------------------|
| <code>babet.setMode(path, mode)</code>                  | <code>(true, nil)</code> |
| <code>babet.setAttributes(path, uid, gid, mode?)</code> | <code>(true, nil)</code> |
| <code>babet.symlinkAttr(path, uid, gid)</code>          | <code>(true, nil)</code> |

## Existence, type, and size

### `babet.fileExists(path)`

Checks whether path designates a **regular file**.

```
local exists, err = babet.fileExists("report.txt")
```

Results:

- regular file: `(true, nil)`;
- directory, dangling link, or missing path: `(false, nil)`;
- valid link to a regular file: `(true, nil)`;
- genuine inspection error: `(nil, err)`.

Despite its historical name, `fileExists` is not true for every kind of path. Use `isDir` for a directory.

```
local is_file = assert(babet.fileExists("archive.tar"))
local is_dir = assert(babet.isDir("backups"))
```

### `babet.isFile(path)` and `babet.isfile(path)`

`isFile` is the canonical name. `isfile` is a deprecated compatibility alias.

It tests the same kind of path as `fileExists`: a regular file, following a valid symlink if present.

```
local regular, err = babet.isFile("program.lua")
if err then
    print("Cannot inspect path:", err)
elseif regular then
    print("This is a regular file")
end
```

A missing path returns `(false, nil)`, not an error.

### `babet.isDir(path)` and `babet.isdir(path)`

`isDir` is the canonical name. `isdir` is a deprecated alias.

```
local directory, err = babet.isDir("assets")
```

Results:

- real directory: `(true, nil)`;
- valid link to a directory: `(true, nil)`;
- file, dangling link, or missing path: `(false, nil)`;
- access or inspection error: `(nil, err)`.

### **babet.fileSize(path)**

Returns the size of a regular file in bytes.

```
local size, err = babet.fileSize("video.mp4")
if not size then
    print("Cannot read size:", err)
else
    print(size, "bytes")
end
```

The function follows a valid symlink to a file. Directories and other file types are rejected.

```
local size, err = babet.fileSize("documents")
-- size == nil; err says the path is not a regular file
```

## **Current working directory**

### **babet.currentDir()**

Returns the process current working directory.

```
local cwd, err = babet.currentDir()
assert(cwd, err)
print(cwd)
```

The result is the path supplied by the filesystem and is normally absolute.

### **babet.chdir(path)**

Changes the current working directory of the entire process.

```
local previous = assert(babet.currentDir())
assert(babet.chdir("/tmp"))
print(assert(babet.currentDir()))
assert(babet.chdir(previous))
```

The path is resolved to its canonical form: . and .. components and symlinks are resolved before changing directory.

The working directory is shared by all threads. Consequently, chdir is permanently rejected after the first babet.workers.spawn(), even after that worker has completed.

```
assert(babet.chdir("/srv/my-app")) -- do this before any spawn
local worker = assert(babet.workers.spawn("return true"))

local ok, err = babet.chdir("/tmp")
-- ok == nil; err explains that chdir is forbidden after workers.spawn
```

## **Lexical path manipulation**

These helpers only process strings. They do not check whether the path exists.

### **babet.getBasename(path)**

Returns the last component, including its extension.

```
local name = assert(babet.getBasename("/var/log/app.log"))
print(name) -- app.log
```

A path with no usable last component, such as /, returns (nil, err).

### **babet.getFilename(path)**

Returns the last component without its final extension.

```
print(assert(babet.getFilename("report.tar.gz"))) -- report.tar
print(assert(babet.getFilename("README")))       -- README
```

### **babet.getExtension(path)**

Returns the final extension, including the dot.

```
print(assert(babet.getExtension("report.tar.gz"))) -- .gz
```

A name without an extension returns (nil, err), not an empty string.

```
local ext, err = babet.getExtension("README")
-- ext == nil
```

### **babet.getPath(path)**

Returns the lexical parent directory.

```
print(assert(babet.getPath("/var/log/app.log"))) -- /var/log
```

A simple name with no parent component, such as app.log, returns (nil, err) rather than ".".

### **babet.joinPath(...)**

Two forms are accepted:

```
local path = babet.joinPath("var", "lib", "my-app", "data.db")
local path = babet.joinPath({ "var", "lib", "my-app", "data.db" })
```

At least two non-empty segments are required. On success, joinPath returns **one value**, the joined string. On failure it returns (nil, err).

The function only adjusts the / separator at segment boundaries:

```
print(babet.joinPath("/var/", "/log", "app.log"))
-- /var/log/app.log
```

It does not normalize . or .., and a later absolute-looking segment does not reset the path:

```
print(babet.joinPath("base", "..", "other")) -- base/../other
print(babet.joinPath("base", "/other"))      -- base/other
```

Do not mix the table form with separate arguments. When the first argument is a table, that table is the complete segment list.

## **Create, remove, rename, and link**

### **babet.touch(path)**

Creates an empty file when the path does not exist, or updates the last-write time of an existing path.

```
assert(babet.touch("cache/ready.flag"))
```

The parent directory must already exist; touch does not create it.

```
local ok, err = babet.touch("missing/nested/file.txt")
-- ok == nil if missing/nested does not exist
```

Create parents first when needed:

```
assert(babet.mkdir("cache/images"))
assert(babet.touch("cache/images/index.dat"))
```

For an existing path, the function updates its timestamp; like the Unix touch command, it can therefore also touch a directory. A valid final symlink is followed to its target.

touch never opens an existing path with a truncating mode. The parent and existing target are pinned before the operation, so a file that appears during a concurrent create race is reused without losing its contents. A dangling final symlink is refused: Babet does not create the missing target through a mutable symlink path.

### **babet.mkdir(path)**

Creates a directory with mkdir -p semantics:

- creation is **always recursive**;
- missing parents are created;
- calling it again on an existing directory succeeds;
- a file or another non-directory blocking the path returns (nil, err).

```
assert(babet.mkdir("data/cache/images"))
-- creates data, data/cache, and data/cache/images as needed

assert(babet.mkdir("data/cache/images"))
-- also succeeds when the directory already exists
```

There is **no option to disable recursion**. To require an already-existing parent, check it before calling mkdir:

```
local parent = "data/cache"
assert(babet.isDir(parent), "parent must already exist")
assert(babet.mkdir(babet.joinPath(parent, "images")))
```

New directories use mode 0777 filtered through the process umask. No source directory mode is involved.

### **babet.remove(path)**

Removes only:

- a regular file;
- a symbolic link, valid or dangling.

```
assert(babet.remove("temp.txt"))
-- the link is removed, not its target
assert(babet.link("important-file.txt", "shortcut"))
assert(babet.remove("shortcut"))
```

Directories, FIFOs, sockets, devices, and other special types are rejected.

```
local ok, err = babet.remove("my-directory")
-- ok == nil; use rmdir or rmdirAll
```

### **babet.rmdir(path)**

Removes a **real empty directory**.

```
assert(babet.mkdir("empty-cache"))
assert(babet.rmdir("empty-cache"))
```

It rejects:

- a non-empty directory;

- a file;
- a symlink, even when it points to a directory;
- a missing path.

```
local ok, err = babet.rmdir("non-empty-directory")
```

### **`babet.rmdirAll(path)`**

Recursively removes a **real directory root** and all of its contents.

```
assert(babet.rmdirAll("temporary-build"))
```

The root must exist and be a real directory. A file or a symlink to a directory is rejected.

Symlinks *inside* the removed directory are deleted as directory entries; their external targets are not traversed.

```
local is_dir = babet.isDir("temporary-build")
if is_dir then
    assert(babet.rmdirAll("temporary-build"))
end
```

### **`babet.rename(source, destination)`**

Renames or moves an entry on the same filesystem.

```
assert(babet.rename("old.txt", "new.txt"))
```

Files, directories, and symlinks are accepted, including dangling links. The destination parent directory is not created.

On Linux, the function follows `rename(2)` semantics: an existing destination may be replaced when the entry types are compatible. Crossing filesystems fails; rename does not perform a copy-and-delete fallback.

```
local ok, err = babet.rename("/tmp/a.txt", "/other-mount/a.txt")
-- may fail with cross-device link
```

Use `moveTree` to move a directory tree with a cross-filesystem fallback.

### **`babet.link(target, linkpath)`**

Always creates a **symbolic link**, never a hard link.

The target is stored exactly as supplied:

```
assert(babet.link("../shared/config.toml", "app/config.toml"))
```

A relative target stays relative, and a missing target is allowed:

```
assert(babet.link("future-file.txt", "dangling-link"))
```

Unlike `touch` and `copy`, `link` recursively creates explicit parent directories of `linkpath`:

```
assert(babet.link("../../data.db", "a/b/c/database"))
-- creates a/, a/b/, and a/b/c/ as needed
```

The link path must not already be occupied.

## **List, iterate, and search**

The three APIs do not return paths in the same form:

| Function                                   | Returned path form                                        |
|--------------------------------------------|-----------------------------------------------------------|
| <code>listFiles(root, ...)</code>          | relative to root                                          |
| <code>createFileIterator(root, ...)</code> | as produced from root: prefixed by the supplied root path |
| <code>find(root, ...)</code>               | as produced from root: prefixed by the supplied root path |

### **`babet.listFiles(path, recursive?)`**

Returns an array containing regular files only. Directories are never included.

#### **Non-recursive listing — the default**

```
local files, err = babet.listFiles("assets")
assert(files, err)

for _, relative_path in ipairs(files) do
    print(relative_path)
end
```

Only files directly contained in assets are returned.

#### **Recursive listing**

```
local files, err = babet.listFiles("assets", true)
assert(files, err)

table.sort(files)
for _, relative_path in ipairs(files) do
    print(relative_path)
end
```

Paths are relative to the requested root:

```
logo.png
icons/open.png
icons/close.png
```

not:

```
assets/logo.png
assets/icons/open.png
```

#### **Symlinks**

- a valid symlink to a regular file is included under the link name;
- a dangling link is skipped;
- a symlink to a directory is neither included as a file nor traversed;
- directory-symlink loops therefore cannot cause infinite recursion.

```
local files = assert(babet.listFiles("collection", true))
-- collection/external -> /other/directory is not followed
```

The second argument is optional and defaults to false. When present, it must be a strict Lua boolean; numbers and strings are not coerced. The function accepts exactly one or two arguments.

### **babet.createFileIterator(path, recursive?)**

Creates a userdata with two methods:

- `iterator:next()` — next path, or nil at end;
- `iterator:close()` — explicitly releases the iterator.

#### **Non-recursive**

```
local iterator, err = babet.createFileIterator("assets")
assert(iterator, err)
```

```
while true do
    local path = iterator:next()
    if path == nil then
        break
    end
    print(path) -- assets/logo.png, etc.
end

iterator:close()
```

#### **Recursive**

```
local iterator = assert(babet.createFileIterator("assets", true))
while true do
    local path = iterator:next()
    if not path then break end
    print(path)
end
iterator:close()
```

Despite its name, the current implementation builds the complete list during `createFileIterator`. Access and traversal failures are therefore returned at creation time, before the first `next()` call.

Symlink behavior:

- valid link to a regular file: included under the link path;
- dangling link, loop, or inaccessible target: skipped;
- link to a directory: not traversed;
- genuine entry-inspection or traversal-increment error: creation fails with `(nil, err)`.

Calling `next()` after `close()` raises a Lua error. Omitting `close()` is not fatal: the garbage collector eventually releases the object. The optional recursive argument must be a strict Lua boolean, and the constructor accepts exactly one or two arguments.

### **babet.find(path, opts?)**

Recursively searches the entries of a directory and returns an array of paths.

`path` must be a strict Lua string. `opts` must be a table or nil, and the function accepts exactly one or two arguments. Option fields are also strict: depths are Lua integers, and every `name/path/regex/glob` field is a Lua string.

These calls are equivalent:

```
local entries = assert(babet.find("src"))
local entries = assert(babet.find("src", nil))
```

Without options, both files **and** directories are included. The root itself is not included.

## Options

| Option     | Type    | Default               | Effect                                                             |
|------------|---------|-----------------------|--------------------------------------------------------------------|
| type       | string  | no filter             | "f" for files, "d" for directories                                 |
| glob       | string  | absent                | bounded glob on the complete basename, case-sensitive              |
| iglob      | string  | absent                | bounded glob on the complete basename, ASCII case-insensitive      |
| path_glob  | string  | absent                | bounded glob on the complete returned path, case-sensitive         |
| path_iglob | string  | absent                | bounded glob on the complete returned path, ASCII case-insensitive |
| name       | string  | absent                | RE2 regex on the complete basename, case-sensitive                 |
| iname      | string  | absent                | RE2 regex on the complete basename, case-insensitive               |
| path       | string  | absent                | RE2 regex searched within the full path                            |
| mindepth   | integer | 0                     | minimum included depth                                             |
| maxdepth   | integer | practically unlimited | maximum included depth                                             |

All supplied options are combined with logical **AND**. An empty pattern acts like an absent filter, matching the historical regex behavior.

**Safe glob filters** Use the glob fields for ordinary filename filtering, especially when a pattern may come from an untrusted user:

```
local lua_files = assert(babet.find("src", {
  type = "f",
  glob = "*.lua",
}))
```

```
local images = assert(babet.find("assets", {
  type = "f",
  iglob = "*.jpg",
}))
```

The glob language is deliberately small:

- \*: zero or more bytes except /;
- \*\*: zero or more bytes, including /;
- ?: exactly one byte except /;
- \x: literal byte x.

Matching is anchored to the complete basename or path. Brackets, braces, and shell extglob syntax have no special meaning and are matched literally. Each pattern is limited to 4096 bytes. The

matcher uses no recursive backtracking; its work is bounded polynomially by the pattern and candidate lengths.

`path_glob` and `path_iglob` match the complete returned path with `/` as the separator. Use `**` when a wildcard must cross directories:

```
local tests = assert(babet.find(".", {
    type = "f",
    path_glob = "**/tests/**/*lua",
}))
```

The insensitive variants fold ASCII letters only. UTF-8 bytes outside ASCII are compared exactly, which keeps behavior independent of the process locale.

**Bounded RE2 regular expressions** The historical `name`, `iname`, and `path` fields now use the statically linked RE2 engine. `name` and `iname` match the complete basename:

```
local images = assert(babet.find("assets", {
    type = "f",
    iname = ".*\\.(png|jpg|jpeg)$",
}))
```

`path` performs a partial regex search within the complete path:

```
local tests = assert(babet.find(".", {
    type = "f",
    path = "/tests/",
}))
```

Each regex is limited to 4096 bytes and compiled with a 1 MiB RE2 memory budget. Matching is linear-time; when the DFA cache reaches its budget, RE2 falls back to its bounded NFA instead of backtracking exponentially. Babet uses RE2's Latin-1 mode so every Linux pathname byte remains matchable, including names that are not valid UTF-8. `iname` disables case sensitivity through RE2; no process locale is consulted.

RE2 deliberately rejects constructs that require backtracking or non-regular state, notably back-references and look-around assertions. Such a pattern makes the whole call fail with `(nil, err)`. Use the safe glob fields for simple filename filters and RE2 when grouping, alternation, character classes, or bounded repetitions are genuinely useful.

**Limit depth** Depth 0 is the root's immediate children:

```
local direct_children = assert(babet.find("src", {
    maxdepth = 0,
}))
```

Children of a subdirectory are at depth 1:

```
local one_level_below = assert(babet.find("src", {
    mindepth = 1,
    maxdepth = 1,
}))
```

`mindepth` filters results only; shallower directories are still traversed to reach requested depths.

**Symlinks and errors** Traversal does not recurse into directory symlinks. Type checks do follow a valid target, however: a symlink to a file may match `type = "f"`, and a symlink to a directory may appear with `type = "d"` without its contents being traversed.

An invalid or oversized regex, invalid or oversized glob, or traversal error fails the entire call with `(nil, err)`. There is no "continue after errors" mode for `find`.

## Copy and move

### **babet.copy(source, destination)**

Copies one file using `std::filesystem::copy_file` and overwrites an existing file destination.

```
assert(babet.copy("config/default.toml", "config/local.toml"))
```

The destination parent directory must already exist:

```
assert(babet.mkdir("backup/config"))
assert(babet.copy("config/app.toml", "backup/config/app.toml"))
```

Important behavior:

- a valid source symlink is followed and produces a new regular file;
- a valid destination symlink is followed: the link target is overwritten;
- source and destination directories are rejected;
- ordinary source permissions are copied according to filesystem semantics; owner, group, and timestamps are not guaranteed;
- this operation does not use `copyTree`'s hardened destination confinement.

Use `copyTree` for a tree or when destination symlink redirection must be prevented.

### **babet.copyTree(source, destination, continue\_on\_error?)**

Recursively copies a source directory to a destination directory.

```
assert(babet.copyTree("site", "backup/site"))
```

**Default value of `continue_on_error`** The optional third argument defaults to **true**. When present it must be a strict Lua boolean; the function accepts exactly two or three arguments.

For an error limited to one entry:

1. a warning is written to `stderr`;
2. copying continues with other entries when possible;
3. the final call returns `(nil, "completed with warnings ...")`.

A partial success is therefore never reported as a full success.

```
local ok, err = babet.copyTree("source", "destination")
if not ok then
    print(err) -- may report warnings after a partial copy
end
```

**Strict mode** Pass `false` explicitly to stop at the first error:

```
local ok, err = babet.copyTree("source", "destination", false)
assert(ok, err)
```

Pass `true` explicitly when you want the continuation choice to be visible:

```
local ok, err = babet.copyTree("source", "destination", true)
```

## Destination and merging

- the destination is recursively created when absent;
- existing destinations are merged;
- existing regular files are replaced;
- a pre-existing destination symlink, either as an intermediate component or final file, is rejected;
- the destination cannot be inside the source;
- source and destination roots must not be symlinks.

**Metadata** When a new file is created:

- ordinary source rwx bits are preserved;
- setuid, setgid, and sticky bits are removed;
- the new inode's owner and group are determined by the system for the caller;
- timestamps are not preserved.

Created directories use 0777 filtered through the umask. Source directory mode, owner, group, and timestamps are not reproduced.

**Internal symlinks** Links inside the source tree are recreated:

- relative targets stay unchanged;
- dangling links stay dangling;
- an absolute target inside the source is rewritten to its corresponding destination path;
- an absolute external target stays unchanged.

Directory symlinks are never traversed as directories.

**babet.moveTree(source, destination)**

Recursively moves a directory tree.

```
assert(babet.moveTree("staging/site", "public/site"))
```

**Fast path** When the destination does not exist, the move remains on one filesystem, and no internal absolute link needs rewriting, Babet renames the whole tree. This is fast and preserves inodes and their metadata.

**Merge or cross-filesystem fallback** When the destination exists, the move crosses a filesystem, or an internal link needs rewriting, Babet uses an entry-by-entry fallback:

- scan the complete source before the first modification;
- create destination directories;
- create destination symlinks;
- move other entries;
- remove the residual source tree.

In this fallback, a file copied to a new inode preserves ordinary rwx bits but not special bits, owner, group, or timestamps. Created directories follow the umask.

## Safety checks

- source must be a real directory, not a symlink;
- destination root must not be a symlink;
- destination cannot be inside source;
- a pre-existing symlink in a merge destination is rejected;
- internal links use the same rewriting rules as copyTree.

moveTree reduces partial-state risk by scanning first and preparing symlinks before moving files. It is not a general transaction, however: an error after several file moves may leave some entries in the source and others in the destination. Such an error requires checking both trees manually.

## Unix permissions and attributes

Modes are integers. Lua has no 0o755 literal; use an octal string where the API accepts it, or `tonumber("755", 8)`.

```
local mode_755 = tonumber("755", 8)
```

### **babet.getMode(path) and babet.setMode(path, mode)**

getMode returns ordinary permissions and special bits in the range 0000..07777.

```
local mode, err = babet.getMode("run.sh")
assert(mode, err)
print(string.format("%04o", mode))
```

setMode accepts two forms:

```
assert(babet.setMode("run.sh", "755"))
assert(babet.setMode("run.sh", tonumber("755", 8)))
```

A string is interpreted in base 8. A number is taken as the direct integer value. Non-integers, negative values, and values above 07777 are rejected.

Special bits may be requested explicitly:

```
assert(babet.setMode("tool", "4755"))
local mode = assert(babet.getMode("tool"))
assert(mode == tonumber("4755", 8))
```

Both functions follow a symlink to its target.

### **babet.getAttributes(path)**

Returns exactly these three fields:

```
local attrs, err = babet.getAttributes("file.txt")
assert(attrs, err)
```

```
print(attrs.mode) -- integer 0000..07777
print(attrs.owner) -- UID
print(attrs.group) -- GID
```

The table contains no size, mtime, or file type.

The function follows a symlink to its target.

### **babet.setAttributes(path, uid, gid, mode?)**

Changes owner and group, then optionally the mode.

Without a mode:

```
local attrs = assert(babet.getAttributes("file.txt"))
assert(babet.setAttributes(
    "file.txt",
    attrs.owner,
    attrs.group
))
```

With a mode:

```
assert(babet.setAttributes(
    "file.txt",
    1000,
    1000,
    tonumber("640", 8)
))
```

Unlike setMode, the fourth argument must be an **integer**; a string such as "640" is not accepted.

Validation before mutation:

- UID and GID must be non-negative integers in the `uid_t` and `gid_t` range;
- mode must be between 0 and 07777.

POSIX provides no atomic operation combining `chown` and `chmod`. Babet resolves the path once, pins that target, and performs the initial metadata read, `chown`, optional `chmod`, and any rollback on the same inode. Replacing the pathname during the call therefore cannot redirect the later phase to a new target. If `chmod` fails after a successful `chown`, Babet attempts to restore the original owner, group, and mode; an incomplete rollback is reported in the error.

The call follows a final symlink, but the target selected when the operation starts remains pinned even if the link is changed concurrently. Required privileges depend on the system and process identity; changing owner normally requires root privileges.

### **`babet.symlinkAttr(path, uid, gid)` and `babet.symlinkattr(...)`**

`symlinkAttr` is the canonical name. `symlinkattr` is a deprecated alias.

The function uses `lchown` and therefore does not follow a symbolic link:

```
assert(babet.symlinkAttr("shortcut", 1000, 1000))
```

It changes the UID and GID of the link itself, not of its target. It does not change mode.

UID and GID are validated as in `setAttributes`.

## **File checksums**

All functions below stream the file and return lowercase hexadecimal text:

| Function                               | Algorithm   | Hex length | Recommended use                    |
|----------------------------------------|-------------|------------|------------------------------------|
| <code>babet.crc32sum(path)</code>      | CRC-32 IEEE | 8          | accidental-error detection only    |
| <code>babet.md5sum(path)</code>        | MD5         | 32         | legacy compatibility, not security |
| <code>babet.sha1sum(path)</code>       | SHA-1       | 40         | legacy compatibility, not security |
| <code>babet.sha256sum(path)</code>     | SHA-256     | 64         | common cryptographic integrity     |
| <code>babet.sha384sum(path)</code>     | SHA-384     | 96         | cryptographic integrity            |
| <code>babet.sha512sum(path)</code>     | SHA-512     | 128        | cryptographic integrity            |
| <code>babet.sha3_256sum(path)</code>   | SHA3-256    | 64         | cryptographic integrity            |
| <code>babet.sha3_384sum(path)</code>   | SHA3-384    | 96         | cryptographic integrity            |
| <code>babet.sha3_512sum(path)</code>   | SHA3-512    | 128        | cryptographic integrity            |
| <code>babet.blake2b512sum(path)</code> | BLAKE2b-512 | 128        | cryptographic integrity            |
| <code>babet.blake2s256sum(path)</code> | BLAKE2s-256 | 64         | cryptographic integrity            |

Example:

```
local digest, err = babet.sha256sum("release.tar.gz")
assert(digest, err)
print(digest)
```

Check against an expected value:

```
local expected = "..."  
local actual = assert(babet.sha256sum("release.tar.gz"))  
assert(actual == expected, "checksum mismatch")
```

Common contract:

- regular files only;
- a valid symlink to a regular file is followed;
- directories, FIFOs, sockets, devices, and non-regular pseudo-files are rejected;
- a read error returns (nil, err) and never a misleading digest of partial data;
- the complete content is not loaded into memory.

CRC-32, MD5, and SHA-1 must not authenticate data against an attacker. Prefer SHA-256, SHA-3, or BLAKE2 for new integrity checks.

## In-memory CRC-32

### `babet.crc32(data)`

Computes CRC-32 for a Lua string already in memory. Lua strings are binary-safe, so embedded NUL bytes are accepted.

```
print(babet.crc32("abc")) -- 352441c2  
print(babet.crc32("")) -- 00000000
```

```
local bytes = "a\0b\0c"  
local digest = babet.crc32(bytes)
```

On success this function returns one string, without a second nil value. A bad argument count or type raises a Lua error.

## Error contract

### Raised Lua errors

A bad signature or incompatible type generally raises a Lua error:

```
babet.fileSize({}) -- Lua error  
babet.mkdir("a", true) -- Lua error: mkdir accepts exactly one argument
```

Paths containing a NUL byte are rejected before any system call.

### Returned errors

Filesystem failures are normally returned:

```
local ok, err = babet.remove("missing.txt")  
if not ok then  
    print(err)  
end
```

The `fileExists`, `isFile`, and `isDir` predicates treat a simply missing path as the normal result (false, nil). A genuine inspection failure remains (nil, err).

### Partial operations

- `copyTree` in continuation mode may copy some entries and then return a summary error reporting warnings;
- `moveTree` may leave state split between source and destination when a late error occurs in its entry-by-entry fallback;

- `setAttributes` attempts rollback if `chmod` fails after `chown`, and explicitly reports an incomplete rollback.

## Design decisions and limitations

- **Historical flat namespace:** `babet.fileExists`, not `babet.fs.fileExists`.
- **Canonical names:** `isFile`, `isDir`, and `symlinkAttr`. The `isfile`, `isdir`, and `symlinkattr` forms are deprecated.
- **`mkdir` is always recursive:** no non-recursive API variant exists.
- **copy for one file, copyTree for a tree:** no implicit overload based on source type.
- **Glob is a find filter, not a standalone function:** use `glob`, `iglob`, `path_glob`, or `path_iglob` in `babet.find`.
- **No automatic sorting:** call `table.sort` when stable ordering matters.
- **No complete metadata preservation during copy:** regular file permission bits are handled, but owners, groups, timestamps, and directory modes are not preserved.
- **No asynchronous I/O:** use [workers](#) to parallelize independent work.

## writeFileAtomic — atomic and durable file writing

`babet.writeFileAtomic()` publishes a binary Lua string to a regular file without exposing partially written content under the final name. Data is first written to a private same-directory temporary file, final permissions are applied, then the temporary file is published atomically.

It is intended for configuration, tokens, JSON state, small binary files, and screenshots already held in memory. It does not replace streaming APIs such as `babet.http.download()` for very large payloads.

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## API

```
local ok, err = babet.writeFileAtomic(path, data, opts?)
```

| Argument          | Type         | Purpose                                                 |
|-------------------|--------------|---------------------------------------------------------|
| <code>path</code> | string       | relative or absolute destination, with no NUL byte      |
| <code>data</code> | string       | complete binary content; embedded NUL bytes are allowed |
| <code>opts</code> | table or nil | strict options                                          |

Results:

- success: true, nil;
- filesystem or publication failure: nil, err;
- bad arity, bad type, or unknown option: raised Lua error.

## Defaults

```
{  
  overwrite = false,  
  permissions = tonumber("644", 8),  
  durable = true,  
}
```

| Option      | Default | Effect                                               |
|-------------|---------|------------------------------------------------------|
| overwrite   | false   | rejects an existing destination                      |
| permissions | 0644    | exact permissions for the new final file             |
| durable     | true    | synchronizes the temporary file and parent directory |

Unknown and non-string keys are rejected. Booleans are not coerced from 0, 1, or strings. permissions must be an integer from 0 through 0777.

## Basic write

```
local ok, err = babet.writeFileAtomic("state.json", [{["ready":true}]))  
assert(ok, err)
```

The destination becomes visible only after the temporary file is complete. If state.json already exists, the call fails and preserves its contents.

Parent directories are not created automatically:

```
assert(babet.mkdir("runtime"))  
assert(babet.writeFileAtomic("runtime/state.json", "{}"))
```

## Explicit replacement

```
local ok, err = babet.writeFileAtomic("state.json", new_state, {  
  overwrite = true,  
})  
assert(ok, err)
```

With overwrite = true:

- an absent destination is created;
- an existing regular file is replaced by a same-directory rename;
- a symbolic link, directory, FIFO, socket, or device is rejected;
- the old file remains intact after any failure before publication.

Without overwrite, Babet uses a publication operation that atomically fails if another task wins the race and creates the destination first.

## Permissions

```
local ok, err = babet.writeFileAtomic("token.bin", token, {
    permissions = tonumber("600", 8),
})
assert(ok, err)
```

The temporary file is always created with private mode 0600, or more restrictive because of the process umask. Before publication Babet applies the requested permissions exactly using `fchmod()`. Setuid, setgid, and sticky bits cannot be requested: the accepted range is 0000..0777.

On replacement, the new file gets the requested permissions rather than those of the previous file.

## Durability

With default `durable = true`, the sequence is:

1. create a unique temporary file in the destination directory;
2. write all bytes while handling EINTR and partial writes;
3. apply final permissions;
4. `fsync()` the temporary file;
5. publish atomically;
6. `fsync()` the parent directory.

This aims to preserve the update after a crash, subject to filesystem and hardware guarantees.

For a rebuildable cache where speed matters more:

```
local ok, err = babet.writeFileAtomic("cache/index.bin", cache, {
    overwrite = true,
    durable = false,
})
assert(ok, err)
```

`durable = false` retains atomic publication but does not confirm persistence after a crash or power loss.

If publication succeeds and parent-directory synchronization then fails, the call returns an explicit error stating that the destination has already been published atomically but its persistence could not be confirmed.

## Binary data

data is a binary Lua string. No UTF-8 validation is performed:

```
local payload = "\0\1\2PNG\255\254"
assert(babet.writeFileAtomic("capture.bin", payload))
```

An empty string creates a zero-length regular file.

All data must already be in memory. Prefer a streaming API for downloads or transformations measured in gigabytes.

## Paths and symbolic links

Paths may be relative or absolute. They are interpreted literally: there is no `~`, `$HOME`, shell, or glob expansion.

To confine the write:

- every parent component is opened with `openat()` and `O_NOFOLLOW`;
- symlink parent components are rejected;
- `..` is rejected;

- a final symbolic link is rejected;
- missing parent directories are not created;
- the destination must be absent or a regular file when `overwrite = true`.

```
local ok, err = babet.writeFileAtomic("link/config.json", "{}", {
    overwrite = true,
})
assert(not ok)
print(err)
```

A process that can itself write the parent directory remains inside the same trust boundary: POSIX has no portable conditional rename tied to one exact inode. Babet never follows the final link and reinspects the destination immediately before publication, but directory permissions remain the security boundary against a hostile concurrent actor.

## Atomicity and concurrency

Readers never observe a mixture of old and new data: they see one complete file or the other.

With `overwrite = false`, several producers may attempt the same creation:

```
local ok, err = babet.writeFileAtomic("once.dat", value)
if not ok and err:find("already exists", 1, true) then
    -- Another producer won.
end
```

Exactly one call publishes the destination. Others fail without replacing it.

With `overwrite = true`, the last successful rename determines the final contents. Ordering between concurrent writers is unspecified.

## Using it in a worker

The function is available in every worker Lua state:

```
local job = assert(babet.workers.spawn([
    local ok, err = babet.writeFileAtomic(worker.args.path, worker.args.data, {
        overwrite = true,
        permissions = tonumber("600", 8),
    })
    assert(ok, err)
    return true
]), {
    path = "runtime/result.bin",
    data = "worker-result",
}))

assert(job:join())
```

Values carried in `worker.args` remain subject to the workers JSON serializer. A binary string containing NUL cannot therefore be passed directly through `worker.args`; generate it in the worker, retain it as Base64, or read it from another source.

## Combined Selenium example

Decode a WebDriver screenshot and publish it without a partial final file:

```
local png, err = babet.base64.decode(response.value, {
    max_output = 32 * 1024 * 1024,
})
```

```

assert(png, err)

assert(babet.mkdir("screenshots"))
local ok
ok, err = babet.writeFileAtomic("screenshots/latest.png", png, {
    overwrite = true,
    permissions = tonumber("640", 8),
    durable = true,
})
assert(ok, err)

```

Readers of `latest.png` see either the complete previous screenshot or the complete new one, never a partially decoded or written image.

## Error contract

Raised Lua errors:

```

babet.writeFileAtomic()           -- arity
babet.writeFileAtomic(42, "data") -- path is not a string
babet.writeFileAtomic("x", {})    -- data is not a string
babet.writeFileAtomic("x", "data", 42) -- opts is not a table
babet.writeFileAtomic("x", "data", {
    permissions = "600",
})
babet.writeFileAtomic("x", "data", {
    unknown = true,
})

```

Failures returned as `nil`, `err` include:

- absent, inaccessible, symlink, or non-directory parent;
- existing destination without `overwrite`;
- final symlink or non-regular destination;
- temporary creation, write, `fchmod`, `fsync`, close, or publication failure;
- inability to allocate a unique temporary name.

After a failure before publication, the old destination remains intact and the temporary file is removed on a best-effort basis.

## Limitations

The first version does not provide:

- automatic parent-directory creation;
- append mode;
- input from an `io.file` object or numeric descriptor;
- a streaming API;
- application-level locking between writers;
- automatic rotation or size limits;
- atomic publication of several files as one transaction.

For multiple related files, publish a new version directory and switch an application pointer, or use SQLite when a transaction is required.

# HTTP — synchronous web requests, headers, bodies, TLS, and limits

`babel.http` is a synchronous HTTP/HTTPS client built on `cpp-httplib` and `OpenSSL`. It is intended for APIs, webhooks, simple REST calls, bounded in-memory responses, and direct-to-file downloads.

It supports:

- HTTP and HTTPS;
- GET, HEAD, OPTIONS, POST, PUT, PATCH, and DELETE;
- encoded and normalised query parameters;
- request headers;
- text or binary request bodies;
- binary responses;
- repeated response headers;
- optional redirects;
- TLS verification by default;
- connection and global request timeouts;
- a maximum in-memory response size;
- streamed GET downloads committed atomically to a destination file.

It does not expose streaming uploads, response callbacks, persistent sessions, a cookie jar, multipart/form-data, form helpers, proxies, an HTTP server, WebSockets, HTTP/2, or HTTP/3.

## Module contents

- [Core conventions](#)
- [API overview](#)
- [request\(opts\)](#)
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## Core conventions

### Synchronous client

The call blocks the current thread until response or error. Use independent [workers](#) for parallel requests.

### HTTP status is not a transport error

A received 404 or 500 is a normal response table:

```

local response, err = babet.http.get(url, { timeout = 10 })
assert(response, err)

if response.status >= 200 and response.status < 300 then
    print(response.body)
else
    io.stderr:write("HTTP ", response.status, "\n", response.body)
end

```

DNS, TCP, TLS, timeout, and body-limit failures return (nil, err).

## Binary bodies

Request bodies and response.body may contain NUL bytes:

```

local response = assert(babet.http.post(url, "AB\0CD", {
    timeout = 10,
}))

```

URLs, methods, header names, CA paths, and other C-facing strings reject NUL. URLs and headers also reject CR/LF to prevent HTTP-line injection.

## Unknown option fields

Unknown fields are currently ignored. A typo such as timeot = 5 does not activate a timeout and does not fail. Use the documented names exactly.

## API overview

```

local response, err = babet.http.request(opts)
local response, err = babet.http.get(url, opts?)
local result, err = babet.http.download(url, destination, opts?)
local response, err = babet.http.post(url)
local response, err = babet.http.post(url, body)
local response, err = babet.http.post(url, opts)
local response, err = babet.http.post(url, body, opts)
local response, err = babet.http.post(url, nil, opts)

```

| Field            | Type              | Default             | Meaning                             |
|------------------|-------------------|---------------------|-------------------------------------|
| url              | string            | required by request | absolute http:// or https:// URL    |
| method           | string            | "GET"               | supported method, case-insensitive  |
| headers          | table             | {}                  | string names; string/number values  |
| body             | binary string     | absent              | body for body-capable methods       |
| query            | table             | absent              | string keys; string/number values   |
| timeout          | finite number > 0 | library defaults    | seconds, connect plus global budget |
| verify           | strict boolean    | true                | HTTPS certificate verification      |
| ca_cert          | string            | OpenSSL trust store | CA file path                        |
| follow_redirects | strict boolean    | false               | follow redirects automatically      |

| Field         | Type             | Default | Meaning                            |
|---------------|------------------|---------|------------------------------------|
| max_body_size | integer 1..2 GiB | 64 MiB  | in-memory cap for request/get/post |
| max_file_size | positive integer | 8 GiB   | received-file cap for download     |

get and post always overwrite url and method after shallow-copying options.

### **babet.http.request(opts)**

General form for every method:

```
local response, err = babet.http.request({
    url = "https://api.example.com/items/42",
    method = "DELETE",
    headers = {
        ["Authorization"] = "Bearer " .. token,
    },
    timeout = 10,
})
```

The first argument must be a table. opts.url must be a strict string. Invalid option fields return (nil, err); a wrong first-argument type raises.

### **babet.http.get(url, opts?)**

The wrapper shallow-copies opts, then forces:

```
opts.url = url
opts.method = "GET"

local response, err = babet.http.get(
    "https://api.example.com/health",
    { timeout = 5 }
)
```

The second argument must be absent, nil, or a table. GET rejects opts.body instead of silently dropping it.

### **babet.http.download(url, destination, opts?)**

Downloads a GET response directly to a file without accumulating the body in Lua or in a C++ response string:

```
local result, err = babet.http.download(
    "https://example.com/archive.tar.gz",
    "downloads/archive.tar.gz",
    {
        timeout = 120,
        follow_redirects = true,
        max_file_size = 4 * 1024 * 1024 * 1024,
    }
)
assert(result, err)
assert(result.saved, "HTTP " .. result.status)
print(result.path, result.bytes)
```

The call is synchronous, but received chunks are written progressively to a same-directory temporary file. The final destination is replaced with one renameat only after the complete final response has been received and the temporary file has been synced and closed.

The wrapper always forces GET. It accepts the common headers, query, timeout, verify, ca\_cert, and follow\_redirects options. body is rejected. max\_file\_size replaces max\_body\_size for this call:

- strict positive Lua integer;
- default: 8 GiB;
- configurable up to math.maxinteger;
- enforced on the actual bytes delivered to the file receiver.

Only a final 2xx response is committed. A received 1xx, unfollowed 3xx, 4xx, or 5xx response is not a transport failure: download returns a result table with saved = false, removes the temporary file, and leaves any existing destination unchanged.

The destination contract is intentionally strict:

- the parent directory must already exist;
- Babet does not create parent directories;
- ... path components are rejected;
- parent-directory symlink components are rejected;
- a symlink at the final destination is replaced as an inode; its target is never followed or modified;
- a temporary file is created in the same directory with O\_EXCL and is removed on every failure;
- a pre-existing destination remains untouched on DNS, TCP, TLS, timeout, size-limit, HTTP non-2xx, or disk-write failure.

The atomic rename prevents readers from observing a partially downloaded final path. It does not provide resume support, progress callbacks, crash-proof parent-directory durability, or preservation of metadata from an older file. The new file is created with mode 0666 filtered by the process umask.

## **babet.http.post call forms**

No explicit body:

```
babet.http.post(url)
babet.http.post(url, { timeout = 5 })
babet.http.post(url, nil, { timeout = 5 })
```

Positional body:

```
babet.http.post(url, "payload", { timeout = 5 })
```

Body in options:

```
babet.http.post(url, {
    body = "payload",
    timeout = 5,
})
```

A table in the second position is interpreted as opts. Any other non-string, non-nil second value raises. When both a positional body and opts.body exist, the positional body wins.

## **URLs, fragments, and query**

Only absolute HTTP/HTTPS URLs are accepted. A non-empty host is required; bracketed IPv6 literals are supported:

```
local r, err = babet.http.get("http://[::1]:8080/", { timeout = 2 })
```

Fragments are removed and never sent.

Add query parameters with a table:

```
local response = assert(babet.http.get("https://example.com/search", {
    query = {
        q = "lua socket",
        page = 2,
    },
    timeout = 10,
}))
```

Keys must be strings; values may be strings or numbers. Babet protects query delimiters and non-ASCII bytes first, then cpp-http lib 0.45.0 normalises the query before it is sent. The effective wire format therefore follows these conventions:

- spaces become +;
- + becomes %2B;
- / and ? remain literal inside a value;
- non-ASCII UTF-8 bytes are percent-encoded as %HH;
- &, =, and % are percent-encoded when they belong to a key or value.

For example, q = "lua socket" becomes q=lua+socket, while path = "a/b" remains path=a/b.

When the URL already contains ?a=b, new parameters are appended with &. The existing query is normalised as well: %20 may become + and %2F may become /. The #... fragment is removed before transmission.

Lua table iteration order is not stable, so do not build canonical signatures by assuming the order produced from query. For canonical signing, construct an already ordered URL yourself and account for the normalisation described above.

NUL bytes in query keys/values are sent as %00; make sure the server accepts that semantic.

## HTTP methods

Supported methods:

- GET;
- HEAD;
- OPTIONS;
- POST;
- PUT;
- PATCH;
- DELETE.

Method names are uppercased internally:

```
local response = assert(babet.http.request({
    url = "https://api.example.com/items/42",
    method = "patch",
    body = '{"enabled":true}',
    headers = { ["Content-Type"] = "application/json" },
    timeout = 10,
}))
```

GET, HEAD, and OPTIONS reject a body. POST, PUT, PATCH, and DELETE accept an absent, empty, or non-empty body.

There are no put, patch, or delete shortcuts; use request.

## Request headers

```
headers = {  
  ["Accept"] = "application/json",  
  ["Authorization"] = "Bearer " .. token,  
  ["X-Retry"] = 3,  
}
```

Header names must be non-empty HTTP token strings: alphanumeric plus `!#$%&'*+-.^_`|~`. Spaces, tabs, colon, CR/LF, NUL, and non-string keys are rejected.

Values may be strings or numbers. Numbers are converted using Lua's textual conversion. NUL, CR/LF, booleans, tables, and other types are rejected.

A Lua table cannot represent repeated exact request-header names. The binding does not expose a multi-value request-header form. Combine values only when the specific header grammar permits it.

Names are sent with the provided case; HTTP matching is case-insensitive. Content-Type detection is case-insensitive.

## Request body and Content-Type

Absent and explicitly empty bodies differ:

```
babet.http.post(url, { timeout = 5 }) -- no body field  
babet.http.post(url, "", { timeout = 5 }) -- present empty body
```

When a body exists on POST/PUT/PATCH/DELETE and no Content-Type is supplied, Babet uses `application/octet-stream`. No default Content-Type is added when the body is absent.

JSON example:

```
local payload = assert(babet.json.encode({ name = "babet" }))  
local response = assert(babet.http.post(url, payload, {  
  headers = {  
    ["Content-Type"] = "application/json",  
    ["Accept"] = "application/json",  
  },  
  timeout = 10,  
}))
```

There is no form helper. Encode URL-encoded form data yourself and set `application/x-www-form-urlencoded`. The query table changes the URL only; it does not build a form body.

No multipart builder or streaming upload is exposed. For large/complex uploads, use a specialised tool through `babet.exec` without unsafe shell concatenation.

## opts.timeout

```
local response, err = babet.http.get(url, { timeout = 10 })
```

The value must be finite and strictly positive. Positive values below 1 ms are rounded up. Babet configures both a connection timeout and `cpp-httpplib`'s global maximum request timeout; the first limit reached wins.

Synchronous DNS may still happen outside the controllable library budget and can make wall-clock duration exceed the requested timeout.

Without an explicit timeout, `cpp-httpplib`/system defaults apply. Always set one for remote services.

## HTTPS, verification, and CA

Verification defaults to true:

```
local response = assert(babet.http.get("https://example.com/", {
    timeout = 10,
}))
```

Use a private CA file:

```
local response = assert(babet.http.get("https://internal.example/", {
    ca_cert = "/etc/myapp/internal-ca.pem",
    timeout = 10,
}))
```

HTTP exposes `ca_cert`, not `ca_path`, and does not run the socket TLS module's manual distro-path probing. It relies on the embedded OpenSSL defaults, environment, and the supplied CA file.

```
local response = assert(babet.http.get("https://127.0.0.1:8443/", {
    verify = false,
    timeout = 3,
}))
```

Use `verify = false` only in controlled tests. HTTP has no separate hostname override; the URL host is used.

## follow\_redirects

Redirects are visible by default:

```
local r = assert(babet.http.get(url, { timeout = 10 }))
if r.status == 302 then
    print(r.headers.location)
end
```

Enable automatic following:

```
local r = assert(babet.http.get(url, {
    follow_redirects = true,
    timeout = 10,
}))
```

The result describes the final response. Redirect history and `max_redirects` are not exposed; cpp-httplib's internal limit applies.

When a redirect changes scheme, host, or port, cpp-httplib 0.45 removes `Host`, `Authorization`, and `Proxy-Authorization` before issuing the next request. A same-origin redirect keeps `Authorization`. Other script-provided headers, notably `Cookie` or custom secrets, are not covered by a general stripping rule, so do not follow an untrusted destination with such headers.

For untrusted URLs, also consider HTTPS-to-HTTP downgrade, internal or cloud metadata addresses, and the final destinations explicitly allowed by the application. The local regression suite verifies that `Authorization` is not forwarded across origins.

## max\_body\_size

The entire response is buffered in memory:

```
local response, err = babet.http.get(url, {
    timeout = 30,
    max_body_size = 8 * 1024 * 1024,
})
```

- default: 64 MiB;
- minimum: 1 byte;
- maximum: 2 GiB;
- strict positive Lua integer.

When actual received chunks would exceed the cap, Babet aborts and returns:

```
nil, "http: response body exceeds max_body_size"
```

No partial body is exposed. The guard applies to actual received bytes, not only Content-Length, including chunked or misleading responses.

Babet supports the three HTTP/1.1 response-body framings relevant here: Content-Length, Transfer-Encoding: chunked, and a body terminated by connection close. Babet 2.9.1 specifically fixes the latter two with `cpp-httplib 0.45.0`; limits still apply to bytes actually delivered to Babet's receiver.

For large GET payloads, use `babet.http.download` instead of raising this in-memory limit.

## max\_file\_size and file safety

`max_file_size` applies only to download. It limits received body bytes, not the value advertised by Content-Length. If the stream would exceed the cap, Babet aborts, removes the temporary file, preserves an existing destination, and returns:

```
nil, "http: response body exceeds max_file_size"
```

The default 8 GiB is a safety guard, not a recommendation to accept every file of that size. Set the smallest limit compatible with the expected artifact. Free-space exhaustion and filesystem errors are reported as `(nil, err)` with the same cleanup guarantee.

## Response tables

### In-memory response

```
{
  status = 200,
  body = "...",
  headers = {
    ["content-type"] = "application/json",
    ["set-cookie"] = "b=2",
  },
  headers_multi = {
    ["content-type"] = { "application/json" },
    ["set-cookie"] = { "a=1", "b=2" },
  },
}
```

- status: integer HTTP status;
- body: complete binary string, possibly empty; HEAD returns an empty body;
- headers: lowercase names, last occurrence wins;
- headers\_multi: lowercase names, every occurrence in an array, even one.

Use `headers_multi` for repeated fields such as Set-Cookie:

```
for _, cookie in ipairs(response.headers_multi["set-cookie"] or {}) do
  print(cookie)
end
```

## Download result

```
{
    status = 200,
    saved = true,
    bytes = 123456,
    path = "downloads/archive.tar.gz",
    headers = { ... },
    headers_multi = { ... },
}
```

- status, headers, and headers\_multi have the same meaning as above;
- saved is true only when the final status is 2xx and the atomic commit succeeded;
- bytes is the number of received body bytes, including for a non-2xx result;
- path exists only when saved is true;
- there is deliberately no body field.

## Complete examples

### GET JSON

```
local response, err = babet.http.get("https://api.example.com/v1/status", {
    headers = { ["Accept"] = "application/json" },
    query = { verbose = 1 },
    timeout = 10,
    max_body_size = 1024 * 1024,
})
assert(response, err)
assert(response.status == 200, response.body)
local data = assert(babet.json.decode(response.body))
```

### Authenticated JSON POST

```
local body = assert(babet.json.encode({ title = "hello" }))
local response, err = babet.http.post(
    "https://api.example.com/v1/items",
    body,
    {
        headers = {
            ["Authorization"] = "Bearer " .. token,
            ["Content-Type"] = "application/json",
        },
        timeout = 15,
        max_body_size = 2 * 1024 * 1024,
    }
)
assert(response, err)
```

### Binary PUT

```
local file = assert(io.open("image.bin", "rb"))
local payload = file:read("a")
file:close()

local response = assert(babet.http.request({
    url = "https://upload.example.com/blob/42",
    method = "PUT",
}))
```

```

    body = payload,
    headers = { ["Content-Type"] = "application/octet-stream" },
    timeout = 30,
    max_body_size = 1024 * 1024,
}))

```

This loads the file into memory and is not suitable for very large uploads.

### Download a release artifact

```

assert(babet.mkdir("downloads"))
local result, err = babet.http.download(
    "https://example.com/releases/tool.tar.gz",
    "downloads/tool.tar.gz",
    {
        timeout = 120,
        follow_redirects = true,
        max_file_size = 512 * 1024 * 1024,
    }
)
assert(result, err)
if not result.saved then
    error("unexpected HTTP status " .. result.status)
end
print("saved", result.bytes, "bytes to", result.path)

```

### Separate network and HTTP failures

```

local response, err = babet.http.get(url, { timeout = 5 })
if not response then
    io.stderr:write("transport failure: ", err, "\n")
elseif response.status == 404 then
    print("not found")
elseif response.status >= 400 then
    io.stderr:write("HTTP ", response.status, "\n")
else
    print(response.body)
end

```

### Parallel requests with WORKERS

```

local jobs = {}
for i, url in ipairs(urls) do
    jobs[i] = assert(babet.workers.spawn([
        local r, err = babet.http.get(worker.args.url, {
            timeout = 10,
            max_body_size = 1024 * 1024,
        })
        if not r then error(err) end
        return { status = r.status, size = #r.body }
    ]), { url = url }))
end

for i, job in ipairs(jobs) do
    local ok_result, value = job:join()
    if ok_result then
        print(urls[i], value.status, value.size)
    end
end

```

```

else
    io.stderr:write(urls[i], ": ", value, "\n")
end
end

```

## Error contract

Wrong positional types raise:

```

babet.http.request("not a table")
babet.http.get(42)
babet.http.get(url, "not a table")
babet.http.post(url, 42)
babet.http.download(url, 42)
babet.http.download(url, "file", "not a table")

```

Invalid values and runtime failures return `(nil, err)`: missing/bad URL, unsupported method, forbidden body, invalid options/query/headers, DNS/TCP/TLS, timeout, response/file limit, destination-path or filesystem failure, and converted internal exceptions. Allocation failure returns `(nil, "http: out of memory")`; other internal exceptions return `(nil, "http: internal failure")`.

Any received HTTP status, including 3xx/4xx/5xx, returns a normal response. For download, only 2xx is saved; all other statuses return `saved = false`.

## Security and limitations

- Validate untrusted URLs against SSRF to loopback, LAN, cloud metadata, and administrative services.
- Treat redirects as new destinations and protect sensitive headers.
- Keep TLS verification enabled; use private CA trust instead of disabling it.
- Set the smallest practical `max_body_size` or `max_file_size`.
- `request`, `get`, and `post` keep response bodies in memory. `download` streams only GET responses to a file; uploads are still in memory.
- Atomic replacement protects the final path from partial content, but callers remain responsible for validating expected status, size, and cryptographic checksum.
- Every call constructs a client; no exposed cookie jar or connection pool.
- No form/multipart helper, proxy configuration, HTTP/2/3, WebSocket, or server.
- DNS is synchronous and may exceed the requested timeout.

## babet.inotify — filesystem watching

`babet.inotify` provides a small interface to Linux `inotify(7)`. Notifications come from the kernel, so watched directories do not need to be scanned periodically.

The module is **Linux-specific** and watching is not recursive.

## Module contents

- [API](#)
- [Creating a watcher](#)
- [Adding a watch](#)
  - [onlydir option](#)
- [Reading events](#)
  - [Queue overflow](#)
  - [Signal interruption](#)
- [Removing and closing](#)
- [Example](#)

- [Error contract](#)
- [Limits](#)

## API

| Function                                  | Result                                                                                                                 |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <code>babet.inotify.new()</code>          | <code>watcher</code>   <code>(nil, err)</code>                                                                         |
| <code>w:add(path, events [, opts])</code> | <code>integer wd</code>   <code>(nil, err)</code>                                                                      |
| <code>w:read([timeout])</code>            | <code>event array</code>   <code>(nil, "timeout")</code>   <code>(nil, "interrupted")</code>   <code>(nil, err)</code> |
| <code>w:remove(wd)</code>                 | <code>(true, nil)</code>   <code>(nil, err)</code>                                                                     |
| <code>w:close()</code>                    | <code>(true, nil)</code>                                                                                               |

Signatures are strict: extra arguments raise a Lua error. `close()` is idempotent, and garbage collection also closes a forgotten watcher.

## Creating a watcher

```
local watcher, err = babet.inotify.new()
if not watcher then
    error(err)
end
```

Each call creates an independent instance. The inotify descriptor uses `IN_NONBLOCK` and `IN_CLOEXEC`: `read()` controls waiting itself, and the descriptor is not inherited by programs launched through `exec`.

## Adding a watch

```
local wd, err = watcher:add(path, events [, opts])
```

`path` must be a string without an embedded NUL byte. It may refer to an existing directory or file.

`events` is a strict, non-empty 1..N array with no holes or extra keys. Every element must be one of these names:

| Name            | Requested event                                                        |
|-----------------|------------------------------------------------------------------------|
| "access"        | file read or accessed                                                  |
| "modify"        | content modified                                                       |
| "attrib"        | attributes or metadata modified                                        |
| "close_write"   | write-open descriptor closed                                           |
| "close_nowrite" | non-write descriptor closed                                            |
| "close"         | combination of <code>close_write</code> and <code>close_nowrite</code> |
| "open"          | file opened                                                            |
| "moved_from"    | entry moved out of the watched directory                               |
| "moved_to"      | entry moved into the watched directory                                 |
| "move"          | combination of <code>moved_from</code> and <code>moved_to</code>       |
| "create"        | entry created in the watched directory                                 |
| "delete"        | entry deleted from the watched directory                               |
| "delete_self"   | watched target deleted                                                 |
| "move_self"     | watched target moved                                                   |

## onlydir option

```
local wd = assert(watcher:add(path, { "create" }, {  
    onlydir = true,  
}))
```

When present, onlydir must be a boolean:

- true adds Linux's IN\_ONLYDIR mask and therefore rejects a target that is not a directory;
- false keeps the normal behavior;
- other fields in opts are currently ignored.

Adding a watch again for the same target follows native inotify semantics: the existing mask is replaced because IN\_MASK\_ADD is not used.

## Reading events

```
local events, err = watcher:read([timeout])
```

The timeout is in seconds and may be fractional:

- omitted or nil: wait indefinitely;
- 0: non-blocking read of events already available;
- positive value: bounded wait.

A negative, NaN, infinite, or excessively large timeout returns (nil, err). A value of the wrong type raises a Lua error.

A successful read returns an array of tables:

```
{  
  {  
    wd = 1,  
    name = "photo.jpg",  
    events = { create = true, close_write = true },  
    is_dir = false,  
    cookie = 0,  
  },  
}
```

- wd is the watch descriptor returned by add;
- name is the affected entry name, or "" for an event on the watched target itself;
- events contains a true boolean for every received bit;
- is\_dir reflects IN\_ISDIR;
- cookie pairs moved\_from with moved\_to; outside moves it is usually 0.

Besides events requestable through add, the kernel may produce:

- events.ignored when a watch is removed, automatically or through remove;
- events.unmount when the watched filesystem is unmounted.

One call may return several events because the module reads and decodes a whole batch from the kernel queue.

## Queue overflow

If the inotify queue overflows, events have been lost. The module surfaces this explicitly:

```
{  
  wd = -1,  
  name = "",  
  events = { overflow = true },  
  is_dir = false,
```

```
    cookie = 0,
}
```

The program must then rescan watched paths to rebuild reliable state.

## Signal interruption

If a signal handled by `babot.signal` arrives while waiting, its Lua callback runs and `read()` then returns `(nil, "interrupted")`.

## Removing and closing

```
assert(watcher:remove(wd))
assert(watcher:close())
```

After `remove(wd)`, the kernel normally queues an ignored event. Removing an invalid descriptor returns `(nil, err)`.

After `close`, `add`, `read`, and `remove` return an error saying the watcher is closed. `close` may be called more than once.

## Example

```
local watcher = assert(babot.inotify.new())
local wd = assert(watcher:add("/srv/incoming", {
    "close_write",
    "moved_to",
}), {
    onlydir = true,
}))

while true do
    local events, err = watcher:read()
    if not events then
        if err == "interrupted" then
            break
        end
        error(err)
    end

    for _, event in ipairs(events) do
        if event.events.overflow then
            rescan_directory()
        elseif event.wd == wd then
            process_entry(event.name)
        end
    end
end

watcher:close()
```

## Error contract

- wrong argument count or type: Lua error;
- invalid value, system error, or closed watcher: `(nil, err)`;
- no event before the deadline: `(nil, "timeout")`;
- handled signal during read: `(nil, "interrupted")`;

- successful remove and close: (true, nil).

System errors are prefixed with "inotify: " and retain the description provided by the operating system.

## Limits

- Linux only;
- no recursive watching;
- no IN\_DONT\_FOLLOW, IN\_ONESHOT, or IN\_MASK\_ADD;
- one instance may hold several watches, and several independent instances may coexist in one process.

## babet.json — JSON encoding and decoding

The `babet.json` module is backed by **nlohmann/json 3.11.3**. It converts Lua values to JSON text and back, with two sentinels that represent JSON null and the empty array [] without ambiguity.

## Module contents

- [API](#)
- [Type mapping](#)
  - [Lua to JSON](#)
  - [JSON to Lua](#)
- [null](#)
- [Empty object, empty array, and as\\_array](#)
- [encode options](#)
- [Strings, UTF-8, and NUL bytes](#)
- [Numbers](#)
- [Object order and duplicate keys](#)
- [Error contract](#)
- [Limits](#)

## API

| Element                                        | Contract                                          |
|------------------------------------------------|---------------------------------------------------|
| <code>babet.json.encode(value [, opts])</code> | Returns text, nil or nil, err                     |
| <code>babet.json.decode(text)</code>           | Returns value, nil or nil, err                    |
| <code>babet.json.null</code>                   | Sentinel representing JSON null                   |
| <code>babet.json.empty_array</code>            | Constant sentinel that encodes as []              |
| <code>babet.json.as_array(t)</code>            | Tags t as a JSON array and returns the same table |

`encode` accepts exactly one or two arguments. `decode` requires exactly **one Lua string**: a number is not implicitly converted to text. `as_array` requires exactly one table.

## Type mapping

### Lua to JSON

| Lua                           | JSON |
|-------------------------------|------|
| nil passed directly to encode | null |

| Lua                                    | JSON                                    |
|----------------------------------------|-----------------------------------------|
| <code>babet.json.null</code>           | <code>null</code>                       |
| <code>babet.json.empty_array</code>    | <code>[]</code>                         |
| <code>boolean</code>                   | <code>boolean</code>                    |
| <code>Lua integer</code>               | <code>JSON integer number</code>        |
| <code>finite Lua float</code>          | <code>JSON floating-point number</code> |
| <code>valid UTF-8 string</code>        | <code>JSON string</code>                |
| <code>untagged empty table</code>      | <code>empty object {}</code>            |
| <code>integer keys exactly 1..n</code> | <code>JSON array</code>                 |
| <code>string keys only</code>          | <code>JSON object</code>                |

Inside a Lua table, assigning `nil` removes the key before encoding ever starts. Use `babet.json.null` to retain a key whose JSON value must explicitly be `null`.

A table mixing string and integer keys, a sparse array, or a table with an unrepresentable key (0, a negative integer, a float, a boolean, a table, and so on) cannot be converted: `encode` returns `nil`, `err`.

## JSON to Lua

| JSON                                                             | Lua                                |
|------------------------------------------------------------------|------------------------------------|
| <code>null</code>                                                | <code>babet.json.null</code>       |
| <code>boolean</code>                                             | <code>boolean</code>               |
| <code>signed integer in the Lua range</code>                     | <code>Lua integer</code>           |
| <code>unsigned integer above <code>math.maxinteger</code></code> | <code>Lua float</code>             |
| <code>JSON floating-point number</code>                          | <code>Lua float</code>             |
| <code>string</code>                                              | <code>Lua string</code>            |
| <code>array</code>                                               | <code>sequential table 1..n</code> |
| <code>object</code>                                              | <code>string-keyed table</code>    |

A JSON integer too large for `lua_Integer` is converted to a float instead of silently wrapping to a negative value. As with any IEEE 754 conversion, precision may be lost.

## `null`

Lua cannot retain a `nil` value in a table. The shared `babet.json.null` sentinel fills that gap:

```
local J = babet.json

local value = assert(J.decode([["answer":null]]))
assert(value.answer == J.null)

local text = assert(J.encode({ answer = J.null }))
-- text represents {"answer":null}
```

Compare the sentinel by identity with `==`. It is implemented as a special table, but must not be used as a data table.

## Empty object, empty array, and `as_array`

An empty Lua table is ambiguous. Babet chooses a JSON object by default:

```
local J = babet.json
```

```
assert(J.encode({}) == "{}")
assert(J.encode(J.empty_array) == "[]")
assert(J.encode(J.as_array({})) == "[]")
```

`empty_array` is a shared sentinel for values that are not meant to be modified. `as_array(t)` is preferable for a dynamically built array: it tags the table with an internal metatable, leaves it mutable, and returns that exact same table.

```
local tags = {}
-- table.insert(tags, ...) may never be called

local text = assert(J.encode({ tags = J.as_array(tags) }))
-- {"tags":[]}
```

Important details:

- `as_array` is idempotent;
- the tag **replaces any existing metatable** on the table;
- the shape is validated only when encode runs: a tagged table with string keys or holes returns `nil, err`;
- `as_array(J.null)` and `as_array(J.empty_array)` raise a Lua error;
- only a decoded **empty** JSON array receives the tag automatically. A decoded non-empty array that is later emptied becomes an ordinary empty table and therefore re-encodes as `{}`; call `as_array(t)` to retain `[]` in that case;
- ordinary writes to the sentinels are rejected. As with every Lua table, `rawset` bypasses metamethods: do not mutate the sentinels.

## encode options

The optional table has one interpreted option:

```
local compact = assert(babet.json.encode({ a = 1 }))
local pretty  = assert(babet.json.encode({ a = 1 }, { indent = 4 }))
```

- `indent = 0..256` enables formatted output with line breaks;
- `indent = 0` adds no indentation spaces but keeps formatted-output line breaks;
- omitted, `nil`, or a negative integer: compact output;
- a non-integer value or a value above 256 raises a Lua error;
- other option fields are ignored. In particular, the previously and incorrectly documented `pretty` option does not exist.

## Strings, UTF-8, and NUL bytes

JSON strings must contain valid UTF-8. Unicode characters are preserved in output; Babet does not force them into `\uXXXX` escapes.

Lua strings may contain NUL bytes. They remain binary-safe on the Lua side and are escaped according to JSON rules during encoding. The same applies to object keys:

```
local J = babet.json
local original = "a\0b"
local text = assert(J.encode({ [original] = original }))
local back = assert(J.decode(text))
assert(back[original] == original)
```

A string or key containing invalid UTF-8 makes encode return `nil, err`. JSON text containing an invalid UTF-8 string is likewise rejected by decode.

## Numbers

Babet preserves the Lua numeric subtype when JSON permits it: 42 decodes as a Lua integer, whereas 42.0 and 1e3 decode as floats. When encoding, a Lua float such as 3.0 remains a JSON floating-point number.

NaN, +Inf, and -Inf are not part of the JSON standard and return nil, err.

## Object order and duplicate keys

JSON object member order has no semantic meaning and must not be treated as an API contract. The current backend generally emits members in lexicographic key order, but callers should compare decoded data rather than raw text whenever an external format does not prescribe ordering.

When decoding a JSON object that repeats the same key, the last value replaces earlier ones.

## Error contract

Conversion and parsing failures are returned and do not raise a Lua exception:

```
local value, err = babet.json.decode("{ invalid")
if not value then
    print(err) -- prefixed with "json:"
end
```

This includes, among other cases:

- invalid JSON, trailing data after the document, or invalid UTF-8;
- mixed tables, sparse arrays, or unrepresentable keys;
- a function, userdata, or thread that cannot be encoded;
- NaN or infinity;
- a cycle or excessive nesting depth.

API misuse does raise a Lua error:

- wrong argument count;
- a non-string argument to decode;
- an encode second argument that is neither a table nor nil;
- a non-integer opts.indent or one above 256;
- an invalid argument to as\_array.

Parsing errors produced by nlhmann/json include the failure position, usually as a line and column.

## Limits

Conversion depth is capped at **1000 levels** in both directions. This also acts as the guard against cyclic Lua tables. The module builds the whole document in memory and does not provide a streaming API.

## logging — leveled logging

logging is a pure-Lua module bundled with Babet. Load it with require("logging"); it is not a member of the babet table.

```
local log = require("logging")
```

It exposes five levels:

| Level     | Value |
|-----------|-------|
| log.TRACE | 10    |
| log.DEBUG | 20    |
| log.INFO  | 30    |
| log.WARN  | 40    |
| log.ERROR | 50    |

There is no fatal level.

## Module contents

- [Emitting a message](#)
  - [No-propagation guarantee](#)
- [Output format](#)
- [Threshold](#)
- [Destination](#)
- [ANSI colors](#)
- [Module state](#)
- [Error contract](#)
- [Limitations](#)

## Emitting a message

```
log.trace(...)
log.debug(...)
log.info(...)
log.warn(...)
log.error(...)
```

Each function accepts zero or more values and returns no values. A message is emitted when the function's level is greater than or equal to the current threshold.

The initial threshold is log.INFO, so trace and debug are filtered by default.

Arguments are converted with `tostring` and joined with one space:

```
log.info("user=", 42, "active=", true)
-- ... [INFO ] user= 42 active= true
```

Calling a log function with no arguments emits a line with an empty message. Embedded NUL bytes and newlines are neither removed nor escaped. A multiline message therefore produces several physical lines, but only the start of the call has the logger prefix.

## No-propagation guarantee

The five emission functions never raise. The whole pipeline is protected:

- argument conversion through `tostring`;
- timestamp creation through `os.date`;
- the sink's write call.

If any of these operations fails, the message is silently lost and the script continues. Arguments of a filtered message are not passed to `tostring`.

## Output format

Without colors, each line uses this format:

YYYY-MM-DD HH:MM:SS [LEVEL] message\n

Example:

2026-07-13 21:45:02 [WARN ] attempt 3 of 5

The timestamp uses the machine's **local time**. It includes neither a time zone nor milliseconds. Labels occupy five characters: [TRACE], [DEBUG], [INFO ], [WARN ], [ERROR].

## Threshold

```
log.set_level(level)
local level = log.get_level()
```

set\_level requires exactly one argument and returns no values. get\_level accepts no arguments and returns the current threshold.

The level may be:

- one of "trace", "debug", "info", "warn", "error", case-insensitively;
- any finite number, allowing an intermediate threshold such as 25.5.

```
log.set_level("DEBUG")
log.set_level(log.WARN)
log.set_level(25.5)
```

Numeric strings are not coerced and whitespace is not trimmed: "30" and " info " are rejected. NaN, +inf, and -inf are rejected as well.

## Destination

```
log.set_output(out)
local out = log.get_output()
```

The initial destination is io.stderr.

set\_output requires exactly one table or userdata exposing a callable write method. A method supplied through \_\_index is accepted. The method is then called as out:write(line).

```
local file, err = io.open("app.log", "a")
if not file then
    log.error("cannot open log file:", err)
else
    log.set_output(file)
end
```

The module:

- performs no write during set\_output;
- never closes the destination;
- never calls flush;
- ignores the value returned by write;
- swallows write errors during emission.

A closed file still exposes a write method and can therefore be installed; subsequent writes will fail silently. Sink lifetime management belongs to the script.

get\_output accepts no arguments and returns the exact installed object.

## ANSI colors

```
log.set_color(true)
local enabled = log.get_color()
```

Colors are disabled by default. `set_color` requires exactly one boolean and returns no values. `get_color` accepts no arguments and returns a boolean.

When colors are enabled:

- trace: ANSI dim;
- debug: cyan;
- info: no color;
- warn: yellow;
- error: red.

The ANSI reset is written before the `\n`. The module does not detect whether the sink is a terminal; enabling colors is always explicit.

## Module state

`require("logging")` returns the same table until `package.loaded.logging` is removed. The threshold, sink, and color setting are therefore shared by every user of the module within **one Lua state**.

Each Babet worker owns a separate Lua state, so its logging settings are independent from the main thread and from other workers.

The initial values are:

```
log.set_level(log.INFO)
log.set_output(io.stderr)
log.set_color(false)
```

The module also exposes:

```
log._VERSION      -- "babet logging 1.1.0"
log._DESCRIPTION  -- textual description
```

## Error contract

Setters and getters raise a Lua error on wrong arity or an invalid value. They do not return (`nil`, `err`).

`trace`, `debug`, `info`, `warn`, and `error` never raise and return no values.

## Limitations

The module does not provide:

- named or hierarchical loggers;
- per-module thresholds;
- multiple destinations;
- custom or JSON formatting;
- UTC timestamps or sub-second precision;
- file rotation;
- buffered or asynchronous writes;
- protection against a recursive sink that calls the logger again;
- automatic terminal detection.

Use an external tool such as `logrotate` for rotation. For more advanced needs, wrap the module or replace its functions in the script.

## SIGNAL — graceful shutdown, reloads, and POSIX signals

`babet.signal` lets the main script install a Lua callback for a small set of POSIX signals, ignore a signal, or restore its operating-system default.

The module covers:

- graceful shutdown on TERM and INT;
- configuration reload requests on HUP;
- two application-defined signals, USR1 and USR2;
- explicitly ignoring or restoring a signal;
- interruption of selected blocking Babet calls;
- deferred, safe callback dispatch in the main Lua thread.

It does not send signals, expose a pending-signal queue, configure POSIX masks, or allow handlers to be installed from a worker.

## Module contents

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## Essential conventions

### Success returns one value

All three functions return only the boolean `true` on success. They do not return a second `nil`.

```
local ok = babet.signal.ignore("PIPE")
assert(ok == true)
```

A rare `sigaction(2)` failure instead returns two values:

```
local ok, err = babet.signal.ignore("PIPE")
if not ok then
    io.stderr:write(err, "\n")
end
```

### Names are strict

A signal name must be an actual Lua string, with no NUL byte, in the exact case shown by this documentation.

```
assert(babet.signal.handle("TERM", function() end))
```

These forms are invalid:

```
babet.signal.handle("term", function() end) -- wrong case
babet.signal.handle(15, function() end)     -- number, not string
babet.signal.handle("TERM\0X", function() end)
```

### A callback receives no arguments

The callback is invoked with no parameters. Install a distinct function for each signal when code needs to know which signal arrived.

```
babet.signal.handle("USR1", function()
    print("USR1")
end)
```

### API overview

```
local ok, err = babet.signal.handle(name, callback_or_nil)
local ok, err = babet.signal.ignore(name)
local ok, err = babet.signal.default(name)
```

| Function          | Success | Rare system failure | Invalid call       |
|-------------------|---------|---------------------|--------------------|
| handle(name, fn)  | true    | (nil, err)          | raises a Lua error |
| handle(name, nil) | true    | (nil, err)          | raises a Lua error |
| ignore(name)      | true    | (nil, err)          | raises a Lua error |
| default(name)     | true    | (nil, err)          | raises a Lua error |

handle(name) without a second argument is intentionally rejected. Write handle(name, nil) explicitly to remove a callback.

### Supported signals

| Babet name | POSIX signal | Common use                                      | Usual OS default            |
|------------|--------------|-------------------------------------------------|-----------------------------|
| "TERM"     | SIGTERM      | graceful shutdown request, notably from systemd | terminate the process       |
| "INT"      | SIGINT       | keyboard interruption, usually Ctrl-C           | terminate the process       |
| "HUP"      | SIGHUP       | configuration reload request                    | often terminate the process |
| "USR1"     | SIGUSR1      | free application event                          | terminate the process       |
| "USR2"     | SIGUSR2      | second free application event                   | terminate the process       |
| "PIPE"     | SIGPIPE      | writing to a closed pipe or socket              | terminate the process       |

KILL and STOP cannot be exposed: POSIX does not allow them to be caught or ignored.

Dangerous signals or signals reserved for other internal mechanisms — such as SEGV, BUS, FPE, ILL, CHLD, and ALRM — are not in the allowlist.

## **handle(name, fn) — install or replace**

```
local ok, err = babet.signal.handle(name, fn)
```

- name is one of the six supported names;
- fn must be a Lua function;
- the previous POSIX disposition is replaced;
- an existing Lua callback for the same signal is replaced;
- the callback is retained in the Lua registry until removed.

```
local stopping = false
```

```
assert(babet.signal.handle("TERM", function()  
    stopping = true  
end))
```

Replacing a callback:

```
assert(babet.signal.handle("USR1", function()  
    print("first version")  
end))
```

```
assert(babet.signal.handle("USR1", function()  
    print("new version")  
end))
```

After the second call, only the new function can run.

## **handle(name, nil) — remove a callback**

```
local ok, err = babet.signal.handle(name, nil)
```

This form:

1. removes the associated Lua callback;
2. restores SIG\_DFL, the operating-system default.

```
assert(babet.signal.handle("HUP", reload_config))
```

```
-- Later: explicit removal.
```

```
assert(babet.signal.handle("HUP", nil))
```

handle("TERM") without a second argument is **not** shorthand: it raises an error, preventing accidental removal caused by a missing argument.

## **ignore(name) — ignore**

```
local ok, err = babet.signal.ignore(name)
```

ignore installs SIG\_IGN and removes any previously associated Lua callback. The kernel then ignores that signal.

```
assert(babet.signal.ignore("PIPE"))
```

This is useful when an application prefers normal write errors over process termination caused by SIGPIPE.

The ignore disposition remains active until a later handle, default, or handle(name, nil) call for the same signal.

## default(name) — restore the OS default

```
local ok, err = babet.signal.default(name)
```

default installs SIG\_DFL and removes any Lua callback. For TERM, INT, HUP, USR1, USR2, and usually PIPE, the next matching signal terminates the process according to operating-system rules.

```
assert(babet.signal.ignore("PIPE"))
-- ... section where SIGPIPE should be ignored ...
assert(babet.signal.default("PIPE"))
```

handle(name, nil) and default(name) have the same system effect. default is more explicit when no callback is involved.

## Callback execution

The real POSIX handler never invokes Lua. It only sets a sig\_atomic\_t flag, which is safe in asynchronous signal context.

The Lua callback runs later from a safe point:

1. a supported signal arrives;
2. the C handler marks that signal pending;
3. the main thread notices the flag from the instruction hook or when an interrupted blocking call returns;
4. Babet clears the flag;
5. Babet invokes the Lua callback with zero arguments.

## Main thread and safe context

The callback always runs in the main Lua state, never in the POSIX handler and never in a worker. It can therefore use ordinary Lua and Babet APIs.

```
babet.signal.handle("USR1", function()
    -- Runs in normal Lua context.
    local now = babet.time.now()
    print("USR1 at", now)
end)
```

That freedom does not mean a callback should be long. A short function that sets a flag or schedules an action remains easier to reason about.

## Dispatch order

When several distinct signals are pending at the same dispatch point, Babet processes them in the fixed order of its allowlist:

1. TERM;
2. INT;
3. HUP;
4. USR1;
5. USR2;
6. PIPE.

This is neither callback registration order nor a guarantee of exact kernel arrival order.

For example, if USR1 and then TERM arrive before the next dispatch, the TERM callback runs before the USR1 callback.

## Occurrence coalescing

Babet stores one flag per signal type, not a counter or queue. Multiple identical occurrences received before the next dispatch are therefore coalesced into one callback invocation.

USR1, USR1, USR1 before dispatch -> one USR1 callback invocation

A new identical signal received **while** its callback runs can set the flag again and be processed during a later dispatch.

Do not use this module as a reliable event counter. Use a queue, pipe, socket, or another message mechanism when every business event must be retained.

## Callback errors

Babet invokes callbacks through `lua_pcall`. An uncaught error is removed from the stack and silently ignored; it is not propagated to the code that was running.

```
babet.signal.handle("USR1", function()
    error("invisible to the caller")
end)
```

Protect the body explicitly to log failures:

```
babet.signal.handle("USR1", function()
    local ok, err = pcall(function()
        refresh_metrics()
    end)
    if not ok then
        io.stderr:write("USR1: ", tostring(err), "\n")
    end
end)
```

## Interaction with blocking calls

Babet installs handlers without `SA_RESTART`. Selected internal blocking calls can therefore be interrupted by a handled signal. When the main thread sees a pending Babet signal, it dispatches callbacks and generally returns a typed "interrupted" error.

Integrated call families currently include, among others:

- `babet.sleep(...)`;
- `Inotify watcher:read(...)`;
- TCP connection, `accept`, and receive waits;
- TLS handshakes and waits.

Example:

```
local data, err = peer:recv(4096, 30)
if not data then
    if err == "interrupted" then
        -- A callback has already run: re-check application state.
    elseif err == "timeout" then
        -- No bytes arrived before the deadline.
    else
        -- Close or network error.
    end
end
```

Not every Babet function is interruptible through this mechanism. `babet.exec`, for example, applies its own timeout and does not expose an "interrupted" result for handled signals.

A syscall can also receive EINTR while no Babet-managed signal is pending. In that case internal loops resume waiting instead of reporting a false "interrupted".

## Interaction with `debug.sethook`

Outside integrated blocking functions, Babet uses a Lua count hook, triggered about every 10,000 instructions, to inspect pending flags.

Lua supports only one active hook per state. Consequently:

- the first `signal.handle(...)` replaces an existing user hook;
- a later `debug.sethook(...)` replaces Babet's hook;
- after that replacement, callbacks are no longer dispatched by instruction count, though they may still be dispatched when integrated blocking calls return;
- removing every callback with `handle(name, nil)`, `ignore(name)`, or `default(name)` does not uninstall Babet's hook and does not restore a user hook that was previously replaced. The hook remains active, but does nothing while no Babet callback is registered.

Avoid combining `babet.signal` and a custom debug hook in the same Lua state unless this interaction is deliberate and understood.

## Complete examples

### Graceful shutdown on TERM and INT

```
local running = true

local function request_stop()
    running = false
end

assert(babet.signal.handle("TERM", request_stop))
assert(babet.signal.handle("INT", request_stop))

while running do
    do_one_iteration()

    local slept, err = babet.sleep(250, "ms")
    if not slept and err ~= "interrupted" then
        io.stderr:write(tostring(err), "\n")
        break
    end
    -- On "interrupted", the callback has already run and the loop
    -- re-checks running.
end

close_resources()
```

For a network loop, use the socket's direct "interrupted" result instead of `pcall`.

### Reload configuration with HUP

It is often better for the callback to set a flag only. Full loading then happens in the normal loop.

```
local running = true
local reload_requested = false

assert(babet.signal.handle("TERM", function()
    running = false
```

```

end))

assert(babet.signal.handle("HUP", function()
    reload_requested = true
end))

while running do
    if reload_requested then
        reload_requested = false
        local new_config, err = load_config()
        if new_config then
            config = new_config
        else
            io.stderr:write("reload: ", err, "\n")
        end
    end

    do_one_iteration(config)
end

```

### Interrupt a socket wait

```

local running = true

assert(babet.signal.handle("TERM", function()
    running = false
end))

while running do
    local client, err = server:accept(60)
    if client then
        serve(client)
    elseif err == "interrupted" then
        -- The TERM callback may have set running to false.
    elseif err ~= "timeout" then
        io.stderr:write("accept: ", err, "\n")
    end
end

server:close()

```

### Temporarily ignore PIPE

```

assert(babet.signal.ignore("PIPE"))

local sent, err = socket:send(payload)
if not sent then
    io.stderr:write("send: ", err, "\n")
end

assert(babet.signal.default("PIPE"))

```

Warning: restoring default SIGPIPE behavior means a later write to a closed pipe may terminate the process.

## Observe callback errors

```
local function safe_handler(label, fn)
    return function()
        local ok, err = xpcall(fn, debug.traceback)
        if not ok then
            io.stderr:write(label, ": ", tostring(err), "\n")
        end
    end
end

assert(babet.signal.handle("USR2", safe_handler("USR2", function()
    rotate_logs()
end))))
```

## Error contract

### Raised Lua errors

These situations immediately raise a Lua error:

- call from a worker;
- missing, non-string, or NUL-containing name;
- unsupported name or wrong case;
- handle without a second argument;
- handler other than a function or nil.

```
local ok, err = pcall(function()
    babet.signal.handle("TERM")
end)
assert(not ok)
```

An unsupported name also raises; it does not return (nil, err).

```
local ok, err = pcall(function()
    babet.signal.ignore("QUIT")
end)
```

### Returned system error

If `sigaction(2)` fails, the function returns:

```
nil, "signal: sigaction failed: ..."
```

This is rare with the six allowed signals, but callers may retain the (ok, err) convention when they need to handle it.

## Workers, concurrency, and limitations

Signal dispositions are process-wide, not thread-local. Babet therefore enforces these rules:

- handle, ignore, and default are restricted to the main thread;
- using them from a worker raises an explicit error;
- workers block all six supported signals so they remain owned by the main thread;
- workers cannot observe or consume the parent's pending flags.

Control a worker through message queues, not `babet.signal`:

```
local job = assert(babet.workers.spawn([
    while true do
        local ok, message = worker.recv()
```

```

        if not ok then
            return "closed"
        end
        if message == "stop" then
            return "stopped"
        end
    end
end
]))

-- The main handler converts the signal into an application message.
babet.signal.handle("TERM", function()
    job:send("stop", 0)
end)

```

Keep this callback short. A full queue can make the non-blocking send fail; a robust application also keeps a fallback flag.

## Features not provided

The module currently does not provide:

- `kill(pid, name)` to send a signal;
- `list()` to retrieve supported names;
- `is_pending()` or an occurrence counter;
- `sigprocmask` or per-thread masks;
- `signalfd`;
- signals beyond the six listed names;
- a debug hook that can be chained with the user's hook.

An external program can send a signal from Lua:

```

local result, err = babet.exec("kill", {
    "-TERM",
    tostring(pid),
})
assert(result, err)
assert(result.code == 0, result.stderr)

```

Reception remains subject to ordinary POSIX rules and the dispatch model described on this page.

## SOCKET — TCP clients and servers, binary streams, lines, and timeouts

`babet.socket` provides synchronous TCP sockets for custom protocols: IRC, plain SMTP before STARTTLS, line-oriented services, binary streams, and small internal daemons.

It covers:

- connecting to a hostname or IP address;
- listening on one interface or all interfaces;
- accepting clients;
- sending a complete binary Lua string;
- receiving chunks, lines, or everything until remote EOF;
- per-socket and per-call deadlines;
- local and peer addresses;
- explicit close and garbage-collector cleanup;
- interruption by signals handled through Babet.

It does not expose UDP, Unix-domain sockets, asynchronous I/O, select/poll loops, or an application protocol. Use [HTTP](#) for web requests and [TLS](#) for direct TLS or STARTTLS.

## Module contents

- [Core conventions](#)
- [API overview](#)
- [Connecting a client](#)
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## Core conventions

### Synchronous TCP

Every call runs in the current thread. Without a timeout, connect, accept, send, and receive methods can wait indefinitely. Use [workers](#) for independent concurrent connections; Babet does not expose a Lua event loop.

### Binary-safe data

Payload strings may contain NUL bytes:

```
assert(sock:send("AB\0CD") == 5)
local data = assert(sock:recv(5, 2))
assert(#data == 5 and data:byte(3) == 0)
```

Hostnames are passed to C/POSIX APIs and may not contain NUL.

### Timeout rules

Timeouts are seconds and may be fractional:

- omitted or nil: use the socket default;
- 0: wait forever;
- positive finite number: one deadline for the whole call;
- positive values below 1 ms: rounded up to 1 ms;
- negative, NaN, infinity, or excessively large: error.

The deadline is not restarted for every byte or internal retry.

### Return values

Constructors and reads return a value or (nil, err). set\_timeout, starttls, and close return exactly (true, nil) on success.

Wrong positional types raise a Lua error. Invalid values and runtime failures normally return (nil, err).

## API overview

```
local sock, err = babet.socket.connect(host, port, timeout?)
local server, err = babet.socket.listen(host, port, backlog?)

local count, err = sock:send(data)
local data, err = sock:recv(count, timeout?)
local line, err, partial = sock:recv_line(timeout?)
local data, err = sock:recv_all(timeout?, max_bytes?)
local ok, err = sock:set_timeout(seconds)
local addr, err = sock:peer()
local addr, err = sock:sockname()
local ok, err = sock:close()

local client, err = server:accept(timeout?)
```

| Argument        | Type                             | Default  | Meaning                            |
|-----------------|----------------------------------|----------|------------------------------------|
| client host     | strict non-empty string          | required | DNS name or IP address             |
| listen host     | strict string                    | required | interface; "" means all interfaces |
| port            | strict Lua integer<br>0..65535   | required | TCP port                           |
| connect timeout | finite number >= 0               | 0        | connection deadline in seconds     |
| backlog         | strict Lua integer<br>1..INT_MAX | 16       | requested listen queue size        |

Numeric strings such as "443" and floats such as 443.0 are rejected for integer fields.

A listening socket rejects stream operations and peer; a connected socket rejects accept.

### **babet.socket.connect(host, port, timeout?)**

```
local sock, err = babet.socket.connect("example.net", 9000, 5)
assert(sock, err)
```

Babet calls getaddrinfo() and tries returned IPv4/IPv6 addresses in order. All connection attempts share the same deadline. Synchronous DNS resolution itself is outside that deadline and may add delay.

Use a numeric address to avoid DNS:

```
local sock = assert(babet.socket.connect("127.0.0.1", 9000, 2))
```

The constructor timeout only covers connection establishment. The new socket's default I/O timeout is still zero:

```
local sock = assert(babet.socket.connect(host, port, 3))
assert(sock:set_timeout(10))
```

Typical results:

- timeout: (nil, "timeout");
- handled signal: (nil, "interrupted") after callback dispatch;

- DNS/TCP failure: (nil, "socket: ...");
- empty host or out-of-range port: (nil, err);
- wrong positional type: Lua error.

## **babet.socket.listen(host, port, backlog?)**

Creates a TCP listening socket with SO\_REUSEADDR.

Loopback only:

```
local server = assert(babet.socket.listen("127.0.0.1", 9000))
```

All interfaces:

```
local server = assert(babet.socket.listen("", 9000))
```

An empty host uses AI\_PASSIVE. Use "0.0.0.0" or ":::" to request a specific address family. Listening on all interfaces exposes the service to the network; add authentication and firewall rules.

Ask the kernel for a free test port:

```
local server = assert(babet.socket.listen("127.0.0.1", 0))
local addr = assert(server.sockname())
print(addr.host, addr.port)
```

Backlog is a kernel request, not a maximum number of active clients:

```
local server = assert(babet.socket.listen("127.0.0.1", 9000, 128))
```

## **server:accept(timeout?)**

```
local client, err = server:accept(10)
```

The positional timeout overrides server:set\_timeout() for this call only. The accepted socket has a default I/O timeout of zero; it does not inherit the server timeout:

```
local client = assert(server:accept(10))
assert(client:set_timeout(30))
```

Accepted descriptors are close-on-exec.

## **sock:set\_timeout(seconds)**

Sets the default for send, recv, recv\_line, recv\_all, and accept:

```
assert(sock:set_timeout(3.5))
assert(sock:set_timeout(0)) -- infinite again
```

It does not apply to a completed constructor and does not control STARTTLS; starttls uses opts.timeout.

Per-call overrides:

```
assert(sock:set_timeout(10))
local a, err = sock:recv(4096, 0.2) -- 200 ms
local b, err = sock:recv(4096)      -- 10 s
local c, err = sock:recv(4096, 0)   -- infinite
```

## **sock:send(data)**

Sends the whole string or fails:

```

local count, err = sock:send("PING\r\n")
assert(count, err)
assert(count == 6)

```

Babet loops over partial OS/OpenSSL writes, so a successful count equals #data. Empty and binary strings are accepted. send has no positional timeout and uses the socket default.

If a transport error happens after some bytes were written, no partial count is returned. Protocols needing safe retries must use application-level IDs or acknowledgements.

### sock:recv(count, timeout?)

Reads **at most** count bytes and may return fewer bytes while the connection remains open:

```

local chunk, err = sock:recv(4096, 5)

```

count must be a strict Lua integer from 1 through 16 MiB. This is not “read exactly N”. A helper can accumulate:

```

local function recv_exact(sock, wanted, timeout)
    local chunks, total = {}, 0
    while total < wanted do
        local chunk, err = sock:recv(wanted - total, timeout)
        if not chunk then return nil, err end
        chunks[#chunks + 1] = chunk
        total = total + #chunk
    end
    return table.concat(chunks)
end

```

Here the timeout is per recv call. Compute a monotonic application deadline when one budget must cover the whole helper.

EOF before data returns (nil, "closed"); expiration returns (nil, "timeout"); a handled signal returns (nil, "interrupted").

### sock:recv\_line(timeout?)

Reads through LF and returns the line without LF. A CR immediately before LF is also stripped:

```

hello\n    -> "hello"
hello\r\n  -> "hello"

```

The line limit is 8 MiB. Oversized lines return a line too long error and the protocol should generally be considered desynchronised.

EOF before LF returns three values:

```

local line, err, partial = sock:recv_line(5)
if not line and err == "closed" then
    print(partial) -- always a string, possibly empty
end

```

On timeout/interruption, bytes already consumed are retained internally and are delivered first by the next recv, recv\_line, or recv\_all.

### sock:recv\_all(timeout?, max\_bytes?)

Reads until remote EOF; EOF is the successful terminator:

```

local body, err = sock:recv_all(10)

```

Use this only when connection close frames the message. A persistent peer that keeps the connection open makes this call wait until timeout.

Memory limit:

- default: 64 MiB;
- strict positive integer;
- maximum: 2 GiB.

```
local body, err = sock:recv_all(10, 4 * 1024 * 1024)
```

Limit, timeout, and interruption failures expose no partial result, but already consumed bytes remain buffered. A later call can resume with a larger limit or new deadline:

```
local body, err = sock:recv_all(2, 1024)
if not body and err:find("max_bytes", 1, true) then
    body = assert(sock:recv_all(2, 4096))
end
```

## Mixing recv, recv\_line, and recv\_all

All three methods share one pending-byte buffer. Switching methods after a timeout does not lose or reorder the TCP stream:

```
-- Peer sent "abc" without LF.
local line, err = sock:recv_line(0.1)
assert(line == nil and err == "timeout")
local prefix = assert(sock:recv(3, 1))
assert(prefix == "abc")
```

Still prefer one framing strategy per protocol phase. Mixing helpers is useful for a line header followed by a fixed-size body, but the application must know the exact boundary.

Pending plaintext must be consumed before starttls; Babet rejects the upgrade otherwise.

## sock:peer() and sock:sockname()

Both return numeric address tables:

```
{ host = "127.0.0.1", port = 9000 }
```

- peer() returns the remote endpoint of a connected socket;
- sockname() returns the local endpoint and also works on listeners.

No reverse DNS is performed. peer() is invalid on a listening socket.

## sock:close() and garbage collection

close() is idempotent and returns (true, nil):

```
assert(sock:close())
assert(sock:close())
```

Other methods then report a closed socket. The userdata owns its FD and SSL object, so GC eventually cleans forgotten sockets, but GC timing is not a lifecycle guarantee. Close explicitly.

Created and accepted sockets are close-on-exec, so `babet.exec` children do not silently inherit them.

## Signals and interruption

A wait interrupted by a signal registered through `babet.signal` may return (nil, "interrupted") after the Lua callback is dispatched. Pending bytes from line/all reads are retained.

```

local stopping = false
babet.signal.handle("TERM", function() stopping = true end)
while not stopping do
    local client, err = server:accept(1)
    if client then
        client:close()
    elseif err ~= "timeout" and err ~= "interrupted" then
        error(err)
    end
end
end

```

## Complete examples

### Line-oriented echo server

```

local S = babet.socket
local server = assert(S.listen("127.0.0.1", 9000, 64))
assert(server:set_timeout(1))

while true do
    local client, err = server:accept()
    if client then
        assert(client:set_timeout(30))
        local line, read_err = client:recv_line()
        if line then
            assert(client:send("echo: " .. line .. "\n"))
        elseif read_err ~= "closed" then
            io.stderr:write(read_err, "\n")
        end
        client:close()
    elseif err ~= "timeout" then
        error(err)
    end
end
end

```

### Length-prefixed binary client

```

local sock = assert(babet.socket.connect("127.0.0.1", 9000, 3))
assert(sock:set_timeout(5))
assert(sock:send(string.pack(">I4", 5) .. "hello"))
local raw_len = assert(recv_exact(sock, 4, 5))
local len = string.unpack(">I4", raw_len)
local payload = assert(recv_exact(sock, len, 5))
print(payload)
sock:close()

```

### Response framed by EOF

```

local sock = assert(babet.socket.connect("127.0.0.1", 9001, 3))
assert(sock:send("request\n"))
local response = assert(sock:recv_all(10, 8 * 1024 * 1024))
print(#response)

```

## Error contract

Wrong positional types raise:

```
babet.socket.connect("localhost", "9000")
sock:send(42)
sock:recv(4096.0)
sock:set_timeout("5")
```

Invalid ranges and runtime failures return `(nil, err)`: invalid ports, backlog, timeouts, receive sizes, incompatible methods, closed sockets, DNS/TCP errors, and the stable typed states "timeout", "closed", and "interrupted".

Only compare complete strings for stable typed errors. System messages may vary across libc and kernel versions.

Internal C++ exceptions never cross into Lua. Allocation failure is converted to `(nil, "socket: out of memory")`; other internal exceptions receive a fixed literal diagnostic. This boundary covers every public `babet.socket` function and method.

## Security and limitations

- Raw TCP has no encryption or peer authentication.
- Set deadlines for peer-controlled input.
- Bind loopback unless remote access is required.
- TCP is a byte stream; send boundaries are not preserved.
- DNS is synchronous and outside the controlled connection deadline.
- No half-close, configurable keepalive, UDP, Unix sockets, or Lua event loop.
- One userdata should not be used concurrently; socket userdata cannot cross a WORKERS message boundary.
- `connect`, `listen`, `accept`, and `connect_tls` create the Lua userdata owner before acquiring an FD or OpenSSL object, so a Lua memory-error longjmp cannot abandon an unowned resource. `starttls` uses a separate guard and closes the stream if an exception occurs after the handshake has started.
- OOM exception safety is established by structural review, strict compilation, and static analysis rather than deterministic `bad_alloc` injection; the integration suite covers observable lifecycle and failure paths.

## babet.sqlite — embedded SQL database

`babet.sqlite` embeds SQLite 3.53.1 and exposes a compact but complete Lua API: connections, direct SQL execution, lazy iterators, reusable prepared statements, managed transactions, and explicit BLOB binding.

### Module contents

- [API](#)
  - [Opening and closing](#)
  - [Direct execution](#)
  - [Reusable prepared statements](#)
  - [Managed transactions](#)
  - [SQL parameters](#)
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## API

### Opening and closing

| Function                                    | Returns                  |
|---------------------------------------------|--------------------------|
| <code>babot.sqlite.open(path, opts?)</code> | db userdata   (nil, err) |
| <code>babot.sqlite.blob(data)</code>        | opaque BLOB wrapper      |
| <code>db:close()</code>                     | (true, nil) — idempotent |
| <code>db:in_transaction()</code>            | boolean   (nil, err)     |

`open` opens or creates a read/write database. The special `":memory:"` path creates a temporary in-memory database. The path must be a string without a NUL byte.

Optional `opts` fields:

| Field                     | Type                                       | Default |
|---------------------------|--------------------------------------------|---------|
| <code>wal</code>          | boolean — requests PRAGMA journal_mode=WAL | false   |
| <code>busy_timeout</code> | integer from 0 to 3600000 ms               | 0       |

`wal = true` requests WAL mode, but SQLite may keep another journal mode when WAL is not applicable, notably for `":memory:"`.

`busy_timeout` asks SQLite to retry for the requested duration when the database is locked. Zero reports SQLITE\_BUSY immediately.

`db:in_transaction()` returns true whenever the connection is inside a transaction, whether it was opened by `db:transaction()` or by manual SQL.

### Direct execution

| Function                            | Returns                                                |
|-------------------------------------|--------------------------------------------------------|
| <code>db:exec(sql)</code>           | (true, nil)   (nil, err) — multiple statements allowed |
| <code>db:exec(sql, params)</code>   | (true, nil)   (nil, err) — <b>one</b> statement        |
| <code>db:query(sql, params?)</code> | callable temporary stmt   (nil, err)                   |
| <code>stmt:close()</code>           | (true, nil) — idempotent                               |

**db:exec** Without `params`, `exec` accepts multiple semicolon-separated statements. They are prepared and executed in order. Execution stops on the first error; earlier statements remain committed unless they were inside an explicit transaction.

With `params`, exactly one statement is accepted. Trailing whitespace, semicolons, and SQL comments do not count as another statement.

A SELECT passed to `exec` is stepped to completion and its rows are ignored. Empty or comment-only SQL is a successful no-op with `exec(sql)` or `exec(sql, {})`.

**db:query** `query` prepares one statement immediately and returns a **single-use callable iterator**. Execution starts on the first iterator call:

```
for row in db:query("SELECT id, name FROM users ORDER BY id") do
  print(row.id, row.name)
end
```

Each row is a table keyed by column name. The iterator returns `nil` when it is exhausted and automatically finalizes its statement. `stmt:close()` releases it earlier, and garbage collection finalizes an iterator abandoned after `break`.

`query` also accepts DDL and DML. Such a statement runs on the first call and then ends without yielding a row.

Multiple statements are rejected. Empty or comment-only SQL returns an already exhausted iterator.

## Reusable prepared statements

| Function                             | Returns                                              |
|--------------------------------------|------------------------------------------------------|
| <code>db:prepare(sql)</code>         | <code>prepared</code>   <code>(nil, err)</code>      |
| <code>prepared:exec(params?)</code>  | <code>(true, nil)</code>   <code>(nil, err)</code>   |
| <code>prepared:query(params?)</code> | the same prepared userdata   <code>(nil, err)</code> |
| <code>prepared:reset()</code>        | <code>(true, nil)</code>   <code>(nil, err)</code>   |
| <code>prepared:close()</code>        | <code>(true, nil)</code> — idempotent                |
| <code>prepared:finalize()</code>     | alias of <code>close()</code>                        |

`db:prepare()` accepts exactly **one non-empty SQL statement**. Preparation is immediate, so an unknown table or column is reported before the first run.

A statement can then be reused as many times as needed:

```
local insert = assert(db:prepare(
    "INSERT INTO files(path, digest) VALUES(?, ?)"
))

for _, file in ipairs(files) do
    assert(insert:exec({ file.path, file.digest }))
end

assert(insert:finalize())
```

`prepared:exec()` resets the statement before binding, steps it to `SQLITE_DONE`, then clears its bindings. A `SELECT` is fully consumed and its rows are ignored.

`prepared:query()` resets and binds the statement, then returns **the same userdata**, whose metatable makes it callable:

```
local select_by_age = assert(db:prepare([
    SELECT id, name FROM users
    WHERE age >= ?
    ORDER BY id
]))

for row in select_by_age:query({ 18 }) do
    print(row.id, row.name)
end

for row in select_by_age:query({ 65 }) do
    print("senior", row.name)
end
```

Natural exhaustion automatically resets the statement and clears all bindings. After breaking early, either call `prepared:reset()`, start another `prepared:query(...)`, or call `prepared:exec(...)`. The latter two operations also reset the previous partial iteration and discard its remaining rows. A

prepared statement can have only one active iteration: reusing the same userdata from a nested loop therefore resets the outer loop.

Calling `prepared()` directly without an active `prepared:query()` simply returns `nil`; it never executes SQL with cleared bindings.

`prepared:reset()` aborts the current run and clears bindings. `close()` and `finalize()` permanently finalize the statement. Calling a closed prepared userdata directly returns `nil`; execution methods return `(nil, "sqlite: statement closed")`.

## Managed transactions

```
local ok, result = db:transaction(function(tx)
    assert(tx:exec(
        "INSERT INTO accounts(name, balance) VALUES(?, ?)",
        { "alice", 100 }
    ))

    assert(tx:exec(
        "INSERT INTO audit(message) VALUES(?)",
        { "account created" }
    ))

    return "created"
end, "immediate")
```

```
assert(ok, result)
print(result) -- "created"
```

Signature:

```
ok, ...callback_results = db:transaction(callback [, mode])
-- or
nil, err = db:transaction(callback [, mode])
```

The callback receives the database connection as its only argument. `mode` is one of:

- "deferred" — default;
- "immediate";
- "exclusive".

When the callback returns normally, Babet commits and returns true followed by every callback result. A normal `nil` or false result **does not request a rollback**. Only a Lua error triggers automatic rollback.

SQLite operations usually return `(nil, err)`, so use `assert` inside the callback to convert operational failures into Lua errors:

```
local ok, err = db:transaction(function(tx)
    assert(tx:exec("INSERT INTO unique_names VALUES(?)", { "alice" }))
    assert(tx:exec("INSERT INTO unique_names VALUES(?)", { "alice" }))
end)
```

*-- The second INSERT fails, assert raises, and both writes are rolled back.*

Callback errors are caught: Babet attempts `ROLLBACK` and returns `(nil, "sqlite: transaction callback failed: ...")` rather than rethrowing. A failed `COMMIT` also returns `(nil, err)` after a best-effort rollback. Any values returned by the callback are discarded in that case.

After a successful `BEGIN`, Babet explicitly owns the transaction until `COMMIT` or `ROLLBACK`. The Lua stack space required to call the callback is reserved before `BEGIN`; an internal C++ exception also

triggers an immediate rollback attempt. Callback diagnostics are formatted only after that attempt, so an unusual Lua error object cannot leave the connection in an intermediate state.

If the normal rollback fails, the returned message contains both the original failure and the rollback failure, followed by one final emergency rollback attempt. After an error, `db:in_transaction()` can be used to verify explicitly that the connection has returned to autocommit mode before continuing.

**Example: failure during commit** A deferred constraint may be checked only by COMMIT:

```
assert(db:exec("PRAGMA foreign_keys = ON"))
assert(db:exec([[
    CREATE TABLE parent(id INTEGER PRIMARY KEY);
    CREATE TABLE child(
        id INTEGER PRIMARY KEY,
        parent_id INTEGER NOT NULL,
        FOREIGN KEY(parent_id) REFERENCES parent(id)
            DEFERRABLE INITIALLY DEFERRED
    )
]]))

local ok, err = db:transaction(function(tx)
    assert(tx:exec(
        "INSERT INTO child(id, parent_id) VALUES(?, ?)",
        { 1, 999 }))
    return "this result will not be forwarded"
end)

assert(ok == nil)
assert(type(err) == "string")
assert(db:in_transaction() == false)
```

The INSERT is rolled back and the connection remains reusable when rollback succeeds.

**Example: reusing a prepared statement after rollback**

```
local insert = assert(db:prepare(
    "INSERT INTO log(id, value) VALUES(?, ?)"))

local ok, err = db:transaction(function()
    assert(insert:exec({ 1, "rolled back" }))
    error("intentional failure")
end)

assert(ok == nil and type(err) == "string")
assert(insert:exec({ 2, "kept" }))
insert:finalize()
```

Rollback resets SQLite's transactional state and the prepared statement can be reused normally afterwards.

Intentional limitations:

- nested `db:transaction()` helpers are rejected;
- the helper is rejected while a manual SQL transaction is already active;
- `db:close()` is rejected during the callback;
- the callback must not issue its own BEGIN, COMMIT, or ROLLBACK.

Savepoints remain available through raw SQL.

## SQL parameters

The following forms are supported by direct and prepared execution:

| Placeholder          | Lua value                                  |
|----------------------|--------------------------------------------|
| ?                    | params[1], params[2], and so on in ? order |
| :name, @name, \$name | params.name with the prefix removed        |

Positional and named parameters may be mixed:

```
assert(db:exec(
    "INSERT INTO events VALUES (?, :kind, ?)",
    { 42, 1700000000, kind = "start" }
))
```

The table must match all placeholders exactly. Missing or extra values, sparse numeric indices, non-integer numeric indices, and unsupported value types raise a Lua error. Explicit nil as the params argument is the same as omitting it.

Numbered ?NNN placeholders are outside the public contract.

Accepted bind values:

| Lua type                | SQLite value                       |
|-------------------------|------------------------------------|
| boolean                 | INTEGER 0 or 1                     |
| integer                 | INTEGER                            |
| non-integer number      | REAL                               |
| string                  | TEXT, including embedded NUL bytes |
| babet.sqlite.blob(data) | binary-safe BLOB                   |

## Explicit BLOB values

A normal Lua string is **always** bound with `sqlite3_bind_text`, even for a column with BLOB affinity. `babet.sqlite.blob(data)` explicitly marks the same bytes as a BLOB:

```
local insert = assert(db:prepare(
    "INSERT INTO assets(name, payload) VALUES(?, ?)"
))

assert(insert:exec({
    "icon",
    babet.sqlite.blob("\x89PNG\r\n\x1a\n...")
}))
```

The constructor accepts exactly one Lua string and returns an opaque userdata. Embedded NUL bytes are preserved. `babet.sqlite.blob("")` creates a real zero-byte BLOB rather than NULL.

The wrapper is only used for binding. Reading a BLOB column still returns a binary-safe Lua string. There is no explicit NULL sentinel yet; use the SQL literal NULL where needed.

## SQL text and NUL bytes

SQL text must be a string without a NUL byte. SQLite would treat a NUL as the end of the SQL string and could execute only a prefix, so Babet rejects it before preparation. SQL text is also limited to INT\_MAX bytes to match `sqlite3_prepare_v2`.

This restriction applies only to **SQL text**. Bound strings and BLOB values are binary-safe.

## Row reading and type mapping

| SQLite type | Lua type                                                              |
|-------------|-----------------------------------------------------------------------|
| INTEGER     | integer                                                               |
| REAL        | number (float)                                                        |
| TEXT        | string                                                                |
| BLOB        | binary-safe string, including an empty BLOB                           |
| NULL        | absent table key ( <code>row.col == nil</code> and absent from pairs) |

When multiple columns use the same name, the last column wins. Use aliases to preserve every value.

SQLite `length()` counts characters for TEXT and may stop at an embedded NUL. For diagnostic byte counts, use for example `length(CAST(col AS BLOB))`.

## Lifetime

`db:close()` is idempotent. After closing, new connection operations return `(nil, "sqlite: connection closed")`.

An iterator or prepared statement created **before** `db:close()` remains usable. Babet uses `sqlite3_close_v2`, so SQLite keeps a zombie connection until the last active statement is finalized.

```
local stmt = assert(db:prepare("SELECT id FROM users ORDER BY id"))
assert(db:close())
```

```
for row in stmt:query() do
    print(row.id)
end
```

```
stmt:finalize()
```

Temporary `db:query` iterators and prepared statements both finalize their handles from `__gc`. Explicit close/finalize is still preferable because it releases locks and resources immediately.

## Error contract

Errors fall into two groups:

- operational failures (open, preparation, step, transaction, closed handle) generally return `(nil, "sqlite: <description>")`;
- wrong types, invalid arity, and invalid parameter tables raise a Lua error.

Errors raised while calling a query iterator (`db:query` or `prepared:query`) are Lua errors because the iterator protocol cannot also return a separate `(nil, err)` result.

When no params table is provided, placeholders are detected before execution. Babet never lets SQLite silently bind them as NULL.

`db:transaction()` is the controlled exception: it catches callback Lua errors, rolls back, and converts them to `(nil, err)`.

## Complete examples

### Prepared inserts and BLOB values

```
local db = assert(babet.sqlite.open("state.db", {
    wal = true,
```

```

        busy_timeout = 2000,
    )))

assert(db:exec([[
    CREATE TABLE IF NOT EXISTS files (
        path TEXT PRIMARY KEY,
        digest BLOB NOT NULL
    )
]]))

local insert = assert(db:prepare([[
    INSERT INTO files(path, digest)
    VALUES(?, ?)
    ON CONFLICT(path) DO UPDATE SET digest = excluded.digest
]]))

for _, file in ipairs(files) do
    assert(insert:exec({
        file.path,
        babet.sqlite.blob(file.digest),
    }))
end

insert:finalize()

```

### Reusable query and managed transaction

```

local by_prefix = assert(db:prepare([[
    SELECT path, digest
    FROM files
    WHERE path LIKE ?
    ORDER BY path
]]))

local ok, count = db:transaction(function(tx)
    local n = 0
    for row in by_prefix:query({ "images/%" }) do
        assert(tx:exec(
            "INSERT INTO audit(path) VALUES(?)",
            { row.path }
        ))
        n = n + 1
    end
    return n
end, "immediate")

assert(ok, count)
print(count, "rows audited")

by_prefix:finalize()
assert(db:close())

```

### Functions not exposed

The following are not implemented:

- an explicit `sqlite.NULL` bind sentinel;

- `db:last_insert_rowid()` and `db:changes()`;
- `opts.readonly` and `opts.foreign_keys`;
- a nested savepoint helper;
- the `sqlite3_blob_open` streaming BLOB API;
- the `sqlite3_backup_init` backup API.

`last_insert_rowid`, `changes`, foreign keys, savepoints, and `VACUUM INTO` remain available through raw SQL. FTS5 and R-Tree are not enabled in the current embedded build.

## babet strings — splitting strings

Babet exposes one string-manipulation function: `babet.split`. It is registered directly on the `babet` table, without a sub-namespace, to preserve the historical API.

### Module contents

- [API](#)
- [Separator mode](#)
  - [Limiting the number of cuts](#)
- [Byte mode](#)
- [Binary strings and NUL bytes](#)
- [Empty subject](#)
- [Error contract](#)
- [Limits and design choices](#)

### API

```
local parts = babet.split(s [, sep [, max_splits]])
```

The function requires **one to three arguments** and returns exactly one value: a dense table indexed from 1 to `n`.

| Argument                | Type and behavior                                                                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>s</code>          | Required Lua string. Numbers are not converted implicitly. Embedded NUL bytes are preserved.                                                      |
| <code>sep</code>        | Optional Lua string. Omitted or empty: byte mode. Otherwise it must contain exactly <b>one byte</b> , used literally.                             |
| <code>max_splits</code> | Optional Lua integer greater than or equal to -1. -1 means unlimited and 0 prevents every cut. Used only when <code>sep</code> contains one byte. |

`nil` is not a placeholder for an optional argument: `babet.split("abc", nil)` raises an error. To supply `max_splits` in byte mode, pass `""` explicitly as the second argument.

### Separator mode

When `sep` contains exactly one byte, every occurrence of that byte produces a cut. The separator is **literal**: it is neither a Lua pattern nor a regular expression.

```
babet.split("a,b,c", ",")
-- { "a", "b", "c" }
```

```
babet.split("a.b.c", ".")
-- { "a", "b", "c" } -- the dot is not a wildcard
```

Empty fields are preserved at the beginning, between adjacent separators, and at the end:

```
babet.split(",a", ",")
-- { "", "a" }
```

```
babet.split("a,,b", ",")
-- { "a", "", "b" }
```

```
babet.split("a,", ",")
-- { "a", "" }
```

If the separator does not occur, the result contains the whole subject:

```
babet.split("hello", ",")
-- { "hello" }
```

### Limiting the number of cuts

`max_splits` counts **cuts**, not produced elements. The uncut remainder is stored verbatim in the last element.

```
babet.split("a,b,c,d", ",", 2)
-- { "a", "b", "c,d" }
```

```
babet.split("a,b,c", ",", 0)
-- { "a,b,c" }
```

```
babet.split("a,b,c", ",", -1)
-- { "a", "b", "c" }
```

Any value greater than the number of available separators has the same effect as `-1`.

### Byte mode

If `sep` is omitted or is `"`, the subject is split byte by byte. There is therefore **no default space separator**.

```
babet.split("ab c")
-- { "a", "b", " ", "c" }
```

```
babet.split("abc", "")
-- { "a", "b", "c" }
```

When supplied in this mode, `max_splits` is still validated but is not used:

```
babet.split("abc", "", 0)
-- { "a", "b", "c" }
```

This mode works on **bytes**, not Unicode code points. A multi-byte UTF-8 character is therefore returned as several one-byte strings:

```
local bytes = babet.split("é")
-- #bytes == 2 in UTF-8
-- table.concat(bytes) == "é"
```

Likewise, a multi-byte UTF-8 separator such as `"é"` is rejected because `sep` must contain zero or one byte. Use Lua code (`string.find`, `string.gmatch`, and so on) or a dedicated function to split on a multi-byte string.

## Binary strings and NUL bytes

The subject, separator, and output elements are handled with their exact Lua length. A NUL byte never truncates the data:

```
local parts = babet.split("a\0b,c", ",")
-- { "a\0b", "c" }
```

```
local parts2 = babet.split("a\0b", "\0")
-- { "a", "b" }
```

UTF-8 content also stays intact when split with an ASCII byte:

```
babet.split("été:ok", ":")
-- { "été", "ok" }
```

## Empty subject

Two different historical behaviors are preserved:

```
babet.split("", ",")
-- { "" }
```

```
babet.split("")
-- {}
```

```
babet.split("", "")
-- {}
```

In separator mode, finding zero separators means returning one element with the whole subject, even when it is empty. In byte mode, a zero-byte subject produces no elements.

## Error contract

Every failure is a raised Lua error; `split` has no `(nil, err)` return path.

The function raises in particular when:

- fewer than one or more than three arguments are supplied;
- `s` is not an actual Lua string;
- `sep` is present but is not an actual Lua string;
- `sep` contains more than one byte;
- `max_splits` is not a Lua integer;
- `max_splits` is less than -1.

Implicit number-to-string conversions are not accepted:

```
babet.split(123, ",")           -- error
babet.split("123", 2)           -- error
babet.split("a,b", ",", "1") -- error
```

## Limits and design choices

- This is not a CSV parser: quoting, escaping, and records are not handled.
- Multi-byte separators and Lua patterns are not supported.
- Empty fields are intentionally preserved.
- No join counterpart is exposed because Lua's standard `table.concat` already provides it.

# SYS - process, host, environment, executables, version, and Lua memory

The functions in this chapter are exposed directly on the global `babet` table. There is no `babet.sys` subtable: this flat namespace is historical and is part of the stable API.

SYS covers:

- the identity of the Babet process;
- the hostname and kernel information;
- reading and changing environment variables;
- locating an executable program;
- runtime version constants;
- memory used by the current Lua state.

The current working directory is also process-global state, but `currentDir` and `chdir` are documented under [FS - Current working directory](#), alongside path operations.

## Module contents

- [General conventions](#)
- [API overview](#)
- [Version constants](#)
- [Identify the process and host](#)
  - [pid](#)
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  - [What is and is not measured](#)
- [Error contract](#)
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## General conventions

### Flat functions

All functions are called directly from `babet`:

```
local pid = babet.pid()
local home = babet.env("HOME")
```

The following forms do not exist:

```
-- Incorrect: there is no babet.sys subtable
-- babet.sys.pid()
-- babet.sys.env("HOME")
```

## Strict strings

env, setenv, and which require actual Lua strings. Numbers are not converted automatically.

```
local ok, err = pcall(function()
    return babet.env(42)
end)
```

```
-- ok == false; err describes an invalid call
```

An embedded NUL byte is always rejected. This prevents a Lua string from being silently truncated when passed to a C API.

## Process-global state

The environment and current working directory belong to the **whole process**, not to a single script or worker.

- setenv changes the environment seen by the runtime and future programs launched through babet.exec;
- chdir changes how all relative paths are resolved;
- both mutations are permanently forbidden after the first babet.workers.spawn().

This restriction prevents data races between threads.

## API overview

| Function or constant           | Result                              |
|--------------------------------|-------------------------------------|
| babet.VERSION                  | runtime major.minor.patch string    |
| babet.VERSION_MAJOR            | integer major component             |
| babet.VERSION_MINOR            | integer minor component             |
| babet.VERSION_PATCH            | integer patch component             |
| babet.pid()                    | process PID as an integer           |
| babet.hostname()               | hostname, or (nil, err)             |
| babet.uname()                  | system table, or (nil, err)         |
| babet.which(name)              | executable path, or (nil, err)      |
| babet.env(name)                | variable value, or nil when missing |
| babet.setenv(name, value)      | (true, nil) or (nil, err)           |
| babet.getMemoryUsage()         | current Lua-state memory in bytes   |
| babet.getDetailedMemoryUsage() | two currently identical integers    |

## Version constants

The four constants describe the Babet binary running the script.

```
print(babet.VERSION)      -- for example "2.9.2"
print(babet.VERSION_MAJOR) -- for example 2
print(babet.VERSION_MINOR) -- for example 8
print(babet.VERSION_PATCH) -- for example 0
```

babet.VERSION matches the output of:

```
babet --version
```

The integer components let code compare versions without parsing a string.

```
local function version_at_least(major, minor, patch)
    local current = {
        babet.VERSION_MAJOR,
```

```

        babet.VERSION_MINOR,
        babet.VERSION_PATCH,
    }
    local wanted = { major, minor, patch }

    for i = 1, 3 do
        if current[i] ~= wanted[i] then
            return current[i] > wanted[i]
        end
    end
    return true
end
end

```

```
assert(version_at_least(2, 9, 0), "Babet >= 2.9.0 is required")
```

The following consistency is covered by tests:

```

assert(babet.VERSION == string.format(
    "%d.%d.%d",
    babet.VERSION_MAJOR,
    babet.VERSION_MINOR,
    babet.VERSION_PATCH
))

```

## Identify the process and host

### **babet.pid()**

Returns the Babet process ID as a strictly positive integer. The call cannot fail on a supported POSIX system.

```

local pid = babet.pid()
print("Babet PID:", pid)

```

Workers are threads in the same process. They therefore see the same PID as the main thread; pid() cannot identify an individual worker.

```

local main_pid = babet.pid()
local worker = assert(babet.workers.spawn([
    return babet.pid()
]))

```

```

local joined, worker_pid = worker:join()
assert(joined and worker_pid == main_pid)

```

### **babet.hostname()**

Returns the hostname configured for the machine.

```

local host, err = babet.hostname()
assert(host, err)
print("Host:", host)

```

Signature:

```
local host, err = babet.hostname()
```

Results:

- success: host is a non-empty string and err is nil;
- system failure: (nil, "hostname: ...").

The hostname is not necessarily a fully qualified DNS name. It may be a short locally configured name.

### **babet.uname()**

Returns the five main POSIX `uname(2)` fields in a table.

```
local info, err = babet.uname()
assert(info, err)
```

```
print("System :", info.sysname)
print("Node   :", info.nodename)
print("Kernel :", info.release)
print("Version:", info.version)
print("Machine:", info.machine)
```

Returned table:

```
{
  sysname = "Linux",
  nodename = "workstation",
  release = "6.12.0-arch1-1",
  version = "#1 SMP PREEMPT_DYNAMIC ...",
  machine = "x86_64",
}
```

| Field    | Meaning                                        |
|----------|------------------------------------------------|
| sysname  | operating system or kernel name                |
| nodename | network node name, often close to the hostname |
| release  | kernel release string                          |
| version  | detailed kernel build string                   |
| machine  | hardware architecture reported by the kernel   |

If `uname(2)` fails, the function returns `(nil, "uname: ...")`. This is very rare, but it is part of the real contract.

## **Locate an executable**

Signature:

```
local path, err = babet.which(name)
```

`name` must be a string without an embedded NUL byte.

### **PATH lookup**

When `name` contains no `/`, Babet scans the `PATH` environment variable from left to right and returns the first candidate that is:

- a regular file;
- executable by the current process.

```
local shell, err = babet.which("sh")
assert(shell, err)
print(shell) -- usually /usr/bin/sh or /bin/sh
```

The returned path is normalized to an absolute path whenever the system allows it.

To perform a simple availability check:

```

if babet.which("ffmpeg") then
    print("ffmpeg is available")
end

```

PATH is read at call time. A change made with setenv before workers is therefore visible to later calls.

```

local old_path = babet.env("PATH")
assert(babet.setenv("PATH", "/opt/my-app/bin:/usr/bin"))
local tool = babet.which("my-tool")

```

An empty PATH component, as in `:/usr/bin` or `/bin::usr/bin`, represents the current working directory, following the historical Unix convention.

## Direct path

When the argument contains `/`, PATH is not consulted. The path is tested directly.

```

local sh, err = babet.which("/bin/sh")
assert(sh, err)

```

A relative path containing `/` is accepted too:

```

local tool, err = babet.which("./bin/my-tool")
assert(tool, err)

```

A valid symlink to an executable regular file is accepted because inspection follows the target.

## Missing or non-executable program

When no valid executable is found:

```

local path, err = babet.which("missing-program")
-- path == nil
-- err == "which: 'missing-program' not found in PATH"

```

A file that exists but is not executable is treated as not found:

```

local path, err = babet.which("./script-without-x-bit")
assert(path == nil)
print(err)

```

The following are rejected too:

- directories;
- broken symlinks;
- non-regular files;
- files the current process cannot execute.

## Environment variables

### `babet.env(name)`

Reads a process environment variable.

```

local home = babet.env("HOME")
if home then
    print("HOME:", home)
end

```

Signature:

```

local value = babet.env(name)

```

Results:

- defined variable: its value as a string;
- missing variable: nil only, with no error message.

This enables the common idiom:

```
local port = babet.env("APP_PORT") or "8080"
```

name must be a string. The following forms raise a Lua error:

```
-- babet.env()           -- missing argument
-- babet.env(42)         -- no automatic conversion
-- babet.env({})         -- wrong type
-- babet.env("A\0B")     -- embedded NUL
```

### **babet.setenv(name, value)**

Creates an environment variable or replaces its existing value.

```
local ok, err = babet.setenv("APP_MODE", "production")
assert(ok, err)
assert(babet.env("APP_MODE") == "production")
```

A later call overwrites the value:

```
assert(babet.setenv("APP_MODE", "test"))
assert(babet.env("APP_MODE") == "test")
```

name and value must both be actual Lua strings.

Caller errors, which raise a Lua error:

```
-- babet.setenv()
-- babet.setenv("APP_MODE")
-- babet.setenv(42, "test")
-- babet.setenv("APP_MODE", 42)
```

Runtime validation errors return (nil, err):

```
local ok, err = babet.setenv("BAD=NAME", "x")
-- ok == nil; err reports an invalid name
```

An empty name, a name containing =, or an embedded NUL byte is rejected. A NUL byte in the value is rejected as well.

### **Empty value versus missing variable**

An empty value is still defined:

```
assert(babet.setenv("APP_OPTION", ""))
assert(babet.env("APP_OPTION") == "")
```

It is therefore different from a missing variable:

```
local value = babet.env("MISSING_VARIABLE")
assert(value == nil)
```

Babet does not currently expose unsetenv. A variable cannot be removed with setenv(name, nil): nil is a wrong type and the call raises a Lua error.

### **Interaction with workers**

setenv is allowed only before the first workers.spawn().

```
assert(babet.setenv("APP_MODE", "production"))

local worker = assert(babet.workers.spawn([[
    return babet.env("APP_MODE")
]]))
```

```
local joined, value = worker:join()
assert(joined and value == "production")
```

After the first spawn, even when the worker has already finished:

```
local ok, err = babet.setenv("APP_MODE", "test")
-- ok == nil
-- err explains that the shared environment can no longer be changed
```

The restriction lasts for the whole process, including when the first spawn failed. Prepare PATH, application variables, and the current directory before starting workers.

## Lua VM memory

Both functions first perform a full Lua garbage-collection cycle. The result therefore reflects memory still in use after collection, and the call may cause a pause proportional to the size of the Lua state.

### **babet.getMemoryUsage()**

Returns the number of bytes currently accounted for by the memory manager of the current Lua state.

```
local bytes = babet.getMemoryUsage()
print(string.format("Lua uses %d bytes", bytes))
```

The result is a positive integer.

Example before/after comparison:

```
local before = babet.getMemoryUsage()

local values = {}
for i = 1, 10000 do
    values[i] = string.rep("x", 100)
end
```

```
local after = babet.getMemoryUsage()
print("Lua change:", after - before, "bytes")
```

This is not a precise profiling tool: allocator behavior, string interning, and GC optimizations may change the result.

### **babet.getDetailedMemoryUsage()**

Returns two integers:

```
local used, total = babet.getDetailedMemoryUsage()
```

In the current version, both values are **strictly identical**:

```
assert(used == total)
```

The second return value is kept only for API stability. Lua does not expose a separate reliable measurement of fragmentation or reserved-but-unused memory.

New code that needs one measurement should prefer:

```
local bytes = babet.getMemoryUsage()
```

## What is and is not measured

The measurement covers memory managed by **the Lua state performing the call**.

It does not represent:

- total process RSS;
- the native stack;
- C++ runtime allocations;
- OpenSSL, SQLite, HTTP, or socket buffers;
- executable code and shared libraries;
- Lua states belonging to other workers.

Inside a worker, the function measures only that worker's isolated Lua state. To measure the whole process, use a system facility such as `/proc`, `ps`, `smem`, `Valgrind`, or an appropriate profiler.

## Error contract

| Situation                                                             | Behavior                                                                                                |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| missing argument or wrong type                                        | Lua error, recoverable with <code>pcall</code>                                                          |
| NUL in <code>env</code> , <code>setenv</code> , or <code>which</code> | Lua error for <code>env</code> , <code>(nil, err)</code> for <code>setenv</code> and <code>which</code> |
| missing variable in <code>env</code>                                  | <code>nil</code> only                                                                                   |
| executable not found                                                  | <code>(nil, err)</code>                                                                                 |
| <code>hostname</code> or <code>uname</code> system error              | <code>(nil, err)</code>                                                                                 |
| invalid <code>setenv</code> or mutation after a worker                | <code>(nil, err)</code>                                                                                 |
| <code>pid</code>                                                      | always an integer                                                                                       |
| memory functions                                                      | always one or two integers after a full GC                                                              |

Handling a runtime failure:

```
local tool, err = babet.which("optional-tool")
if not tool then
    print("Feature disabled:", err)
end
```

Handling an invalid call:

```
local ok, err = pcall(function()
    return babet.env(42)
end)

if not ok then
    print("Programming error:", err)
end
```

## Design decisions and limits

- Historical functions remain directly under `babet` to avoid breaking existing scripts.
- `which` returns the first executable found, not every candidate.
- `which` observes the process `PATH` and current-process permissions; its result may differ between users or environments.
- `env` distinguishes a missing variable (`nil`) from a variable defined as an empty string (`""`).
- `setenv` always overwrites an existing value.
- No `unsetenv` function or complete environment dump is exposed.

- `pid` is the process PID shared by all workers.
- Memory helpers deliberately run a full GC and do not measure the whole process.
- For system accounts, see [User - system users](#).

## babet tables — merging and copying Lua tables

Two historical helpers live directly on the `babet` table:

- `babet.mergeTables(...)` builds a new table from several sources;
- `babet.deepCopyTable(t)` recursively copies **values** that are tables.

They operate on entries actually stored in Lua tables. The `__pairs` and `__index` metamethods therefore do not provide virtual fields to merge or copy.

### Module contents

- [API](#)
- `babet.mergeTables(t1, t2, ...)`
  - Positive integer keys: concatenation
  - Every other key: last writer wins
  - Shallow merge
- `babet.deepCopyTable(t)`
  - Values and shared graph structure
  - Keys are not copied
  - Metatables and raw traversal
  - Maximum depth
- [Error contract](#)
- [Limits and design choices](#)
- [Not in v1](#)

### API

| Function                                    | Returns       |
|---------------------------------------------|---------------|
| <code>babet.mergeTables(t1, t2, ...)</code> | one new table |
| <code>babet.deepCopyTable(t)</code>         | one new table |

Both functions return exactly one value on success. They never return `(nil, err)`: call errors and depth errors are raised.

### **babet.mergeTables(t1, t2, ...)**

At least two arguments are required and every argument must be a table. The API sets no additional arity limit beyond Lua's general limits.

Sources are not mutated. The result is a plain table without a metatable, even when source tables have one.

### **Positive integer keys: concatenation**

Every key represented by Lua as an integer  $\geq 1$  is treated as a list position. This includes a key written as `2.0`, which Lua canonicalizes to an integer.

Within each source, such keys are visited in ascending numeric order, then their values are appended to the result. Holes and original indices are compacted:

```

local r = babet.mergeTables(
  { [4] = "d", [2] = "b" },
  { [7] = "g", [1] = "a" }
)

-- r == { "b", "d", "a", "g" }

```

The same list position appearing in several sources is not overwritten: every value is appended.

### Every other key: last writer wins

String, boolean, table, function, thread, userdata and light userdata keys, along with floating, zero and negative numeric keys, keep their original identity. A later source overwrites an earlier value for the same key:

```

local key = {}
local r = babet.mergeTables(
  { mode = "safe", [key] = 1, [0] = "a" },
  { mode = "fast", [key] = 2, [0] = "b" }
)

-- r.mode == "fast"
-- r[key] == 2
-- r[0] == "b"

```

### Shallow merge

Values are never copied. A subtable placed in the result is the same table as in the source:

```

local nested = { enabled = false }
local r = babet.mergeTables({ nested = nested }, {})

r.nested.enabled = true
-- nested.enabled == true

```

To obtain an independent graph for table values afterwards, use:

```

local isolated = babet.deepCopyTable(
  babet.mergeTables(defaults, overrides)
)

```

### **babet.deepCopyTable(t)**

The function requires exactly one table. It creates a new table for the root and for every **value** that is itself a table.

### Values and shared graph structure

Cycles reached through values are supported and graph identity is preserved:

```

local shared = { value = 42 }
local source = { a = shared, b = shared }
source.self = source

local copy = babet.deepCopyTable(source)

-- copy ~= source
-- copy.a ~= shared

```

```
-- copy.a == copy.b
-- copy.self == copy
```

Non-table values — numbers, strings, booleans, functions, threads, userdata and light userdata — are reused as-is.

## Keys are not copied

Every key keeps its original value and identity. In particular, a table used as a key remains the source table:

```
local key = { id = 1 }
local source = {
    [key] = "value",
    key_as_value = key,
}
local copy = babet.deepCopyTable(source)

-- copy[key] == "value"           -- original key retained
-- copy.key_as_value ~= key      -- table value copied
-- copy[copy.key_as_value] == nil
```

The function is therefore a deep copy of **table values**, not a structural copy of table keys.

## Metatables and raw traversal

The actual metatable of every copied table is attached to its copy **by reference**. It is not duplicated. Mutating that metatable therefore affects both source and copy.

Only actually stored pairs are traversed. A field returned through `__index` or invented by `__pairs` is not materialized in the copy. Once the shared metatable is attached, its behavior naturally remains active on the copy.

## Maximum depth

The root is at depth 0. Up to **75 descents into new table values** are accepted, for at most 76 tables along one root-inclusive path. A 76th descent into a previously unseen table raises:

Table is too deep to copy (max depth 75 exceeded)

A reference to an already copied table does not create a new level, so a cycle may close exactly at the boundary.

## Error contract

Errors are raised through the Lua API for:

- fewer than two arguments to `mergeTables`;
- any non-table argument to `mergeTables`;
- an argument count other than one for `deepCopyTable`;
- a non-table argument to `deepCopyTable`;
- exceeding the maximum depth.

```
local ok, err = pcall(babet.deepCopyTable, 42)
-- ok == false; err contains "Argument must be a table"
```

## Limits and design choices

- `mergeTables` is not a deep merge: a later subtable replaces the earlier reference wholesale.
- Source metatables are ignored by `mergeTables`; its result has none.

- `deepCopyTable` shares metatables and table keys with the source.
- A weak metatable (`__mode`) remains weak on the copy because it is shared.
- Map-key order is undefined. Only the positive-integer ordering of `mergeTables` is guaranteed.
- The ordered list-key selection in `mergeTables` favors safety across Lua errors; its cost is quadratic per source in the number of positive integer keys. This helper is therefore intended mainly for moderately sized tables.

## Not in v1

Features such as `deepMergeTables` or deep structural comparison could be added separately. They are not part of the current API.

## babet time - clocks, sleep, dates and durations

Babet exposes two sub-second clocks, interruptible sleep, and small utilities for ISO 8601 timestamps and compact durations.

The historical functions remain available at the root:

- `babet.monotonic()`;
- `babet.now()`;
- `babet.sleep(amount [, unit])`.

The same functions are available under `babet.time`, together with four additional utilities: `iso`, `parse_iso`, `parse_duration`, and `format_duration`.

## Module contents

- [API](#)
- `monotonic()` and `now()`
- `sleep(amount [, unit])`
- `babet.time.iso([ts])`
- `babet.time.parse_iso(text)`
- `babet.time.parse_duration(text)`
- `babet.time.format_duration(seconds)`
- [Error contract summary](#)
- [Limits and design choices](#)

## API

| Function                                         | Result                                                                                               |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <code>babet.monotonic()</code>                   | number, or exceptionally ( <code>nil</code> , <code>err</code> ) if <code>clock_gettime</code> fails |
| <code>babet.now()</code>                         | number, or exceptionally ( <code>nil</code> , <code>err</code> ) if <code>clock_gettime</code> fails |
| <code>babet.sleep(amount [, unit])</code>        | ( <code>true</code> , <code>nil</code> ) or ( <code>nil</code> , <code>err</code> )                  |
| <code>babet.time.iso([ts])</code>                | UTC string                                                                                           |
| <code>babet.time.parse_iso(text)</code>          | (integer, <code>nil</code> ) or ( <code>nil</code> , "parse_iso: ...")                               |
| <code>babet.time.parse_duration(text)</code>     | (integer, <code>nil</code> ) or ( <code>nil</code> , "parse_duration: ...")                          |
| <code>babet.time.format_duration(seconds)</code> | compact string, for example "1d2h3m4s"                                                               |

The `babet.time.now`, `babet.time.monotonic`, and `babet.time.sleep` aliases use exactly the same bindings and contract as their root names.

## monotonic() and now()

```
local started = babet.monotonic()
do_something()
local elapsed = babet.monotonic() - started
print(string.format("elapsed: %.3f s", elapsed))
```

```
local timestamp = babet.now()
print(timestamp)
```

monotonic() uses CLOCK\_MONOTONIC. Its value starts at an arbitrary origin, usually related to system boot. It is intended for measuring durations because it does not suffer wall-clock jumps.

now() uses CLOCK\_REALTIME. It returns POSIX time in seconds since 1970-01-01 UTC, including a fractional part. It is appropriate for timestamps, but not for measuring durations because the wall clock may be adjusted.

Both functions:

- accept no arguments;
- return one number in normal operation;
- may theoretically return (nil, err) if the clock\_gettime system call fails.

## sleep(amount [, unit])

```
assert(babet.sleep(250, "ms"))
assert(babet.time.sleep(0.5))      -- 0.5 second
assert(babet.sleep(1000, "us"))
```

amount must be an actual Lua number, finite and non-negative. Floating-point values are accepted. A numeric string such as "100" is rejected.

unit must be an actual Lua string:

| Unit | Meaning              |
|------|----------------------|
| "s"  | seconds, the default |
| "ms" | milliseconds         |
| "us" | microseconds         |

A zero duration succeeds immediately. An unknown textual unit keeps the historical behavior:

```
local ok, err = babet.sleep(1, "minutes")
-- ok == nil
-- err == "Invalid time unit"
```

If a signal managed by `babet.signal` arrives during the wait, its Lua callback is dispatched and `sleep` then returns (nil, "interrupted"). An unmanaged signal that causes EINTR does not shorten the wait: Babet resumes with the remaining time.

Any other nanosleep failure returns (nil, "sleep: <description>").

## babet.time.iso([ts])

iso formats a Unix second in UTC:

```
print(babet.time.iso(0))
-- 1970-01-01T00:00:00Z

print(babet.time.iso())
-- current time, at whole-second precision
```

Without an argument, `iso()` uses the current time. With an argument:

- `ts` must be an actual Lua number;
- an integer or finite float is accepted;
- a fractional value is rounded toward negative infinity;
- the value must fit in a signed 64-bit integer.

The rounding matters before the Unix epoch:

```
babet.time.iso(0.9)    -- 1970-01-01T00:00:00Z
babet.time.iso(-0.5)   -- 1969-12-31T23:59:59Z
```

For ordinary years, the result has the form `YYYY-MM-DDTHH:MM:SSZ`. Extreme valid `int64` timestamps may produce a signed year wider than four digits. That extended result is not necessarily accepted by `parse_iso`, whose grammar requires an exactly four-digit year.

`iso` applies neither locale nor local timezone and does not retain fractional seconds.

### **babet.time.parse\_iso(text)**

The parser intentionally accepts a small, deterministic subset:

```
YYYY-MM-DDTHH:MM:SSZ
YYYY-MM-DD HH:MM:SSZ
YYYY-MM-DDTHH:MM:SS.fraction+HH:MM
YYYY-MM-DDTHH:MM:SS.fraction-HH:MM
```

Exact rules:

- `text` must be an actual Lua string;
- the year contains exactly four digits and ranges from 0001 to 9999;
- the Gregorian calendar is validated, including leap years;
- the separator is uppercase `T` or one space;
- hours `00`..23, minutes `00`..59, seconds `00`..59;
- leap seconds written as `:60` are rejected;
- a fraction is optional, but the dot must be followed by at least one digit; all fraction digits are ignored, without rounding;
- a timezone is required: uppercase `Z`, `+HH:MM`, or `-HH:MM`;
- accepted offsets range from `00:00` through `23:59`;
- `+0200`, `+02`, lowercase `z`, leading or trailing whitespace, and trailing data are rejected;
- an embedded NUL byte is never truncated: it makes the string invalid.

Example:

```
local ts = assert(babet.time.parse_iso(
    "2026-06-17T10:00:00+02:00"))

assert(ts == babet.time.parse_iso("2026-06-17T08:00:00Z"))
```

A syntactically or calendrically invalid string returns `(nil, "parse_iso: ...")`. A wrong type or argument count raises a Lua error.

### **babet.time.parse\_duration(text)**

The format is a whitespace-free sequence of number + unit chunks:

```
assert(babet.time.parse_duration("45s") == 45)
assert(babet.time.parse_duration("1h30m") == 5400)
assert(babet.time.parse_duration("2d12h") == 216000)
```

Available units:

| Unit | Seconds |
|------|---------|
| d    | 86400   |
| h    | 3600    |
| m    | 60      |
| s    | 1       |

Each unit may occur at most once and units must be in strict order  $d > h > m > s$ . Signs, decimal values, whitespace, and the units *w*, *ms*, *us*, or *ns* are rejected. An embedded NUL byte is not truncated and also produces a parse error.

Components are not normalized: "1h90m" is valid and equals 9000 seconds. Leading zeroes and ordered zero-valued chunks are also accepted, for example "0001s" and "0d0h0m0s".

The total must remain between 0 and `math.maxinteger` on Babet's 64-bit configuration. Overflow returns a parse error.

An invalid string returns `(nil, "parse_duration: ...")`. A wrong type or argument count raises a Lua error.

### **`babet.time.format_duration(seconds)`**

```
babet.time.format_duration(0)      -- "0s"
babet.time.format_duration(90)     -- "1m30s"
babet.time.format_duration(90061)  -- "1d1h1m1s"
```

`seconds` must be an actual Lua number with a non-negative integer value. A floating-point value such as 3.0 is accepted; 3.7, "3", NaN, infinity, and negative values raise a Lua error.

The output is Babet's canonical representation:

- units in d, h, m, s order;
- zero components omitted;
- zero represented as "0s".

For every valid integer  $n \geq 0$ :

```
assert(babet.time.parse_duration(
    babet.time.format_duration(n)) == n)
```

The textual reverse is not guaranteed because the parser also accepts non-normalized forms such as "90m" or "0h".

### **Error contract summary**

| Situation                                                       | Behavior                                                                          |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Wrong type or wrong argument count                              | Lua error                                                                         |
| NaN, infinity, negative duration, or out-of-range numeric value | Lua error                                                                         |
| Unknown textual unit in sleep                                   | <code>(nil, "Invalid time unit")</code>                                           |
| Managed signal during sleep                                     | <code>(nil, "interrupted")</code> after callback dispatch                         |
| Invalid ISO or duration text                                    | <code>(nil, "parse_iso: ...")</code> or <code>(nil, "parse_duration: ...")</code> |
| Clock or sleep system failure                                   | <code>(nil, err)</code>                                                           |

### **Limits and design choices**

- Use `monotonic()` for durations and `now()` for timestamps.

- `iso` is a fixed UTC formatter, not a general replacement for `os.date`.
- No named timezone, daylight-saving rule, locale, or calendar arithmetic is provided.
- ISO fractions are discarded and are not emitted by `iso`.
- `parse_duration` works with whole seconds only.
- `CLOCK_BOOTTIME`, negative durations, and week/month/year units are not exposed.

## TLS — direct encryption, STARTTLS, certificates, and SNI

Babet TLS uses the same userdata as `babet.socket`. After the handshake, `send`, `recv`, `recv_line`, `recv_all`, `set_timeout`, `peer`, `sockname`, and `close` behave like their plain-TCP versions.

Two modes are available:

- `babet.socket.connect_tls()` starts TLS on the first application byte;
- `sock:starttls()` upgrades an existing TCP connection after a plaintext protocol negotiation.

The implementation provides TLS 1.2 minimum, certificate verification by default, hostname checks, SNI independent from verification, system and per-call CA trust, shared TCP+handshake deadlines, and fail-closed STARTTLS. It does not expose TLS servers, client certificates, ALPN, public-key pinning, or peer-certificate inspection.

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- [API overview](#)
- [TLS options](#)
- [Direct TLS connection](#)
- [STARTTLS upgrade](#)
- [Verification, hostname, and SNI](#)
- [Certificate authorities](#)
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### Core conventions

#### Verification is on by default

```
local sock = assert(babet.socket.connect_tls("example.com", 443, {
    timeout = 10,
}))
```

`verify = false` explicitly disables both chain and hostname verification. It may be useful in controlled tests but encryption without authentication does not prevent an active man-in-the-middle attack.

#### Same stream API after handshake

```
local sock = assert(babet.socket.connect_tls("irc.example.net", 6697, {
    timeout = 10,
}))
assert(sock:set_timeout(120))
assert(sock:send("PING :hello\r\n"))
local line = assert(sock:recv_line())
```

Handshake timeout and later I/O timeout are separate.

### SNI is not certificate verification

SNI selects a TLS virtual host. Verification checks trust and reference identity. Babet sends DNS SNI whenever a DNS hostname is available, including when `verify = false`.

### API overview

```
local sock, err = babet.socket.connect_tls(host, port, opts?)
local ok, err = sock:starttls(opts?)

{
    verify = true,
    hostname = "example.com",
    ca_cert = "/path/ca-bundle.pem",
    ca_path = "/path/certs",
    min_version = "1.2",
    timeout = 10,
}
```

| Field       | Type               | Default             | Meaning                         |
|-------------|--------------------|---------------------|---------------------------------|
| verify      | strict boolean     | true                | chain and hostname verification |
| hostname    | string             | connect_tls host    | reference name and DNS SNI      |
| ca_cert     | string             | no custom file      | extra PEM trust file            |
| ca_path     | string             | no custom directory | OpenSSL hashed CA directory     |
| min_version | "1.2" or "1.3"     | "1.2"               | minimum protocol version        |
| timeout     | finite number >= 0 | 0                   | handshake budget in seconds     |

Unknown fields are currently ignored; misspellings do not fail.

### TLS options

`verify` must be an actual boolean. `hostname`, `ca_cert`, `ca_path`, and `min_version` must be strings without NUL. `timeout` must be finite and non-negative; a positive value below 1 ms is rounded up.

`hostname` has two roles:

1. OpenSSL reference identity when verification is enabled;
2. SNI value when it is a DNS name.

Connect to an IP while validating a DNS certificate:

```
local sock = assert(babet.socket.connect_tls("203.0.113.10", 443, {
    hostname = "api.example.com",
    timeout = 10,
}))
```

An IP literal is not sent as SNI. `ca_cert` and `ca_path` augment trust for that connection; they are not exclusive pinning.

`min_version = "1.3"` raises the minimum. It does not cap the maximum.

## **babet.socket.connect\_tls(host, port, opts?)**

Establishes TCP then performs a client handshake:

```
local sock, err = babet.socket.connect_tls("example.com", 443, {
    timeout = 10,
})
assert(sock, err)
```

Without `opts.hostname`, the `host` argument is used for verification and DNS SNI. For an IP connection to a DNS certificate, supply the name explicitly.

After synchronous DNS resolution, one deadline covers every TCP address attempt and the TLS handshake. The handshake does not receive a fresh budget. DNS itself remains outside this controlled deadline.

`opts.timeout` is not retained as the socket I/O default:

```
local sock = assert(babet.socket.connect_tls(host, port, { timeout = 10 }))
assert(sock:set_timeout(30))
```

## **sock:starttls(opts?)**

Upgrades an existing connected plain TCP socket in place:

```
local ok, err = sock:starttls({
    hostname = "mail.example.com",
    timeout = 10,
})
```

On success it returns exactly `(true, nil)`.

The socket must be open, connected, not already TLS, and have no pending plaintext buffered by Babet. `starttls` does not remember the `hostname` passed to the earlier TCP constructor, so `hostname` is mandatory when `verify = true`.

Babet does not send a protocol STARTTLS command. The script must:

1. negotiate in plaintext;
2. request STARTTLS;
3. validate the positive protocol reply;
4. call `starttls`;
5. resume the protocol over TLS, often with a new greeting.

If a prior line/all read timed out after consuming plaintext, finish consuming or otherwise resolve that plaintext before upgrading. Babet rejects STARTTLS while the shared pending buffer is non-empty.

## **Verification, hostname, and SNI**

| Connection      | verify | hostname        | Identity check | SNI         |
|-----------------|--------|-----------------|----------------|-------------|
| DNS example.com | true   | absent          | example.com    | example.com |
| IP              | true   | api.example.com | DNS name       | DNS name    |
| IP              | true   | absent          | IP identity    | none        |
| DNS             | false  | absent          | none           | DNS host    |
| IP              | false  | DNS name        | none           | DNS name    |
| IP              | false  | absent          | none           | none        |

Providing `hostname` can therefore still be required by a virtual host even when verification is disabled.

## Certificate authorities

The socket TLS context attempts OpenSSL defaults, OpenSSL environment variables, and several known Linux/BSD CA locations. Per-call `ca_cert` and `ca_path` are then added.

Custom trust is isolated per connection:

```
local a = assert(babet.socket.connect_tls("internal.example", 443, {
    ca_cert = "/tmp/test-root.pem",
    timeout = 5,
}))
a:close()
-- Later calls do not retain /tmp/test-root.pem.
```

`ca_path` must follow OpenSSL's hashed-directory conventions. A CA file trusts certificates issued by that CA; it is not a leaf-certificate/public-key pin. Babet does not expose the peer certificate for custom pinning.

## TLS versions

Protocols older than TLS 1.2 are always rejected.

- default minimum: TLS 1.2;
- optional minimum: TLS 1.3;
- TLS 1.3 is negotiated automatically with the default when supported.

No maximum-version option is exposed.

## Timeouts

For `connect_tls`, `opts.timeout` covers TCP plus handshake after DNS. For `starttls`, it covers only the upgrade handshake and does not inherit `sock:set_timeout()`.

After success, configure I/O separately:

```
assert(sock:set_timeout(30))
local line, err = sock:recv_line()
```

TLS sockets remain non-blocking internally and OpenSSL `WANT_READ/WANT_WRITE` is coordinated with one poll deadline.

## Socket state after failure

`connect_tls` returns no socket and closes all intermediate resources.

Pre-handshake `STARTTLS` validation/configuration failures leave the plain TCP socket usable: invalid options, missing hostname, pending plaintext, context setup, or pre-handshake `fcntl` failure.

Once `SSL_connect` begins, any failure closes the socket: timeout, interruption, certificate failure, alert, protocol, or I/O error. A `ClientHello` may already have been sent and peer bytes consumed, so falling back to plaintext would be unsafe. This is intentionally fail-closed.

## Complete examples

### Raw HTTPS request

For normal HTTP use `babet.http`; this demonstrates stream use:

```
local sock = assert(babet.socket.connect_tls("example.com", 443, {
    timeout = 10,
}))
assert(sock:set_timeout(10))
```

```

assert(sock:send(
  "GET / HTTP/1.1\r\n" ..
  "Host: example.com\r\n" ..
  "Connection: close\r\n\r\n"
))
local response = assert(sock:recv_all(nil, 8 * 1024 * 1024))
print(response)
sock:close()

```

### IP connection with DNS identity

```

local sock = assert(babet.socket.connect_tls("203.0.113.20", 443, {
  hostname = "api.example.com",
  ca_cert = "/etc/myapp/ca.pem",
  timeout = 5,
}))

```

### Simplified SMTP STARTTLS

```

local sock = assert(babet.socket.connect("mail.example.com", 587, 5))
assert(sock:set_timeout(10))
assert(sock:recv_line():match("^220"))
assert(sock:send("EHLO client.example\r\n"))
-- Parse the full multiline EHLO response and require STARTTLS.
assert(sock:send("STARTTLS\r\n"))
assert(sock:recv_line():match("^220"))
assert(sock:starttls({
  hostname = "mail.example.com",
  timeout = 10,
}))
assert(sock:send("EHLO client.example\r\n"))

```

A complete SMTP client must correctly parse every multiline reply and prevent STARTTLS downgrade.

### Controlled self-signed test

```

local sock = assert(babet.socket.connect_tls("127.0.0.1", 8443, {
  verify = false,
  hostname = "dev.local", -- SNI is still sent
  timeout = 3,
}))

```

Do not use this trust policy in production.

### Error contract

Wrong positional types raise; invalid values and runtime failures return `(nil, err)`: empty host, bad port/options, CA loading, DNS/TCP, timeout, interruption, handshake, certificate/hostname verification, and STARTTLS preconditions.

OpenSSL wording varies. Compare stable states such as "timeout" and "interrupted"; log full messages for diagnosis.

### Security and limitations

- Keep `verify = true` in production.

- Validate against a trusted expected hostname, never one supplied by the peer.
- STARTTLS must be required by application policy to prevent stripping.
- No ALPN, client certificate, TLS server, peer-certificate access, custom cipher policy, or application OCSP/CRL policy.
- Protect and rotate private CA files.

## babet.toml - TOML decoding

The `babet.toml` module uses **toml++ 3.4.0** and decodes documents that conform to **TOML 1.0.0**. Optional extensions outside the TOML 1.0 profile supported by this toml++ version are disabled.

The v1 API only decodes a Lua string. It does not read a file for you and it does not provide an encoder.

### Module contents

- [API](#)
- [Type mapping](#)
  - [Integers and floats](#)
- [Strings, UTF-8, and keys](#)
- [Arrays and tables](#)
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### API

| Function                             | Contract                       |
|--------------------------------------|--------------------------------|
| <code>babet.toml.decode(text)</code> | Returns table, nil or nil, err |

`decode` requires exactly **one argument**, and it must be an actual Lua string. A number is not implicitly converted to text, and extra arguments are rejected.

To read a file:

```
local file = assert(io.open("config.toml", "rb"))
local text = assert(file:read("*a"))
file:close()

local config, err = babet.toml.decode(text)
if not config then
    error(err)
end
```

A TOML document always has a table at its root. An empty string or a document containing comments only therefore returns an empty Lua table.

### Type mapping

| TOML                                 | Lua                              |
|--------------------------------------|----------------------------------|
| string                               | string                           |
| signed 64-bit integer                | Lua integer                      |
| float                                | Lua number                       |
| boolean                              | boolean                          |
| array                                | 1..n sequence table              |
| table, inline table, or dotted table | table with string keys           |
| array of tables                      | sequence table containing tables |
| local date/time and date-time        | normalized string                |

TOML has no value equivalent to Lua nil or JSON null.

## Integers and floats

TOML integers cover the full signed 64-bit range, from `math.mininteger` to `math.maxinteger`, and are preserved exactly. Decimal, hexadecimal, octal, and binary notation all produce a Lua integer. A value outside that range is a parse error.

Floats become Lua numbers. TOML also accepts these special values:

```
local values = assert(babet.toml.decode([[
positive = inf
negative = -inf
invalid = nan
]]))

assert(values.positive == math.huge)
assert(values.negative == -math.huge)
assert(values.invalid ~= values.invalid) -- NaN property
```

As in Lua, NaN compares equal to no value, including itself.

## Strings, UTF-8, and keys

The TOML document must contain valid UTF-8. Basic, literal, and multiline strings are decoded before they are returned to Lua: TOML escape sequences become their corresponding characters.

Lua strings are binary-safe. A TOML sequence such as `\u0000` may therefore produce a NUL byte in a value **or in a quoted key**, without truncation:

```
local value = assert(babet.toml.decode([[
"a\u0000b" = "x\u0000y"
]]))

assert(value["a\0b"] == "x\0y")
```

A raw NUL byte injected into the TOML text is not treated as the end of a C string: the parser receives the whole buffer and rejects the document. The same applies to invalid UTF-8.

Bare, quoted, and dotted keys follow TOML rules:

```
local value = assert(babet.toml.decode([[
"a.b" = 1
site."google.com" = true
physical.color = "orange"
]]))

assert(value["a.b"] == 1) -- literal dot in a quoted key
```

```
assert(value.site["google.com"] == true)
assert(value.physical.color == "orange")
```

An empty quoted key ("") is valid TOML and becomes the Lua key "".

## Arrays and tables

TOML arrays become Lua sequences indexed from 1. TOML 1.0 permits heterogeneous arrays:

```
local value = assert(babet.toml.decode([[
items = [1, "two", true, { name = "three" }]]))

assert(value.items[1] == 1)
assert(value.items[2] == "two")
assert(value.items[3] == true)
assert(value.items[4].name == "three")
```

Sections, inline tables, and dotted keys all become ordinary Lua tables. The original syntax is not retained.

### Empty-container ambiguity

An empty array [] and an empty table {} both become an empty Lua table:

```
local value = assert(babet.toml.decode([[
a = []
b = {}
]]))

assert(type(value.a) == "table" and next(value.a) == nil)
assert(type(value.b) == "table" and next(value.b) == nil)
```

No metatable or sentinel remembers the original TOML type. This information loss is usually harmless when reading configuration, but it prevents a faithful round trip.

## Dates and times

TOML defines four temporal types. Since Lua has no native date type, Babet converts them to strings:

| TOML type        | Lua form                                    |
|------------------|---------------------------------------------|
| local date       | YYYY-MM-DD                                  |
| local time       | HH:MM:SS[.fraction]                         |
| local date-time  | YYYY-MM-DDTHH:MM:SS[.fraction]              |
| offset date-time | YYYY-MM-DDTHH:MM:SS[.fraction]Z or +/-HH:MM |

```
local value = assert(babet.toml.decode([[
date = 1979-05-27
time = 07:32:00.1234
local_dt = 1979-05-27T07:32:00
utc_dt = 1979-05-27T07:32:00Z
]]))
```

The temporal value is retained, but its original **spelling** may be normalized by toml++: canonical separator, fractional part, or UTC notation. Comments and the exact source representation are not available. After conversion, a TOML date is also indistinguishable from a TOML string containing the same text.

## Ordering and discarded information

A TOML table has no guaranteed semantic order, and Lua table iteration must not be used as a proxy for source order. Decoding does not preserve:

- comments;
- lexical key order;
- whitespace and line breaks;
- the choice between sections, inline tables, and dotted keys;
- the original spelling of numbers and dates;
- the type of an empty container.

The module is intended to **read values**, not to edit and reproduce a TOML document byte-for-byte.

## Error contract

A TOML syntax or value error returns:

```
local value, err = babet.toml.decode("answer = ")
assert(value == nil)
print(err) -- toml: ... (line ..., col ...)
```

The message is prefixed with `toml:` and includes the line and column reported by `toml++`.

The following are returned as `(nil, err)`, among others:

- invalid syntax or a redefined key;
- integer outside the signed 64-bit range;
- invalid UTF-8 or a forbidden control character;
- an extension unavailable in the TOML 1.0 profile used by Babet.

API misuse raises a Lua error instead:

- missing argument;
- non-string argument;
- extra argument.

## Limits

The whole document is parsed and converted in memory. There is no streaming API, `decode_file`, schema validation, or access to source positions for individual values. Recursive conversion protects the Lua stack; a depth that cannot be converted is returned as a `toml: error`.

## Not exposed in v1

- `babet.toml.encode`;
- `babet.toml.decode_file`;
- parser options or extensions outside TOML 1.0;
- nodes retaining comments, order, and source positions;
- sentinels distinguishing empty arrays from empty tables;
- schema validation.

## USER - look up system accounts by name or UID

The `babet.user` module queries the system user database through NSS. It can:

- look up an account by name;
- look up an account by UID;
- quickly test whether an account exists;
- retrieve the name, UID, primary GID, GECOS field, home directory, and shell.

It never parses `/etc/passwd` directly. It uses the same resolution mechanisms as `id` or `getent passwd`, so it can see accounts supplied by LDAP, SSSD, NIS, FreeIPA, or other sources configured in `/etc/nsswitch.conf`.

## Module contents

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## General conventions

### Dedicated subtable

Unlike the historical `SYS` or `FS` functions, user helpers live under `babet.user`:

```
local user = babet.user.get("root")
local exists = babet.user.exists("root")
```

### Name or UID, without implicit conversion

The argument accepts exactly:

- a Lua string: lookup by **name**;
- a non-negative Lua integer in the `uid_t` range: lookup by **UID**.

A numeric string remains a name:

```
-- Looks for an account literally named "1000"
local by_name = babet.user.get("1000")

-- Looks up numeric UID 1000
local by_uid = babet.user.get(1000)
```

Floats, booleans, tables, `nil`, negative UIDs, and oversized UIDs raise a Lua error. No silent conversion or truncation is performed.

### Fields are always present

When a user is found, the returned table always contains all six documented fields. Text fields may be empty strings, but never `nil`.

## API overview

| Function                                    | Result                                                |
|---------------------------------------------|-------------------------------------------------------|
| <code>babet.user.get(name_or_uid)</code>    | table, (nil, "user not found"), or (nil, "user: ...") |
| <code>babet.user.exists(name_or_uid)</code> | strict boolean                                        |

## Returned user table

Typical example:

```
{
  name  = "www-data",
  uid   = 33,
  gid   = 33,
  gecos = "www-data",
  home  = "/var/www",
  shell = "/usr/sbin/nologin",
}
```

| Field | Type    | Meaning                                    |
|-------|---------|--------------------------------------------|
| name  | string  | canonical account name returned by NSS     |
| uid   | integer | numeric user identifier                    |
| gid   | integer | account's <b>primary</b> GID               |
| gecos | string  | raw descriptive field, often the full name |
| home  | string  | configured home directory                  |
| shell | string  | configured login shell                     |

gid is not a list of supplementary groups. The module does not perform group lookups and does not expose additional memberships.

The gecos field is returned unchanged. Its historical format can contain comma-separated values, but many systems simply store a free-form name.

```
local u = assert(babet.user.get("root"))
print("Name :", u.name)
print("UID  :", u.uid)
print("GID  :", u.gid)
print("GECOS:", u.gecos)
print("Home :", u.home)
print("Shell:", u.shell)
```

## Look up an account with get

Signature:

```
local info, err = babet.user.get(name_or_uid)
```

Use get when you need account details or must distinguish a missing account from an NSS failure.

### By name

```
local root, err = babet.user.get("root")
assert(root, err)
assert(root.name == "root")
assert(root.uid == 0)
```

Preparing a directory for a service account:

```

local account, err = babet.user.get("my-service")
assert(account, err)

assert(babet.mkdir("/var/lib/my-service"))
assert(babet.setAttributes(
    "/var/lib/my-service",
    account.uid,
    account.gid,
    tonumber("750", 8)
))

```

The name is passed to NSS unchanged. It must be a string without an embedded NUL byte. An empty string is a valid API input, but normally matches no account and returns `user not found`.

### By UID

```

local root, err = babet.user.get(0)
assert(root, err)
assert(root.name == "root")

```

Displaying the account associated with a file owner:

```

local attrs, err = babet.getAttributes("report.txt")
assert(attrs, err)

local owner, user_err = babet.user.get(attrs.owner)
if owner then
    print("Owner:", owner.name)
else
    print("Unresolved owner UID:", attrs.owner, user_err)
end

```

The UID must be a non-negative integer compatible with the system `uid_t` type. On Linux the limit is usually  $2^{32} - 1$ , but the code uses the actual platform limit at compile time.

### Missing account

A missing account is not a Lua error. The function returns:

```

local info, err = babet.user.get("missing-account")
-- info == nil
-- err == "user not found"

```

Typical handling:

```

local account, err = babet.user.get("my-service")
if not account then
    if err == "user not found" then
        print("The account must be created")
    else
        print("NSS lookup failed:", err)
    end
end

```

Avoid this when absence is expected:

```

-- Wrong for an optional account: assert raises immediately
-- local account = assert(babet.user.get("optional-account"))

```

## Test existence with exists

Signature:

```
local present = babet.user.exists(name_or_uid)
```

The function always returns a boolean for a valid argument.

### By name

```
if babet.user.exists("www-data") then
    print("The www-data account exists")
end
```

Simple precondition:

```
if not babet.user.exists("my-service") then
    error("The my-service account must exist before startup")
end
```

### By UID

```
assert(babet.user.exists(0)) -- root on a normal Unix system
```

A missing UID returns false:

```
local present = babet.user.exists(2000000000)
print(present)
```

### When to prefer get

exists deliberately converts an NSS error to false.

This makes it convenient for a simple branch, but it cannot distinguish:

- an account that truly does not exist;
- a temporarily unavailable LDAP directory;
- an I/O error;
- an out-of-memory condition in the resolver.

For an important administrative or security decision, use get:

```
local account, err = babet.user.get("my-service")
if account then
    print("Account available")
elseif err == "user not found" then
    print("Account missing")
else
    error("Unable to query NSS: " .. err)
end
```

## Argument validation

The following cases raise a Lua error, recoverable with pcall:

```
local invalid_calls = {
    function() return babet.user.get() end,
    function() return babet.user.get(nil) end,
    function() return babet.user.get(true) end,
    function() return babet.user.get({}) end,
    function() return babet.user.get(1.5) end,
    function() return babet.user.get(-1) end,
}
```

```

    function() return babet.user.get(8589934592) end,
    function() return babet.user.get("root\0other") end,
}

for _, call in ipairs(invalid_calls) do
    local ok, err = pcall(call)
    assert(not ok)
    print(err)
end

```

exists applies exactly the same validation:

```

local ok = pcall(function()
    return babet.user.exists(-1)
end)
assert(not ok)

```

Rejecting NUL bytes prevents "root\0other" from being interpreted as "root" by getpwnam\_r.

## NSS, workers, and security

### Account sources

The result depends on the machine's NSS configuration. According to /etc/nsswitch.conf, an account may come from:

- /etc/passwd;
- LDAP;
- SSSD;
- NIS;
- FreeIPA;
- systemd-userdb;
- another NSS module.

The same script may therefore see different accounts on different machines without any Babet change.

### Calls from workers

The module uses the reentrant getpwnam\_r and getpwuid\_r functions. It can be used inside workers:

```

local worker = assert(babet.workers.spawn([[
    local root, err = babet.user.get("root")
    if not root then
        error(err)
    end
    return root.uid
]]))

local joined, uid = worker:join()
assert(joined and uid == 0)

```

Each call queries NSS. Babet does not maintain an application-level user cache.

### Security-sensitive decisions

The home and shell fields are configuration data, not proof that a path exists or that a program is executable.

```

local u = assert(babet.user.get("my-service"))
local home_is_dir = assert(babet.isDir(u.home))
local shell_path = babet.which(u.shell)

```

Likewise, the presence of an account does not prove that it can log in, that a password is valid, or that it belongs to a particular supplementary group.

## Error contract

| Situation        | get                     | exists    |
|------------------|-------------------------|-----------|
| account found    | user table              | true      |
| account missing  | (nil, "user not found") | false     |
| NSS error        | (nil, "user: ...")      | false     |
| invalid argument | Lua error               | Lua error |

Generic handling:

```

local ok, info, err = pcall(function()
    return babet.user.get("my-service")
end)

if not ok then
    print("Invalid call:", info)
elseif not info then
    print("Lookup failed:", err)
else
    print("UID:", info.uid)
end

```

## Design decisions and limits

- NSS is used instead of parsing `/etc/passwd`.
- Reentrant `_r` variants are used for worker compatibility.
- Text fields are always present and replaced with "" if NSS supplies a null pointer.
- The NSS buffer grows dynamically up to an internal 64 KiB limit; beyond that, `get` returns an NSS error.
- `exists` favors a simple boolean API and hides NSS errors; use `get` when a diagnosis is required.
- The module does not expose supplementary groups.
- It exposes neither `/etc/shadow`, passwords, nor account expiration data.
- It does not create, remove, or modify users.
- To create an account, a script can invoke a system utility through `babet.exec`, with appropriate privileges.

## WORKERS — OS threads, isolated Lua states, queues, and channels

`babet.workers` executes Lua code in real POSIX threads. Each worker owns a separate `lua_State`, receives a serialized copy of its initial arguments, and communicates with the parent through two bounded message queues. Shared channels can also connect the parent and several workers, or workers directly, without relaying messages through the parent's inbox/outbox.

The module covers:

- starting a Lua chunk in an OS thread;
- passing an initial argument table;

- blocking or non-blocking retrieval of the first return value;
- a parent-to-worker inbox and a worker-to-parent outbox;
- direct bounded FIFO multi-producer/multi-consumer channels;
- timeouts and capacity-based backpressure;
- closing with drainage and wake-up of blocked calls;
- loading Babel, bundled, and user modules in each state;
- Lua-memory isolation between threads.

It does not provide shared Lua memory, forced termination, a global pool, or channels between distinct OS processes. Available cancellation is strictly cooperative.

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## Essential conventions

### Each worker owns a separate Lua state

A table, global, or module changed in the parent is not shared with a worker. Data crosses state boundaries only through serialization:

- args during spawn;
- `job:send` and `worker.recv`;
- `worker.send` and `job:recv`;
- shared channel send and recv methods;
- the chunk's first return value.

There can therefore be no data race on Lua objects themselves. External side effects — files, sockets, databases, and processes — remain system resources and must still be coordinated by the application.

### Results use a pcall-like convention

`join(timeout?)` returns:

```
true, business_value
false, error_message
nil, "timeout"
```

The third state only means the worker is still active. The result is neither joined nor consumed.

This lets a worker legitimately return nil:

```
local ok, value = job:join(0.5)
if ok == true then
    -- value may be nil.
elseif ok == false then
    io.stderr:write(value, "\n")
else
    assert(value == "timeout")
end
```

Queue methods also return a boolean first, so a nil message is unambiguous.

### The final result can be consumed once

As soon as `join()` retrieves the result, or `poll()` returns "done" or "error", that result is consumed. A later `join()` or `poll()` reports result already consumed.

## API overview

### Parent side

```
local job, err = babet.workers.spawn(code, args?, opts?)
local channel, err = babet.workers.channel(opts?)

local state = job:status()
local ok, value_or_reason = job:join(timeout?)
local state, value = job:poll()
```

```

local ok, err = job:cancel()

local ok, err = job:send(value, timeout?)
local ok, value_or_err = job:recv(timeout?)
local ok, err = job:close()

local ok, err = channel:send(value, timeout?)
local ok, value_or_reason = channel:recv(timeout?)
local ok, err = channel:close()
local closed = channel:is_closed()

```

| API                    | Main result                                           | Blocking           |
|------------------------|-------------------------------------------------------|--------------------|
| spawn                  | job or (nil, err)                                     | creation only      |
| job:status()           | "running", "done", or "error"                         | no, never consumes |
| job:join(t?)           | (true, result), (false, err), or (nil, "timeout")     | depends on t       |
| job:poll()             | ("running", nil), ("done", result), or ("error", err) | no                 |
| job:cancel()           | (true, nil)                                           | no                 |
| job:send(v, t?)        | (true, nil), (false, reason), or (nil, err)           | depends on t       |
| job:recv(t?)           | (true, value) or (false, reason)                      | depends on t       |
| job:close()            | (true, nil)                                           | no                 |
| workers.channel(opts?) | channel or (nil, err)                                 | creation only      |
| channel:send(v, t?)    | (true, nil), (false, reason), or (nil, err)           | depends on t       |
| channel:recv(t?)       | (true, value), (false, reason), or (nil, err)         | depends on t       |
| channel:close()        | (true, nil)                                           | no                 |
| channel:is_closed()    | boolean                                               | no                 |

## Worker side

The chunk receives a global worker table:

```

worker.args
worker.channels
worker.send(value, timeout?)
worker.recv(timeout?)
worker.cancelled()

```

| API                | Result                                        |
|--------------------|-----------------------------------------------|
| worker.args        | copy of args, or nil                          |
| worker.channels    | table of handles passed through opts.channels |
| worker.send(v, t?) | (true, nil) or (false, reason_or_error)       |
| worker.recv(t?)    | (true, value) or (false, reason)              |
| worker.cancelled() | boolean, no side effect                       |

## spawn(code, args?, opts?)

```

local job, err = babet.workers.spawn(code, args?, opts?)

```

On success, spawn returns only the job userdata. The thread may still be running, may already have completed, or may already have failed by the time the parent receives the userdata.

Chunk load or runtime errors occur **inside** the worker: `spawn` may succeed, then `join()` or `poll()` reports the error.

```
local job = assert(babet.workers.spawn("this is not valid Lua"))
local ok, err = job:join()
assert(not ok)
```

### code

code is required and must be an actual Lua string. Numbers are not coerced.

```
local job = assert(babet.workers.spawn([[
    return 6 * 7
]]))
```

The source is loaded as a text chunk in a fresh state. It is not a closure that captures parent upvalues.

```
local factor = 3

-- factor does not exist in the worker: pass it through args.
local job = assert(babet.workers.spawn([[
    return worker.args.value * worker.args.factor
]], {
    value = 14,
    factor = factor,
}))
```

The binding does not truncate NUL bytes in code, but a NUL normally makes the source invalid for the Lua parser. Use ordinary Lua source text.

### args and worker.args

args is optional. It must be a table or nil.

```
local job = assert(babet.workers.spawn([[
    return worker.args.left + worker.args.right
]], {
    left = 20,
    right = 22,
}))
```

Inside the worker:

- `worker.args` contains a deserialized copy of the table;
- without args, `worker.args == nil`;
- modifying `worker.args` never changes the parent's table;
- metatables and subtable identity do not cross the boundary.

A direct scalar is not accepted as the second argument:

```
babet.workers.spawn("return worker.args", 42) -- raises a Lua error
```

Wrap it in a table:

```
babet.workers.spawn("return worker.args.value", { value = 42 })
```

### Queue capacities

opts can define:

```
{
    inbox_capacity = 64,
```

```
    outbox_capacity = 64,
}
```

- `inbox_capacity`: maximum queued parent-to-worker messages;
- `outbox_capacity`: maximum queued worker-to-parent messages;
- default: 64 messages for each queue;
- allowed range: integer from 1 to 1,000,000;
- the value must be an actual Lua integer; the string "64" and the floating-point value 64.0 are rejected;
- capacity counts **messages**, not bytes.

```
local job = assert(babet.workers.spawn(code, nil, {
    inbox_capacity = 8,
    outbox_capacity = 32,
}))
```

Every unknown `opts` key raises a Lua error, catching typos immediately:

```
babet.workers.spawn("return 1", nil, {
    inbox_capacity = 16, -- error: unknown option
})
```

Option names must be real strings with no hidden NUL suffix; numeric keys are also rejected.

A larger capacity may consume more memory and does not replace a draining strategy. A small capacity applies backpressure earlier.

## Passing channels through `opts.channels`

A channel is a special shared resource. It never crosses the JSON serialization used for `worker.args` and cannot be sent as a message. Pass it explicitly through the `channels` field:

```
local tasks = assert(babet.workers.channel({ capacity = 16 }))
local results = assert(babet.workers.channel({ capacity = 16 }))

local job = assert(babet.workers.spawn([
    local ok, task = worker.channels.tasks:recv(2)
    if not ok then return task end

    assert(worker.channels.results:send({
        id = task.id,
        result = task.value * 2,
    }, 2))

    return true
]), nil, {
    channels = {
        tasks = tasks,
        results = results,
    },
}))
```

In the new Lua state:

- `worker.channels` always exists and is an empty table when no channel was passed;
- each `opts.channels` key becomes a field of `worker.channels`;
- several names may reference the same channel;
- the parent and all workers hold distinct userdata pointing to the same C++ queue;
- names must be non-empty UTF-8 strings containing no NUL;
- every value must be a channel created by `babet.workers.channel()`.

A channel placed in `args` remains non-serializable userdata and is rejected.

### When `setenv` and `chdir` become locked

After validating types and capacities, the first spawn marks the process as having used workers. From then on:

- `babet.setenv(...)` is permanently rejected;
- `babet.chdir(...)` is permanently rejected;
- the restriction remains after `join()`;
- it remains even if `args` serialization, queue initialization, or `pthread_create` later fails.

A call rejected **before** this mark — for example `spawn(42)` or a non-integer capacity — does not lock the environment.

Configure the current directory and environment before the first worker.

### Transferable values

The internal transport format is JSON. The same rules apply to:

- `args`;
- messages in either direction;
- messages sent through channels;
- the worker's first return value.

### Scalars and strings

| Lua value                                 | Transferable | Note                                             |
|-------------------------------------------|--------------|--------------------------------------------------|
| <code>nil</code>                          | yes          | becomes JSON null, then <code>nil</code> again   |
| <code>boolean</code>                      | yes          | identity preserved                               |
| <code>integer</code>                      | yes          | normally restored as a Lua integer               |
| <code>finite floating-point number</code> | yes          | NaN, +Inf, and -Inf rejected                     |
| <code>text string</code>                  | yes          | must pass the UTF-8 validator and contain no NUL |

Unlike several filesystem and socket APIs, worker messages are **not binary-safe**. A string containing `\0` or bytes rejected by the UTF-8 validator cannot be transferred.

### List and object tables

A table is transferable in one of two shapes.

#### Non-empty dense list

```
{ "a", "b", "c" }
```

All keys must be consecutive integers `1..n`, with no holes and no other key.

#### String-keyed object

```
{  
  name = "alice",  
  age = 30,  
}
```

All keys must be strings accepted by the text validator.

An empty table is serialized as an **empty object** {}, not an empty array. There is no equivalent of `babot.json.empty_array` in the worker transport.

Sparse or mixed tables are rejected:

```
{ [1] = "a", [3] = "c" }      -- hole
{ [1] = "a", label = "x" }    -- list + map
{ [0] = "zero" }              -- numeric key outside 1..n
```

## Rejected values

These values do not cross Lua states:

- functions and closures;
- userdata: channel, socket, SQLite statement, watcher, job, file, and so on;
- Lua coroutines/threads;
- cyclic tables;
- sparse or mixed tables;
- non-string keys in object tables;
- NaN or infinite numbers;
- strings with NUL or bytes rejected as UTF-8;
- structures beyond the depth, node, or byte budgets.

Pass a path, URL, or serializable configuration instead of the system object itself. The worker then opens its own resource.

```
-- Wrong: socket is userdata.
babot.workers.spawn(code, { socket = peer })

-- Correct: pass connection parameters.
babot.workers.spawn(code, {
    host = "127.0.0.1",
    port = 9000,
})
```

## Copies, identity, depth, and budgets

Transport copies values:

- no Lua reference is shared;
- metatables are lost;
- two references to the same subtable become two separate tables;
- cycles are rejected;
- maximum nesting is 32 levels according to the internal serialization counter;
- one operation may expand to at most **1,000,000 JSON values**;
- the estimated representation must remain within a conservative **64 MiB** budget.

```
local shared = { value = 1 }
local args = { a = shared, b = shared }
```

```
-- In the worker, worker.args.a and worker.args.b have equal content,
-- but are not the same table.
```

The depth limit protects the C++ and Lua stacks. The two additional budgets prevent a small Lua structure from expanding into an exponential amount of data when the same subtable or string is referenced repeatedly.

Every expanded value consumes exactly one node and a fixed cost of 32 bytes. Every string occurrence, including an object key, also consumes an upper bound of  $6 \times \text{byte\_length} + 2$ . The

factor of six covers the worst JSON escaping of a control byte. The two ceilings deliberately overlap: together they bound both the number of objects in the JSON DOM and the expanded text payload.

The limits apply independently to each operation: `spawn` arguments, final result, `job:send`, `worker.send`, or `channel:send`. A rejected value is not added to the queue. The identical receive-side check validates message consistency; the primary memory bound is applied by the sender before strings are copied and before the internal JSON is published.

These ceilings are part of the public contract. Lowering them in a later release could reject transfers that used to succeed and should be announced as an incompatible change.

## Final worker result

When the chunk completes normally, only its **first** return value is serialized.

```
return "first", "second", "third"
```

The parent receives only "first".

Return a table to transport several fields:

```
return {
    value = 42,
    elapsed = 0.12,
}
```

With no return, or return `nil`, the business result is `nil`.

An uncaught Lua error becomes the error state. The transferred text is a useful diagnostic, but Babet does not automatically add a traceback. A worker can use `xpcall(..., debug.traceback)` when it wants to build one.

## `status()` — observe without consuming

```
local state = job:status()
```

`status()` returns exactly one of these strings:

| State     | Meaning                                   |
|-----------|-------------------------------------------|
| "running" | the chunk or its cleanup is still running |
| "done"    | the worker completed normally             |
| "error"   | the worker completed with an error        |

The method does not block, join the pthread, or consume the result. It remains usable before and after `join()`:

```
while job:status() == "running" do
    update_interface()
    babet.sleep(10, "ms")
end
```

```
local ok, result = job:join()
assert(ok, result)
assert(job:status() == "done")
```

`status()` intentionally does not expose a separate cancelling state: until the chunk ends, it remains "running".

## **join(timeout?) — wait and consume**

```
local ok, value_or_reason = job:join(timeout?)
```

The timeout uses the same seconds contract as message queues:

- omitted or nil: wait indefinitely;
- 0: immediate test;
- finite positive number up to 86400: bounded wait;
- negative value, NaN, infinity, numeric string, or an extra argument: raises a Lua error.

There are three return states:

```
true, result      -- normal completion, result consumed
false, err        -- failed completion, error consumed
nil, "timeout"    -- worker still active, nothing consumed
```

Retry after an expiry:

```
local ok, value = job:join(0.05)

if ok == nil then
    assert(value == "timeout")
    print("worker is still running")

    -- The same job remains fully usable.
    ok, value = job:join(2)
end

if ok then
    print("result", value)
else
    io.stderr:write("worker: ", value, "\n")
end
```

A timeout:

- closes no queue;
- requests no cancellation;
- does not join the pthread;
- consumes neither result nor error;
- still allows status, send, recv, close, cancel, and another join.

The timeout bounds only this call to join(). If the worker remains active and its last reference is later collected, notably while the Lua state is closing, its \_\_gc must still perform an unbounded pthread\_join() and can therefore block program termination.

Even when the worker legitimately returns nil, success is distinguishable by ok == true.

## **poll() — check and consume**

```
local state, value = job:poll()
```

| state     | value                | Consumed |
|-----------|----------------------|----------|
| "running" | nil                  | no       |
| "done"    | result, possibly nil | yes      |
| "error"   | message              | yes      |

```
while true do
    local state, value = job:poll()
    if state == "running" then
```

```

        babet.sleep(10, "ms")
    elseif state == "done" then
        print("completed", value)
        break
    else
        io.stderr:write("worker: ", value, "\n")
        break
    end
end
end

```

Once the state is no longer running, `poll()` quickly joins the already completed pthread and consumes the result.

### Do not call `join()` after completed `poll()`

This sequence is wrong:

```

local state = job:poll()
if state ~= "running" then
    local ok, result = job:join() -- result already consumed
end

```

Use both values from `poll()` directly:

```

local state, value = job:poll()
if state == "done" then
    use_result(value)
elseif state == "error" then
    report_error(value)
end

```

Alternatively, use `poll()` only while it returns running, and design another mechanism that tells code when to call `join()` without first consuming the result. In most event loops, polling through the final result is simpler.

### Parent-to-worker messages

```

local ok, err = job:send(value, timeout?)

```

`job:send` serializes value, then appends it to the worker inbox in FIFO order.

Results:

- (true, nil): message accepted;
- (false, "full"): full queue with timeout 0;
- (false, "timeout"): queue stayed full until the deadline;
- (false, "closed"): inbox closed;
- (false, "cancelled"): cancellation requested through `cancel()`;
- (nil, err): non-transferable value or internal serialization failure.

```

local ok, err = job:send({ command = "scan", path = "/tmp" }, 1)
if not ok then
    io.stderr:write("send: ", err, "\n")
end

```

Success means the message entered the queue, not that the worker has already processed it.

### Worker-to-parent messages

```

local ok, value_or_err = job:recv(timeout?)

```

`job:recv` removes the oldest message from the outbox.

Results:

- `(true, value)`: message received; value may be `nil`;
- `(false, "empty")`: empty queue with timeout 0;
- `(false, "timeout")`: no message before the deadline;
- `(false, "closed")`: queue closed and fully drained.

When a queue is closed but still contains messages, those messages are returned before "closed".

```
while true do
  local ok, message = job:recv(0)
  if ok then
    process(message)
  elseif message == "empty" then
    break
  elseif message == "closed" then
    break
  else
    error(message)
  end
end
```

### **`worker.send`, `worker.recv`, and `worker.cancelled`**

Inside the chunk:

```
local ok, message_or_err = worker.recv(timeout?)
local ok, err = worker.send(value, timeout?)
local cancelled = worker.cancelled()
```

`worker.recv` reads the inbox filled by `job:send`. `worker.send` writes the outbox read by `job:recv`. `worker.cancelled()` reads, without blocking, the flag set by `job:cancel()`.

After cancellation:

- `worker.cancelled()` returns `true`;
- the next `worker.recv()` returns `(false, "cancelled")` and no longer delivers commands still waiting in the inbox; a command already removed by the worker before cancellation may naturally still be processing;
- `worker.send()` remains available for a final result, diagnostic, or stop acknowledgement;
- the chunk decides when and how to finish.

FIFO ordering, capacities, timeouts, and closing remain symmetric outside this cancellation case.

Serialization-error convention differs slightly:

- `job:send` returns `(nil, err)`;
- `worker.send` returns `(false, err)`.

This keeps a boolean status convention inside worker code.

```
local job = assert(babet.workers.spawn([
  while not worker.cancelled() do
    local ok, message = worker.recv(0.2)

    if ok then
      process(message)
    elseif message == "cancelled" then
      break
    elseif message ~= "timeout" then
      return { error = message }
    end
  end
end])
```

```

        end
    end

    worker.send({ stopped = true })
    return "cancelled"
]))))

```

## Direct shared channels

A channel is a bounded queue shared by the parent and any number of workers. Unlike a job inbox or outbox, it is not owned by one worker: every handle for the same channel may produce or consume messages.

Channels are thread-safe, bounded by message count, multi-producer, multi-consumer, FIFO by effective insertion order, and explicitly closed with draining of already queued messages.

### Creating a channel and choosing capacity

```
local channel, err = babet.workers.channel(opts?)
```

The only option is capacity, which defaults to 64 messages and must be a strict Lua integer between 1 and 1,000,000. Capacity counts messages, not bytes.

```
local tasks = assert(babet.workers.channel({ capacity = 16 })))
```

Unknown options, non-string keys, names containing NUL, and out-of-range capacities raise a Lua error. Allocation or initialization failures return (nil, err).

### send and recv

```
local ok, err = channel:send(value, timeout?)
local ok, value_or_reason = channel:recv(timeout?)
```

send returns (true, nil), (false, "full"|"timeout"|"closed"|"cancelled"), or (nil, err) when serialization or an internal operation fails.

recv returns (true, value), (false, "empty"|"timeout"|"closed"|"cancelled"), or (nil, err) for an internal deserialization anomaly. A nil message is unambiguous because the first result is true.

```
local channel = assert(babet.workers.channel({ capacity = 1 })))
assert(channel:send(nil, 0))
```

```
local ok, value = channel:recv(0)
assert(ok == true and value == nil)
```

Channels use the same JSON transfer rules as workers. A channel or any other userdata cannot itself be sent as a message. For binary Selenium screenshots, prefer a file path, or keep Base64 text and decode it with `babet.base64.decode()` at the consumer.

Inside a worker, `job:cancel()` also wakes a blocked `channel:send()` or `channel:recv()` for **that worker only**. The call returns (false, "cancelled") without closing the shared channel or disturbing other participants. New channel sends and receives from that cancelled worker also return "cancelled"; use `worker.send()` for a final acknowledgement to the parent.

### close and is\_closed

`close()` is global and idempotent. It rejects future sends, makes `is_closed()` return true, wakes blocked senders and receivers, and preserves already queued messages. Once those messages are drained, `recv()` returns (false, "closed").

```

local channel = assert(babet.workers.channel({ capacity = 2 }))
assert(channel:send("one"))
assert(channel:send("two"))
assert(channel:close())
assert(channel:is_closed())

assert(select(2, channel:recv()) == "one")
assert(select(2, channel:recv()) == "two")
local ok, reason = channel:recv()
assert(not ok and reason == "closed")

```

## Concurrency, ordering, and lifetime

All handles for a channel reference the same C++ object, while each Lua state owns a separate userdata. Collecting one local handle does not close the channel for the others. Explicit `close()` affects every handle. When the last reference disappears, Babet closes the queue, wakes waiters, and destroys any remaining messages.

FIFO refers to effective insertion order. Concurrent sends from different producers have no deterministic relative order, while sequential sends from a single producer remain ordered.

Capacity does not limit individual message size. JSON serialization creates copies, so large messages can consume substantial memory. Publishing binary data atomically to a file and sending its path is usually preferable.

## Timeouts and failure reasons

All four message methods and `join(timeout?)` share the same timeout validation rules:

| Lua timeout                      | Behavior                        |
|----------------------------------|---------------------------------|
| omitted or nil                   | wait indefinitely               |
| 0                                | immediate, non-blocking attempt |
| $0 < t \leq 86400$               | wait at most $t$ seconds        |
| negative, NaN, Inf, or $> 86400$ | raises a Lua error              |
| other type, including "1"        | raises a Lua error              |

Internal resolution is one millisecond. A strictly positive duration is rounded up: 0.0005 second waits at least 1 ms and is not turned into non-blocking mode.

Queue reasons are:

| Reason      | Meaning                                                                 |
|-------------|-------------------------------------------------------------------------|
| "full"      | immediate send, queue full                                              |
| "empty"     | immediate receive, queue empty                                          |
| "timeout"   | positive deadline reached                                               |
| "closed"    | send to closed queue, or receive from closed and empty queue            |
| "cancelled" | job:send, worker.recv, or a channel method called in a cancelled worker |

Internal diagnostics such as "out of memory" or "internal mutex error" may exceptionally appear instead.

## close() and queue lifecycle

```
local ok, err = job:close()
```

```
job:close():
```

- closes only the parent-to-worker **inbox**;
- is idempotent;
- returns (true, nil);
- wakes a waiting worker.recv(), which gets (false, "closed") once already queued messages are drained;
- makes future job:send() calls fail with (false, "closed");
- leaves the outbox open so the parent can still receive final worker messages.

It does not forcibly terminate the thread and does not consume its final result.

The worker automatically closes **both** queues when it completes, whether successfully or with an error. The parent can still drain messages already in the outbox before receiving "closed".

To cleanly finish a worker that waits for commands:

```
assert(job:close())
```

```
local ok, result = job:join()
```

The worker must treat worker.recv() == false, "closed" as an exit condition.

## cancel() and cooperative cancellation

```
local ok, err = job:cancel()
```

```
job:cancel():
```

- sets an atomic flag visible through worker.cancelled();
- closes the inbox to wake a blocked worker.recv() immediately;
- also wakes the channel send or recv currently blocked in that worker, without closing the global channel;
- makes future job:send() calls return (false, "cancelled");
- leaves the outbox open so the worker can publish a final message;
- does not join the thread;
- does not consume the final result;
- is idempotent and always returns (true, nil) for a valid job.

```
assert(job:cancel())
```

```
local got, final = job:recv(1)
```

```
if got then
```

```
    print("final message", final)
```

```
end
```

```
local ok, result = job:join(2)
```

```
if ok == nil then
```

```
    error("worker did not cooperate before the deadline")
```

```
end
```

```
assert(ok, result)
```

This mechanism never uses pthread\_cancel() and never abruptly interrupts Lua, SQLite, a C++ lock, or a transaction. A worker must therefore check worker.cancelled() or regularly return to worker.recv().

A worker blocked in a non-interruptible system call, in worker.send() without a timeout on a full outbox, or in a loop that never checks the flag can remain active. Babet channel waits, however, are awakened by cancellation. join(timeout) keeps the parent from blocking in every other case, but Babet does not force termination.

close() and cancel() differ:

- close() means “no more commands” and allows already queued commands to drain; worker.recv() eventually returns "closed";
- cancel() means “abandon current work as soon as possible”; queued inbox commands are no longer delivered and worker.recv() returns "cancelled". A command already removed before cancellation may still be processing: the stop remains cooperative.

## Module loading and worker environment

Each thread creates a fresh Lua state and:

- opens the standard Lua libraries;
- registers the complete babet namespace;
- exposes bundled modules through require;
- configures user-module loading like the parent project, in directory or embedded mode;
- exposes worker.args, worker.channels, worker.send, worker.recv, and worker.cancelled;
- does not copy the parent’s global arg table: arg == nil in the worker.

```
local job = assert(babet.workers.spawn([[
  local inspect = require("inspect")
  local mymod = require("mymod")
  return inspect({ answer = mymod.answer() })
]]))
```

Lua states are isolated, but threads belong to the same process:

- same PID;
- same process-wide current directory;
- same process-wide environment;
- same filesystem and accessible external resources;
- Lua memory measured separately in each state.

Babet rejects cwd and environment mutations after the first spawn, stabilizing those two global states.

## Complete examples

### Simple calculation with join

```
local job, err = babet.workers.spawn([[
  return 21 * 2
]])
assert(job, err)

local ok, result = job:join()
assert(ok, result)
print(result) -- 42
```

### Structured arguments

```
local job = assert(babet.workers.spawn([[
  local total = 0
  for _, value in ipairs(worker.args.values) do
    total = total + value
  end
  return {
    name = worker.args.name,
    total = total,
  }
]]))
```

```

    }
  ], {
    name = "batch A",
    values = { 10, 20, 12 },
  })

  local ok, result = job:join()
  assert(ok, result)
  print(result.name, result.total)

```

### Non-blocking progress with poll

```

local job = assert(babet.workers.spawn([
  babet.sleep(200, "ms")
  return "ready"
]))

```

```

while true do
  local state, value = job:poll()

  if state == "running" then
    update_ui()
    babet.sleep(10, "ms")
  elseif state == "done" then
    print(value)
    break
  else
    error(value)
  end
end

```

Do not call `join()` after the done or error branch: `poll()` has already consumed the result.

### Persistent request/response worker

```

local job = assert(babet.workers.spawn([
  while true do
    local ok, request = worker.recv()
    if not ok then
      return "parent closed inbox"
    end

    if request.op == "stop" then
      return "stopped"
    elseif request.op == "square" then
      local sent, err = worker.send({
        id = request.id,
        value = request.value * request.value,
      })
      if not sent then
        return { error = err }
      end
    end
  end
end, nil, {
  inbox_capacity = 16,
  outbox_capacity = 16,

```

```

)))

for i = 1, 5 do
    assert(job:send({ id = i, op = "square", value = i }))
end

for _ = 1, 5 do
    local ok, response = job:recv(2)
    assert(ok, response)
    print(response.id, response.value)
end

assert(job:send({ op = "stop" }))
local ok, result = job:join()
assert(ok, result)
print(result)

```

### **Sending and receiving nil**

The status boolean distinguishes a real nil message from an empty queue.

```

local job = assert(babet.workers.spawn([
    local ok, value = worker.recv()
    assert(ok)
    assert(value == nil)

    assert(worker.send(nil))
    return "done"
]))

assert(job:send(nil))

local got, value = job:recv()
assert(got == true)
assert(value == nil)

local ok, result = job:join()
assert(ok and result == "done")

```

### **Close, then drain the outbox**

```

local job = assert(babet.workers.spawn([
    assert(worker.send("ready"))

    local ok, reason = worker.recv()
    assert(not ok and reason == "closed")

    assert(worker.send("closing"))
    return "finished"
]))

local ok, first = job:recv(1)
assert(ok and first == "ready")

assert(job:close()) -- closes the inbox only

local got, last = job:recv(1)

```

```

assert(got and last == "closing")

local joined, result = job:join()
assert(joined and result == "finished")

```

### Backpressure and timeout

```

local job = assert(babet.workers.spawn([[
    babet.sleep(500, "ms")
    while true do
        local ok, message = worker.recv()
        if not ok then return "closed" end
        if message == "stop" then return "done" end
    end
]], nil, {
    inbox_capacity = 2,
}))

assert(job:send("one", 0))
assert(job:send("two", 0))

local ok, reason = job:send("three", 0)
assert(ok == false and reason == "full")

local waited, why = job:send("three", 0.1)
assert(waited == false and why == "timeout")

job:close()
job:join()

```

### Parent to worker through a channel

```

local tasks = assert(babet.workers.channel({ capacity = 8 }))

local job = assert(babet.workers.spawn([[
    local ok, task = worker.channels.tasks:recv(2)
    assert(ok, task)
    return task.left + task.right
]], nil, {
    channels = { tasks = tasks },
}))

assert(tasks:send({ left = 20, right = 22 }, 2))
local ok, result = job:join(2)
assert(ok and result == 42)

```

### Worker to worker without parent relay

```

local tasks = assert(babet.workers.channel({ capacity = 16 }))
local results = assert(babet.workers.channel({ capacity = 16 }))

local producer = assert(babet.workers.spawn([[
    for i = 1, 10 do
        assert(worker.channels.tasks:send({ id = i, value = i * 10 }, 2))
    end
    return "producer done"

```

```

]], nil, { channels = { tasks = tasks } })

local consumer = assert(babet.workers.spawn([
  for _ = 1, 10 do
    local ok, task = worker.channels.tasks:recv(2)
    assert(ok, task)
    assert(worker.channels.results:send({
      id = task.id,
      result = task.value * 2,
    }, 2))
  end
  return "consumer done"
]], nil, { channels = { tasks = tasks, results = results } })

for expected = 1, 10 do
  local ok, item = results:recv(2)
  assert(ok and item.id == expected and item.result == expected * 20)
end

assert(producer:join(2))
assert(consumer:join(2))

```

Producer-to-consumer messages never pass through the parent.

## Multiple producers and consumers

The same channel can be passed to several producers and consumers. Each message is removed by exactly one consumer, but distribution among consumers depends on thread scheduling and is not predictable. Close the task channel only after all producers have completed, then let consumers drain it until they receive "closed".

## Native cooperative cancellation

```

local job = assert(babet.workers.spawn([
  local total = 0

  for i = 1, worker.args.limit do
    total = total + expensive_step(i)

    if i % 1000 == 0 and worker.cancelled() then
      worker.send({
        stopped = true,
        partial = total,
      })
      return {
        cancelled = true,
        partial = total,
      }
    end
  end

  return { cancelled = false, total = total }
]], { limit = 1000000 })

-- Later:
assert(job:cancel())

```

```

-- The outbox remains drainable after cancel().
local got, progress = job:recv(0.5)
if got then
    print("partial", progress.partial)
end

local ok, result = job:join(2)
if ok == nil then
    error("cancellation was not observed within two seconds")
end
assert(ok, result)

```

Combined worker.recv() and flag example:

```

local job = assert(babet.workers.spawn([
    while not worker.cancelled() do
        local ok, command = worker.recv(0.1)

        if ok then
            execute(command)
        elseif command == "cancelled" then
            break
        elseif command ~= "timeout" then
            return { error = command }
        end
    end

    worker.send({ stopped = true })
    return "cancelled"
]))

assert(job:cancel())
local ok, result = job:join(2)
assert(ok and result == "cancelled")

```

Cancellation remains cooperative: it does not arbitrarily interrupt a system call or code that never checks the flag.

## Correct bounded pool

This loop bounds simultaneous pthreads and directly uses the poll() result, without calling join() afterward.

```

local function map_parallel(items, code, max_concurrent)
    max_concurrent = max_concurrent or 4

    local next_index = 1
    local active = {}
    local results = {}

    while next_index <= #items or #active > 0 do
        while next_index <= #items and #active < max_concurrent do
            local job, err = babet.workers.spawn(code, {
                item = items[next_index],
            })
            assert(job, err)

            active[#active + 1] = {

```

```

        index = next_index,
        job = job,
    }
    next_index = next_index + 1
end

for i = #active, 1, -1 do
    local entry = active[i]
    local state, value = entry.job:poll()

    if state == "done" then
        results[entry.index] = value
        table.remove(active, i)
    elseif state == "error" then
        results[entry.index] = { error = value }
        table.remove(active, i)
    end
end

if #active > 0 then
    babet.sleep(10, "ms")
end

return results
end

```

Every spawn still creates a new thread and Lua state. This pattern bounds concurrency; it does not reuse persistent workers.

## Deadlocks to avoid

### Worker waits for inbox while parent calls join()

```

-- Worker
local ok, message = worker.recv() -- waits indefinitely

-- Parent
job:join() -- waits for worker

```

Both sides wait. Close the inbox or send a command first:

```

job:close()
job:join()

```

### Worker blocks on a full outbox

If the worker calls `worker.send()` with no timeout, the outbox is full, and the parent calls `join()` without draining it, neither side can progress.

Solutions:

- parent regularly calls `job:recv()`;
- worker uses a finite timeout;
- outbox capacity matches the protocol;
- message and join phases are clearly separated.

## Producers blocked on a full channel

A bounded channel can create the same wait cycle as a full outbox: producers wait for space while the parent calls `join()` before consumers have drained the queue. Define who consumes, who closes the channel, and when. Long-running protocols should use finite timeouts or a shutdown strategy so one failed participant cannot block all others.

## Forgetting that `poll()` consumes

Code that waits for `poll()` `~= "running"` and then calls `join()` does not deadlock, but loses the real result and gets already consumed. Use both values from `poll()`.

## Potentially blocking GC

Dropping the last reference to an active worker can run `__gc`, which closes queues and then waits for the thread. Queue waits are released, but GC cannot interrupt infinite computation or an unbounded external syscall.

## Error contract

### Lua errors raised for invalid use

These include:

- missing or non-string code;
- args other than table or `nil`;
- opts other than table or `nil`;
- non-numeric, non-integer, or out-of-range capacities;
- non-numeric, negative, non-finite, or greater-than-86,400-second timeout;
- method call on a value that is not the expected job or channel;
- invalid `workers.channel()` arity, options, or capacity;
- invalid `opts.channels`, empty/non-UTF-8/NUL name, or non-channel value.

```
local ok, err = pcall(function()  
    babet.workers.spawn(42)  
end)  
assert(not ok)
```

### Failures returned by `spawn`

`spawn` returns `(nil, err)` for runtime failures before or during creation:

- non-transferable args;
- serialization-budget exhaustion;
- JSON serialization failure;
- queue or completion-signal initialization failure;
- `pthread_create` failure.

```
local job, err = babet.workers.spawn("return 1", {  
    fn = function() end,  
})  
assert(job == nil)
```

### Final worker errors

`join()` returns `(false, err)` and `poll()` returns `("error", err)` for:

- invalid Lua syntax;
- uncaught Lua error;

- non-transferable return value;
- caught internal C++ exception;
- internal result deserialization corruption or failure.

The diagnostic does not automatically contain a complete traceback.

## Message serialization

- `job:send: (nil, err)` when the value cannot be serialized;
- `worker.send: (false, err)` for the same case;
- `channel:send: (nil, err)` for the same case;
- `job:recv` and `worker.recv: (false, err)` on an internal anomaly;
- `channel:recv: (nil, err)` on an internal anomaly.

## Garbage collection and destruction

The userdata has a safety-net `__gc`. When collected:

1. the cancellation flag is set;
2. inbox and outbox are closed;
3. queue waiters are awakened;
4. Babet calls `pthread_join` when needed;
5. `pthread` primitives, the completion signal, and internal strings are destroyed.

This prevents freeing state still used by a thread. It can nevertheless block when the worker cannot terminate.

Do not use GC as the normal synchronization mechanism. Keep the job, finish the protocol, then call `join()` or consume the result through `poll()`.

Channel handles also have a `__gc`, but its scope is local: it only releases the reference owned by that Lua state. It never replaces `channel:close()` and does not close the resource while other handles remain.

`tostring(channel)` returns `WorkerChannel(open)` or `WorkerChannel(closed)` from the instantaneous global state.

`tostring(job)` returns an indication such as:

```
Worker(running)
Worker(done)
Worker(error)
```

This reflects instantaneous internal status, not whether the result has already been consumed.

## Signals, environment, and concurrency

### Signals

All six signals managed by `babet.signal` are blocked in worker pthreads. `babet.signal.handle`, `ignore`, and `default raise` inside a worker. The main thread remains the only owner of POSIX callbacks.

### Environment and cwd

`setenv` and `chdir` change process-wide state that cannot safely be mutated while other threads use it. Babet therefore rejects both after the first spawn.

## External resources

Two workers can open distinct files, SQLite databases, or sockets. When they target the same external resource, worker serialization provides no automatic lock.

For SQLite, each worker should open its own connection. `wal = true` and a suitable `busy_timeout` can improve concurrency, but do not replace sound transaction design.

## Cost

Each spawn creates:

- a pthread;
- a complete Lua state;
- Babet modules;
- two queues and their buffers;
- handles for explicitly passed channels;
- JSON copies of transferred data.

Avoid a worker for tiny operations. Batch work or use a few persistent workers when the protocol allows it.

## Features not provided

The module currently does not provide:

- forced worker termination;
- `job:done()` — use `poll()`;
- `babet.workers.cpu_count()`;
- a reusable global pool;
- channels between distinct OS processes;
- shared Lua memory;
- transfer of functions, userdata, or coroutines;
- a binary message format;
- automatic transfer of multiple return values.

Cooperative cancellation and a bounded pool can be built in Lua using the examples on this page. CPU count can be queried through an external program such as `babet.exec("nproc")`, after checking its result table.

## Cookbook

Concrete recipes for common scripting tasks. Each recipe is short and self-contained ; copy, adapt, ship.

### Watch a directory and email new files

A drop directory that mails out every file as it arrives. Uses `babet.inotify` for instant wakeup (no polling), and listens on `close_write` + `moved_to` so the file is guaranteed complete.

```
local w = assert(babet.inotify.new())
assert(w:add("/srv/incoming", { "close_write", "moved_to" }))

babet.signal.handle("TERM", function()
    w:close()
    os.exit(0)
end)
```

```

while true do
    local events, err = w:read()
    if not events then
        if err == "interrupted" then break end
        io.stderr:write("inotify: ", err, "\n"); break
    end
    for _, ev in ipairs(events) do
        if ev.events.overflow then
            -- queue overflowed: rescan the directory
            -- (events were lost)
        else
            local path = "/srv/incoming/" .. ev.name
            send_email(path)
            os.remove(path)
        end
    end
end
end

```

## Ensure a service user exists, then drop to it

A typical install-time check before launching a daemon as a non-root user.

```

local function ensure_user(name)
    if babet.user.exists(name) then return true end
    local ok, err = babet.exec("useradd", {
        "--system",
        "--home-dir", "/var/lib/" .. name,
        "--create-home",
        "--shell", "/usr/sbin/nologin",
        name,
    })
    if not ok then
        return nil, "useradd failed: " .. tostring(err)
    end
    return true
end

assert(ensure_user("myapp"))
local u = assert(babet.user.get("myapp"))
babet.chdir(u.home)

```

## Fetch many URLs in parallel

babet.workers runs Lua on multiple cores. Total wall-clock is close to the slowest single fetch, not the sum.

```

local urls = { "https://a.example/", "https://b.example/",
               "https://c.example/", "https://d.example/" }

local jobs = {}
for i, url in ipairs(urls) do
    jobs[i] = babet.workers.spawn([[
        local url = worker.args.url
        local r, err = babet.http.get(url, { timeout = 10 })
        if not r then return { error = err } end
        return { status = r.status, length = #r.body }
    ]], { url = url })
end

```

```

end

for i, job in ipairs(jobs) do
    local ok, result = job:join()
    print(i, ok, result and result.status or result.error)
end

```

## Graceful shutdown of a long-running script

Catch SIGTERM / SIGINT so the script can close resources properly. Works with `systemctl stop` and Ctrl-C.

```

local log = require("logging")
log.set_level("info")

local running = true
babet.signal.handle("TERM", function()
    log.info("SIGTERM, shutting down")
    running = false
end)
babet.signal.handle("INT", function()
    log.info("Ctrl-C, shutting down")
    running = false
end)

while running do
    -- main loop ; blocking calls return (nil, "interrupted") when
    -- a handled signal arrives, so the loop exits quickly.
    local line, err = sock:recv_line()
    if not line then
        if err == "interrupted" then break end
        log.error("recv: ", err); break
    end
    process(line)
end

sock:close()
log.info("clean shutdown")

```

## Read a TOML config, validate, persist to SQLite

```

-- Load TOML
local f = assert(io.open("config.toml", "r"))
local body = f:read("a"); f:close()
local cfg, perr = babet.toml.decode(body)
if not cfg then error("config.toml: " .. perr) end

-- Validate (cheap manual checks ; for a real schema, build one in Lua)
assert(type(cfg.server) == "table", "missing [server]")
assert(type(cfg.server.host) == "string", "missing server.host")
assert(math.type(cfg.server.port) == "integer", "server.port must be integer")

-- Open db, persist
local db = assert(babet.sqlite.open("state.db", { wal = true }))
db:exec([[CREATE TABLE IF NOT EXISTS config (k TEXT PRIMARY KEY, v TEXT)]])
db:exec(["INSERT OR REPLACE INTO config VALUES (?, ?)",
    { "host", cfg.server.host }])

```

```
db:exec("INSERT OR REPLACE INTO config VALUES (?, ?)",
        { "port", tostring(cfg.server.port) })
db:close()
```

## Cache with a human-readable TTL, expiry as ISO timestamps

Parse a TTL from config ("15m", "2h30m", "1d"), compute a deadline, and log it in a uniform UTC format that grep'ing across log files actually works on.

```
-- Config from TOML, CLI, env, anywhere.
local raw_ttl = "15m"

local ttl, err = babet.time.parse_duration(raw_ttl)
if not ttl then error("bad TTL '" .. raw_ttl .. "': " .. err) end

local deadline = babet.now() + ttl
print(string.format("[%s] cache entry created, expires at %s (TTL %s)",
    babet.time.iso(),
    babet.time.iso(deadline),
    babet.time.format_duration(ttl)))

-- ...later, somewhere else in the script:
if babet.now() >= deadline then
    print("[ " .. babet.time.iso() .. " ] cache miss: entry expired")
end
```

Useful properties of this pattern :

- `babet.time.iso()` is always UTC, always 20 characters, always the same shape. `grep` and `sort` work on the timestamps directly.
- `parse_duration("15m")` returns an integer, so `now() + ttl` is exact (no float drift over long durations).
- `format_duration(ttl)` round-trips : `parse_duration( format_duration(n) ) == n` for every `n >= 0`. Safe to use as a canonical representation in logs or databases.

## Pretty-print a Lua value

The bundled `inspect` module gives readable output for nested tables.

```
local inspect = require("inspect")
print(inspect({ a = 1, b = { c = 2, d = { 3, 4, 5 } } }))
-- { a = 1, b = { c = 2, d = { 3, 4, 5 } } }
```

## Hello there

The minimal Babet script :

```
babet.helloThere()
```